Regulatory Capture by Superstars: Lobbying, Innovation, and Industry Concentration*

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Abstract

A long-established concern about corporate lobbying is regulatory capture: incumbent firms shaping regulation in their favor. We study which regulations firms lobby for and how this affects concentration, innovation, and welfare. Using LLMs, we classify federal lobbying reports of public firms and trade associations, assigning each request to a policy tool from a new comprehensive taxonomy, with the direction and scope of the requested change. Lobbying for firm-specific benefits such as tax decreases and government contracts is one-sided: nearly every firm asks for more. Lobbying over regulatory costs is instead contested: the largest firms are the most likely to ask for higher costs, while trade associations mostly ask for lower ones. To quantify the consequences of lobbying, we build a dynamic general equilibrium model of oligopolistic competition and innovation in which firms lobby for firm-specific transfers and over industry-wide regulatory costs, set in a contest between strategic incumbents and an association of the competitive fringe. In the model estimated on US data, lobbying raises regulatory costs and lowers welfare through industry composition: each incumbent innovates more, but industries become more concentrated, lowering output and growth. Almost all the welfare loss comes from lobbying over regulation rather than for benefits.

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1 Introduction

Americans rank the role of money in politics among the country's most serious problems (Pew Research Center, 2026). One reason for concern is regulatory capture: firms can use money to shape regulation in their favor at the expense of other firms and the public interest. For example, large AI companies are asking the government to regulate their industry. Some might think the goal is to ensure AI safety, while others might argue it aims to reduce the number of smaller competitors and secure higher, more stable market share and profits, which is regulatory capture. Concerns about regulatory capture extend well beyond AI. In 2015–2016, Altria and Reynolds lobbied on a bill that would have exempted the vaping products already on sale from premarket review by the Food and Drug Administration; in 2020–2021, both lobbied in support of FDA regulation of synthetic-nicotine products. H&R Block, one of the two largest tax preparers, lobbied for minimum competency standards for paid tax preparers. C.H. Robinson, the largest freight broker, lobbied to raise the bond that brokers must post to guarantee their payments to shippers and truckers; in 2012, Congress raised it from $10,000 to $75,000, the same amount for every broker.

Firms shape government policy mainly through lobbying and political donations, and lobbying is their largest political expense (for public firms, it is 16 times their PAC contributions). The main advantage of analyzing corporate lobbying is that mandatory lobbying reports show which policies firms discussed with the government. These policies fall into groups with very different economic effects. For instance, one group of policies benefits firms without directly harming other firms: increasing subsidies, appropriations, and contracts; reducing tax rates. The other group affects industry- or economy-wide regulatory costs of operation: compliance rules, safety/certification standards, data privacy, licensing. Unlike the first group, these act as fixed rather than variable costs, and they can disproportionately hurt smaller firms, increase concentration, and raise barriers to entry (Ryan, 2012; Johnson et al., 2023; Kalmenovitz et al., 2025). These groups therefore can have very different implications for concentration and welfare.

We ask the following *research questions*. First, what policies do firms lobby for? Although a vast literature documents how much firms spend on lobbying, how persistent lobbying is, and which broad topics it covers, we do not know which specific policy tools firms lobby for, which position they take on each, or how often they lobby on them. To answer this question, we measure which policies firms lobby for by using LLM classification across the universe of US federal lobbying reports. We assign each request a firm makes to a specific policy tool from a new, comprehensive taxonomy, together with the direction of the requested change and whether the firm supports it.

Second, how does lobbying shape concentration, innovation, and growth, and does it reduce welfare? The answer is not obvious and requires analyzing not only static concentration but also dynamic effects. For instance, when regulatory costs increase, they increase the profits of large incumbents. This can reduce welfare through an inefficient reduction in supply and misallocation. However, profits fund innovation; thus, regulations could accelerate growth and raise welfare. Additionally, higher regulation due to lobbying might bring higher welfare by correcting externalities, but we abstract from the non-consumption effects of regulation in this paper. To quantify the net effect of lobbying on consumption-equivalent welfare, we build a dynamic general equilibrium model of oligopolistic competition and innovation in which firms lobby for firm-specific transfers and over industry-wide regulatory costs, and we evaluate the counterfactual of banning lobbying.

Empirical results. To learn what firms lobby for, we construct a new dataset of the specific policy tools the firms are lobbying for using the LLM classification of federal mandatory lobbying reports from 2008 to 2022. We analyze the free text of lobbying reports and, for each lobbying entity, extract each lobbying request and assign it to a concrete policy tool (corporate taxes, subsidies, appropriations, R&D credits, privacy regulation, licensing, etc.) from a new taxonomy of 87 instruments. For each instrument requested, we record the direction of change, whether the filer supports it, and which firms and industries the policy applies to. We focus on public firms, which lobby the most and whose size, costs, and innovation we observe, and on trade associations, the nonprofits that represent firms, whose revenues and dues we observe in their tax filings. We classify 65% of lobbying dollars. We find that large firms lobby on their own, and that firms lobby most on appropriations and on regulatory costs such as privacy, licensing, and reporting rules. Almost all firms and associations ask for more appropriations and lower taxes, but they disagree on regulatory costs: 62% of the largest 5% of firms within their NAICS4 industries, 46% of all public firms, and 35% of associations ask for higher costs. Lobbying participation rises steeply with firm size and is persistent.

Motivated by these findings, we focus on three lobbying instruments: regulatory costs (e.g., privacy, licensing, and compliance requirements); reductions in the corporate tax rate; and increases in procurement contracts. We chose these instruments for several reasons. First, they let us study the industry-wide contest over regulatory costs alongside the own-benefit instruments, taxes and procurement, which are the focus of the existing literature. They also let us compare the welfare consequences of the two kinds of lobbying; additionally, these instruments have measurable outcomes. From the data, we measure participation in each instrument by firm size, participation persistence, and returns to lobbying, which we observe in tax rates and procurement contracts. For regulatory costs, we observe their total

level and how firms' lobbying affects the enactment of bill provisions that raise or lower them.

Quantitative model. In the data, we see that large firms lobby on their own, while small firms don't lobby by themselves but can lobby through trade associations. Thus, we build a dynamic quantitative model in which each industry consists of a few large strategic firms, the superstars, and a mass of atomistic fringe firms. Superstars compete à la Cournot, and fringe firms are perfectly competitive. Superstars innovate to raise their productivity, and fringe firms can make a breakthrough innovation and join the oligopoly. Fringe firms differ in their fixed costs of operating, so higher regulatory costs make the marginal fringe firms stop producing.

Large firms lobby for lower corporate taxes and for more procurement contracts, and all firms lobby over regulatory costs. To lobby on an instrument, a firm pays a fixed participation cost, and it can make its lobbying more effective by spending more. Because fringe firms are atomistic, none of them would lobby on its own, so they lobby through a trade association that acts on behalf of its members. Regulatory costs are set in a contest that aggregates the lobbying of firms that want them higher against the lobbying of firms that want them lower, and firms take into account how regulatory costs change the number of fringe firms and how many break through into the oligopoly.

In the estimated model, large firms lobby for higher regulatory costs, while the trade association mostly lobbies against them. The largest firms favor higher regulatory costs mainly because their profits rise as marginal fringe firms exit; the smallest superstars favor them mainly as protection against displacement by innovating entrants. On net, lobbying raises regulatory costs by 65%. This reduces the number of producing fringe firms by 1.8%, raises the profits of large firms, and lowers the rate at which fringe firms break through into the oligopoly. Industries become more concentrated.

Lobbying lowers welfare through an industry composition effect. With higher regulatory costs, each large firm innovates more. But industries become more concentrated: fewer fringe firms break into the oligopoly, resulting in fewer innovating oligopolistic firms and lower aggregate innovation. Overall, lobbying reduces consumption-equivalent welfare by 0.4% through lower output, higher compliance costs, and slower growth. The paper provides a welfare analysis of how lobbying-driven regulatory costs affect consumption and labor supply, and we abstract from any offsetting or complementary effects on the non-consumption utility effects of regulation.

Importantly, we can decompose the welfare effect by policy instrument. Lobbying over regulatory costs accounts for almost the entire welfare loss, while tax and procurement lobbying, the focus of the literature, accounts for almost none: it mostly shifts money from

the household as a taxpayer to the household as a firm owner. Thus, not all lobbying is the same, and we need to think carefully about which types of lobbying should be allowed. The most salient outcome of lobbying, large firms paying lower taxes, is not the one that reduces welfare; it is lobbying over regulatory costs.

Related literature. This paper contributes to three strands of the literature. The first is the empirical literature on lobbying. It documents that lobbying rises with firm size and is persistent over time (Huneeus and Kim, 2024). Arayavechkit et al. (2018) find that taxes are the most frequent lobbying topic, measure lobbying frequency by general issue code, and study tax lobbying by atomistic firms. Closest to our paper, Macedoni and Weinberger (2026) classify lobbying reports by topic, such as investment, trade policy, and labor policy, and find that firms with low capital intensity are more likely to lobby for increases in fixed costs rather than variable costs. We contribute a systematic classification of the specific policy instruments firms and associations lobby for, such as corporate taxes, subsidies, appropriations, R&D tax credits, privacy regulation, and licensing, along with the requested change, the filer's position, and the persistence of lobbying on each instrument.

The second strand is the theoretical literature on lobbying. In static models of lobbying contests, regulation is determined by aggregating the pressures of groups lobbying pro or against the policy (Becker, 1983; Becker and Mulligan, 2003; Grossman and Helpman, 1994; Bombardini, 2008; Kang, 2016). Dynamic models study atomistic firms that lobby for lower taxes or higher sales (Arayavechkit et al., 2018; Huneeus and Kim, 2024; Hossein Dad and Nguyen, 2025; Macedoni and Weinberger, 2026). However, because these firms are atomistic, they can not move industry-wide regulation on their own. We contribute by developing a dynamic quantitative model in which firms lobby both over industry-wide regulation, in a contest between strategic firms and an association of atomistic firms, and for firm-specific benefits such as taxes and procurement.

The third strand is the literature on oligopoly and innovation (Ericson and Pakes, 1995; Akcigit and Ates, 2023). The closest paper, and the model we build on, is Cavenaile et al. (2026), which introduces innovating oligopolists and a competitive fringe in general equilibrium. We add lobbying and adapt the model so that lobbying can operate through the right channels. First, fringe firms differ in their fixed costs, letting us study how regulation affects different types of small firms and adding a selection margin. Second, a breakthrough by fringe firms doesn't just create new firms at the lowest productivity to become strategic; the entrant can displace incumbents and even leapfrog the leader. This creates an additional lobbying motive: large firms lobby not only to raise their own profits and market shares, but also to protect themselves from innovative smaller firms. Thus, we contribute to the oligopoly and innovation literature by enabling the analysis of how lobbying and regulation

affect industry structure and innovation.

The paper is organized as follows. Section 2 describes the data and classification, Section 3 presents the empirical facts, Section 4 develops the model, and Section 5 describes the model estimation. Section 6 explores the effects of lobbying, its welfare cost, and the counterfactuals. Section 7 concludes.

2 Measuring what firms lobby for

To study the welfare consequences of lobbying over regulations, we first need to know what firms lobby for. Existing lobbying literature shows how much each firm spends, the broad issue areas it lobbies on, and its positions on bills, but not which specific policy tool it mentions, how the tool changes, or whether the firm supports that change. We construct these measures from the free text of quarterly federal lobbying reports filed under the Lobbying Disclosure Act (LDA) in 2008-2022 (see the sample report in the Appendix A). It describes the specific issues each client lobbied on, often mentioning the bills and regulations. We turn this text into a dataset of lobbying requests.

Our unit of observation is a lobbying report: one client, one registrant (a lobbying firm or the client's own lobbyists), and one quarter. We match the client to a public firm or trade association. We split the free text of the reports into individual requests and assign each to one of 87 policy instruments, along with the requested change relative to current law, who the policy covers, and the client's position. For example, the request "support a decrease in the corporate tax rate" is recorded as the corporate income tax rate, a decrease, covering all firms, and supported by the client. At the end of the analysis, for every client, quarter, and instrument, we record whether the client lobbies on the instrument and the side the client takes. The rest of this section describes how we build the dataset in more detail and how we check its correctness.

2.1 Reports and clients

Every registered lobbyist files a report (form LD-2) for each client every quarter. The report names the client and the lobbyist, states the amount spent, and, for each general issue area, describes in free text the specific issues lobbied, often citing bills, public laws, and regulations. We use every report filed from 2008 to 2022, obtained from the Senate Office of Public Records and cross-checked against LobbyView (Kim, 2018) and OpenSecrets (Center for Responsive Politics, 2024). Free issue text is largely missing before 2006: among reports with a positive amount, 21% include it in 1999–2006, against 96% in 2007 and 98% in

2008–2022, the latter carrying 99% of the dollars. We start the dataset in 2008 because the Honest Leadership and Open Government Act of 2007 made the issue-reporting requirement stricter, requiring quarterly rather than semiannual reporting; thus, every year in our sample has the same reporting frequency and better issue-text coverage.

We match clients to public firms and trade associations, which account for 65% of lobbying dollars, 36% of reports, and 52% of the issues listed in 2008–2022 (Figure 1). We choose this set of clients because we can observe industry, size, costs, and innovation for public firms, and the revenues and dues for associations, which also represent smaller firms. We match public firms to the Compustat dataset by name, and we match their subsidiaries and merged companies using CRSP, their SEC filings, and EDGAR, so we attribute a subsidiary's lobbying to its parent. We select firms headquartered in the United States and obtain 75,957 firm-years for 9,690 firms across 268 four-digit NAICS industries. Trade associations are matched to the IRS register of business leagues (501(c)(6) organizations) and assigned to the industries they represent.

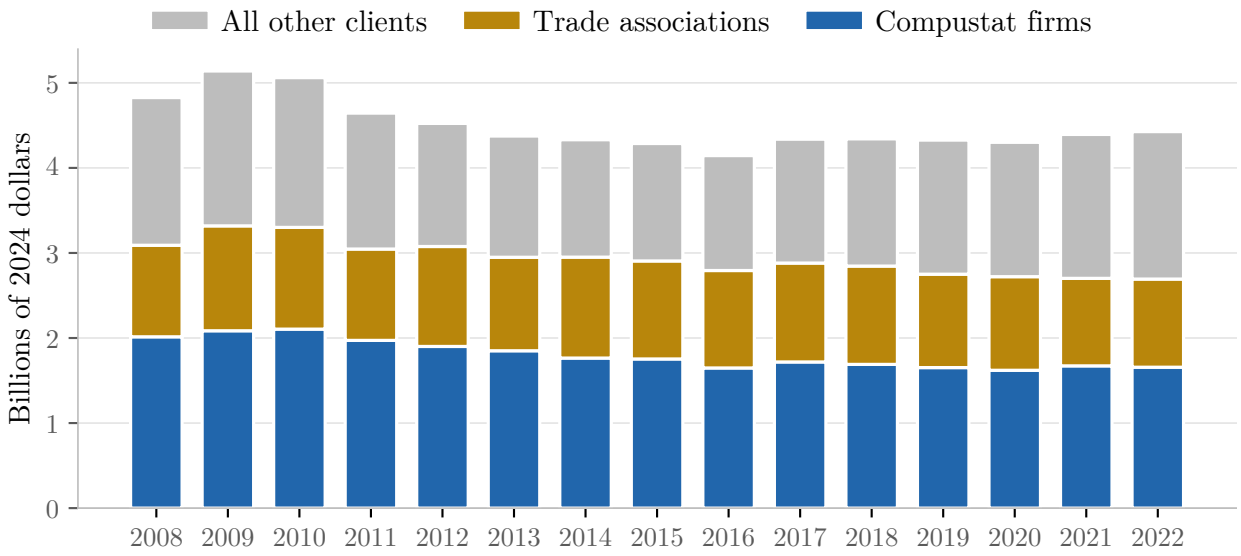


Figure 1. Who Lobbies: Compustat Firms and Trade Associations

Notes: Federal lobbying spending by type of client, billions of 2024 dollars, 2008–2022. Trade associations are clients matched to an IRS 501(c)(6) organization.

2.2 From text to policy instruments

The issue text is the central object of the measurement. A single issue text block often contains several requests on different policies, including bills and regulations. We process it in five steps.

Extracting requests. Each issue text block is split into requests that each address a single policy. We separate text blocks into requests using pattern rules for unambiguous structures, such as enumerated lists; for blocks whose boundaries require judgment, we split them with a large language model and require it to reproduce the original text verbatim (Appendix J).

Bills. Every cited bill and public law is matched to the congressional record and corrected for mistyped numbers using the titles and context of the bills; cited regulations are matched to their titles and summaries on Regulations.gov. We classify all bill provisions the same as issue text.

Instruments. Each request is assigned to an instrument in a taxonomy of 87 policy instruments. We created the taxonomy that unites the government policies that affect firms from the Treasury tax-expenditure list, the taxonomy of regulatory forms of Xie and Perez (2018), the Harmonized Tariff Schedule and the UNCTAD classification of non-tariff measures, the federal Assistance Listings and the Federal Acquisition Regulation, and the patent, exclusivity, and immigration laws, with the tools-of-government catalog of Salamon (2002) as a check on completeness. We map each instrument to the part of the firm's problem it moves, which we call its *wedge* (Table 1), such as affecting labor costs, output, innovation cost, etc. Instruments are defined by their effect on the firm rather than who they apply to: a research credit applying to one firm and one applying to every firm are the same instrument with different eligibility (Appendix K).

Direction, coverage, and side. For each request, the classifier records the change requested relative to current law (introduce, raise, lower, extend, or repeal, with the magnitude when the text states it), whom the policy covers (industry, geography, product, legal form, and thresholds on firm characteristics), and the client's position on it (support, oppose, amend, monitor, or unstated). For each instrument, the codebook records its payoff sign for different types of firms (buyer vs seller of the good, domestic vs foreign firms) and its shape (proportional or quasi-fixed). The *direct payoff* of a request for each firm is the direction of the change times the instrument's sign: increasing a credit is favorable, increasing an excise tax is not. The *side* on the instrument is the position multiplied by the direction of the change: supporting a bill that raises a compliance requirement or opposing its repeal both attribute the client to the side of supporting higher regulatory costs. However, positions are stated rarely in the lobbying reports: the client supports or opposes the policy only for 6% of firms' matched issues and 16% of associations' (Table 2).

From requests to firm measures. We record the client as lobbying on an instrument in a quarter if any of its reports name the instrument, cites a bill or regulation whose provisions move only that instrument or the majority of provisions are about this instrument. A citation of a bill that moves several instruments in different parts of the firm problem doesn't count

Table 1. Wedges of the Firm's Problem and the Instruments That Affect Them

Wedge	Instruments (examples)	Number
Demand, s^d	Appropriations and earmarks, procurement contracts, buyer purchase subsidies, tariffs and quotas, antidumping duties	14
Regulatory cost, s^R	Licenses and charters, product certification, pre-market approval, facility permits, recurring reporting, labeling, data privacy and cybersecurity, liability rules	18
Output, s^y	Per-unit production credits, excise taxes, emission charges, performance standards with per-unit costs, quotas	17
Profit tax, s^T	The corporate income tax, deductions, reduced rates on classes of income, foreign income, losses	7
Innovation cost, s^i	Research credits and expensing, research grants, patents, product exclusivity	5
Capital cost, s^k	Investment credits, accelerated depreciation, tax-exempt bonds, federal lending	6
Labor cost, s^w	Hiring credits, payroll taxes, wage floors, worker protections, visas	8
Set prices, s^p	Government payment rates, mandatory discounts, price ceilings and floors, rate regulation	5
Competition, s^A	Merger control, platform regulation, antitrust exemptions, mandated access	4
Lump-sum transfers, s^t	Grants and direct payments, regulatory user fees	2
Place-based policies	Enterprise zones	1

Notes: The codebook's 87 instruments are categorized by the wedge of the firm's problem they affect. A regulatory cost is a requirement whose cost of compliance does not scale with the firm's output; requirements whose cost scales with output are output wedges. Appendix K lists the instruments.

as lobbying on them; otherwise, citing a defense authorization would make a firm active on procurement, export controls, and cybersecurity. A client's side on an instrument in a year is the share of its requests on the instrument that point to a higher level, averaged over its quarters, counting only requests stated in its own words at high or medium confidence and citations whose provisions move a single instrument; the client asks mostly for a higher level when this share exceeds 50%.

Coverage and validation. Table 2 reports the coverage of the classification by Claude Opus 5.5. Among issues mentioned in the reports, 46% for firms and 47% for associations are matched to a policy instrument in the text itself, and about a third are bare citations of a bill or rule, which enter through the provisions of the legislation; the rest contain no firm policy instrument or are too vague to classify, such as "climate change" or "tax reform". Low confidence issues (10% of firms' and 11% of associations' matched issues) are dropped. Classifying the same text twice gives the same instruments in 94% of cases when neither classification has low confidence, and the same position in 99%. An independent classification

by authors assigns the same instrument to 83% of a sample of 1,000 association issues from high- and medium-confidence matches, and to 89% of a sample of 1,000 bill provisions. The sides we record agree with the positions recorded by MapLight for 95% of the 492 firm–bill pairs both cover. A hand-coded validation sample, which we will use to debias the estimates following Carlson and Dell (2025), is in progress.

Table 2. Classification: Coverage and Confidence

	Compustat firms 2008–2022	Industry associations 2008–2022
Reports with issue text	169,612	105,458
Issues listed in the reports	1,298,944	1,317,793
matched to a policy instrument	46%	47%
bare citation of a bill or rule	31%	35%
no policy instrument	11%	7%
unclear (e.g., "climate change", "tax reform")	13%	10%
Confidence of the matched issues:		
high	51%	46%
medium	37%	40%
low (dropped)	10%	11%
Position stated in the matched issues:		
support or oppose	6%	16%
monitor	5%	4%
unstated	89%	80%

3 What Firms Lobby For

We document four facts about the content and the structure of lobbying. Together they motivate the model in Section 4: lobbying is concentrated on two kinds of policy, firms agree on one and disagree on the other, and the decision to lobby is persistent across instruments.

Participation in lobbying. Lobbying is concentrated among large firms and large trade associations. Figure 2 reports the share of all public firm-years and association-years with a lobbying report, by the firm's sales percentile within its industry: 6.1% of the firm-years below the industry's median of sales have a report, 19% between the median and the 80th percentile, 35% between the 80th and the 90th, 48% between the 90th and the 95th, and 63% among the largest 5%. The same pattern holds for associations. The facts below describe what these clients lobby on.

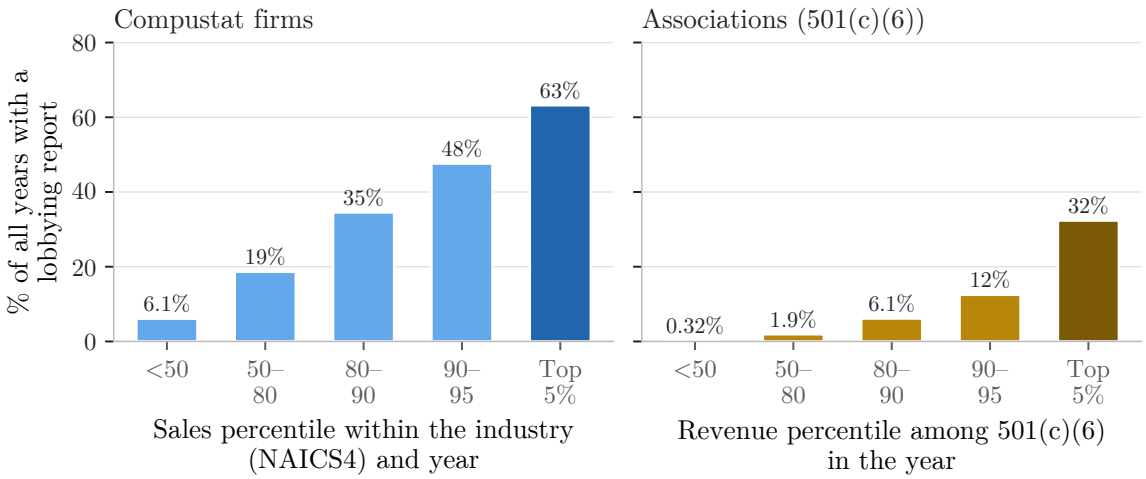


Figure 2. Lobbying by Firm Size

Notes: Share of all firm-years with a lobbying report (a report with a positive amount that names a classified instrument), by the firm's sales percentile within its industry (four-digit NAICS) and year; 75,957 firm-years of Compustat firms headquartered in the United States. Associations: share of the 3,735 industry-years in which at least one industry association of the industry has a lobbying report. 2008–2022.

3.1 Fact 1: Firms lobby most on government demand and on regulatory costs

Figure 3 reports, for each wedge, the share of the quarters with a lobbying report in which the firm (or association) lobbies on it. Two wedges dominate. Lobbying firms lobby on government demand (appropriations, procurement, and buyer subsidies) in 63% of their quarters and on regulatory costs in 50%; for trade associations the shares are 65 and 53%. The profit tax, which dominates public discussion of corporate lobbying, appears in about a fifth of the quarters (20% for firms, 17% for associations). The largest 5% of firms lobby on every wedge more often, on government demand in 75% of their quarters and on regulatory costs in 66%. These shares describe what lobbying clients lobby on.

At the level of instruments, program funding and appropriations lead for every group (32% of firm-quarters, 37% of association-quarters, and 43% for the largest 5% of firms), followed for firms by data privacy and cybersecurity, the most lobbied regulatory cost (21%), technology and performance standards (20%), procurement contracts (19%), buyer subsidies and purchase mandates (19%), and recurring reporting and recordkeeping duties (14%). Associations lobby more than firms on standards, reporting duties, and payment rates, and far less on procurement (9 versus 19%). Each industry lobbies on what it sells to the government and on what its regulator controls, and a regulatory cost is among the three most lobbied instruments in five of the six largest lobbying sectors: conduct rules on transactions in health care services and in finance, pre-market approval in pharmaceuticals and devices,

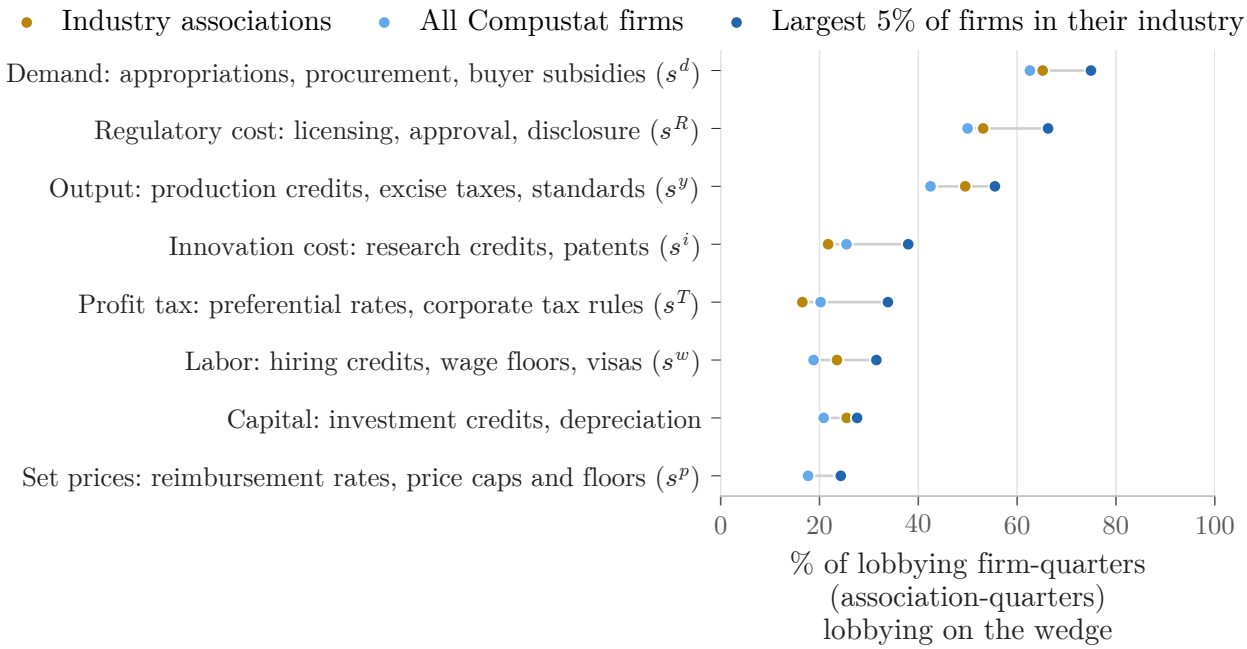


Figure 3. Lobbying by Wedge of the Firm's Problem

Notes: Share of lobbying firm-quarters (association-quarters) in which the client lobbies on the wedge. 52,079 Compustat firm-quarters, of which 14,970 of the largest 5% of firms in their industry by sales; 48,525 association-quarters; 2008–2022.

privacy in technology, and facility permits in energy and utilities (Table 3).

Table 3. What Each Sector Lobbies on Most

	Most common	Second	Third
Aerospace and defense	Procurement (s^d)	Appropriations (s^d)	R&D grants (s^i)
Health care services	Payment rates (s^p)	Appropriations (s^d)	Conduct rules (s^R)
Pharma and devices	Appropriations (s^d)	Payment rates (s^p)	Pre-market approval (s^R)
Technology	Appropriations (s^d)	Procurement (s^d)	Privacy (s^R)
Finance and insurance	Conduct rules (s^R)	Buyer subsidies (s^d)	Privacy (s^R)
Energy and utilities	Technology standards (s^y)	Facility permits (s^R)	Buyer subsidies (s^d)

Notes: The three instruments lobbied in the largest share of the sector's lobbying firm-quarters, Compustat firms, 2008–2022. Conduct rules: rules on how a transaction is made.

3.2 Fact 2: Firms agree on transfers and disagree on regulatory costs

Figure 4 reports, for each instrument, the share of the firms taking a side that ask for more of it. On instruments that benefit the firm directly, firms speak with one voice: 97 to 100% of those taking a side ask for more funding, contracts, grants, and credits (52%

on patents and copyright, an instrument that favors firms holding intellectual property and burdens those using it), 97% of those taking a side on the profit tax ask for a lower one, and 99% ask for more contracts. On regulatory costs, firms split. Depending on the instrument, between 5% (facility permits) and 58% (conduct rules on transactions) of the firms that take a side ask for *higher* regulatory costs; pooling licenses, reporting duties, labeling, and privacy, 46% of firms and 35% of trade associations do.

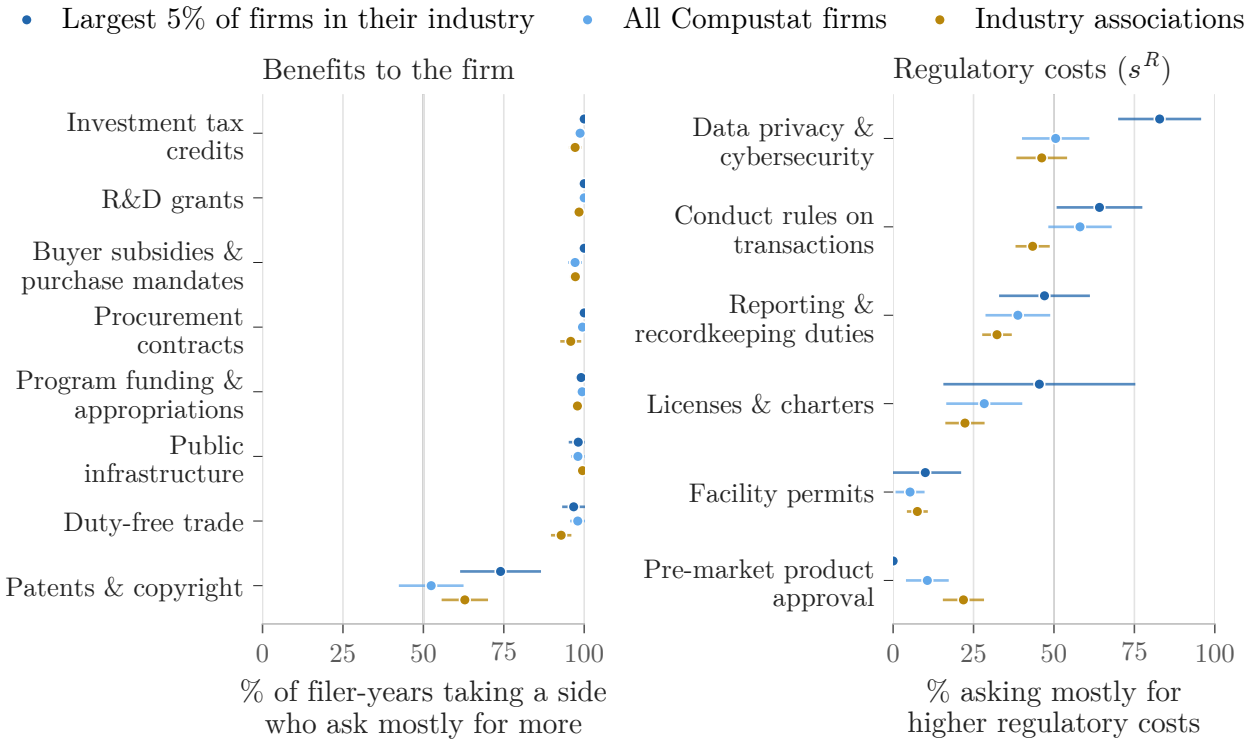


Figure 4. The Side Firms Take: Benefits and Regulatory Costs

Notes: Left: share of the firm-years taking a side that ask mostly for more of the benefit. Right: share asking mostly for higher regulatory costs. A filer-year's side is the share of its requests on the instrument that point to a higher level, averaged over its quarters (Section 2.2). 95% confidence intervals. 2008–2022.

The firms that ask for higher regulatory costs are disproportionately large. The share asking for higher regulatory costs is 27% among firms below the 80th percentile of their industry's sales, 40% between the 80th and 95th percentiles, and 62% among the largest 5%; trade associations ask for higher regulatory costs less often than the largest firms, at 35% (Figure 5). On privacy, the most lobbied regulatory cost, 83% of the largest firms taking a side ask for higher costs, against 51% of all firms. However, this pattern rests on few observations and reflects cross-industry differences: only 271 firm-years take a side on regulatory costs. Across all firms, a log point of sales raises the probability of asking for higher regulatory costs by 4.1 percentage points (standard error 2.8); within an industry,

the estimate is −2.4 points (5.2), and only 11 of 53 industries have firms stating a side both above and below the industry's 80th percentile of sales. We therefore target the overall shares of sides in estimation and treat the size gradient as an untargeted pattern.

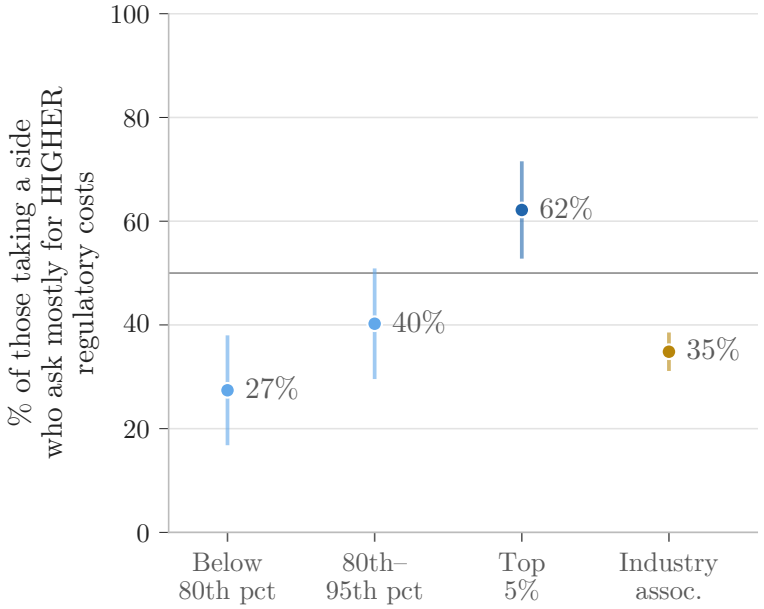


Figure 5. Share Asking for Higher Regulatory Costs, by Firm Size

Notes: Share of the firm-years (association-years) taking a side on regulatory costs (licenses, reporting duties, labeling, privacy) that ask mostly for higher regulatory costs, by the firm's sales percentile within its industry and year. 271 firm-years and 778 association-years take a side; 2008–2022. 95% confidence intervals.

A natural concern is that the sides we observe are selected. Sides are written down rarely: for 1.9% of requests on regulatory costs and 4.2% of requests on other instruments, and 5% of the firm-years that lobby on regulatory costs state a side. Three checks show that the stated sides are representative, with one exception. First, large firms state a side in more years only because they file more requests (a median of 70 a year, against 6 for the smallest firms); at the same number of requests, the largest fifth of firms in an industry and year are no more likely than the smallest fifth to state a side on regulatory costs (−4 percentage points, standard error 2.0). Second, on most instruments, whether a side is written down depends on the lobbying firm that writes the report, not on the client: a client's reports state a side in 11.0% of cases when its lobbying firm states sides for its other clients and in 2.6% otherwise, and clients that state sides often do not choose lobbying firms that do (correlation 0.02). On regulatory costs, it is the client's own habit (5.2 versus 0.2%). Third, and this is the exception, the 27 firms that state a side on regulatory costs in more than a third of their other years lean against them: within an industry and year, they ask for higher regulatory costs 40 percentage points less often than firms that state a side only once (standard error 14). Counting each firm once rather than each firm-year would raise the share of firms asking

for higher regulatory costs from 46 to 50%; we keep the firm-year share.

3.3 Fact 3: Lobbying rises steeply with firm size on every instrument

We now focus on the three instruments that the model will carry: the firm's own profit tax, its own federal procurement contracts, and *regulatory costs*, the requirements whose cost does not grow with the firm's size (licenses, recurring reporting duties, labeling, and privacy and cybersecurity rules). Reports that include these instruments carry 61% of firms' lobbying dollars, and the instruments span both kinds of policy: tax and procurement are one-sided, regulatory costs two-sided. Participation on each instrument rises steeply with size (Figure 6): the share that lobbies on regulatory costs in a year rises from 0.8% of firm-years below the industry's median of sales to 35% among the largest 5%, on the profit tax from 0.4 to 25%, and on procurement from 1.1 to 17%. Most of this gradient is whether a firm lobbies at all: 6% of the firm-years below the median and 63% of those of the largest 5% have a lobbying report (Figure 2). Among the years in which a firm lobbies, the shares still rise with size, from 13 to 55% on regulatory costs, from 6 to 39% on the profit tax, and from 18 to 27% on procurement (Appendix Figure 13). Within an industry and year, each log point of sales raises the probability that a lobbying firm lobbies on regulatory costs by 9.1 percentage points, on the profit tax by 7.8, and on procurement by 2.7 (standard errors 0.4; 13,669 lobbying firm-years, 2008–2022), against averages of 36, 23, and 22%. Among superstars, the gradient continues by rank: in a given quarter, 34% of industry leaders lobby on regulatory costs, against 25, 21, and 18% of the second through fourth firms (Table 4).

Table 4. Superstars' Participation and Persistence

	Lobbying in a quarter, by rank				Next quarter	
Instrument	Leader	#2	#3	#4	Stay	Join
Regulatory costs	0.342	0.253	0.208	0.176	0.928	0.026
Profit tax	0.266	0.182	0.144	0.126	0.910	0.022
Procurement	0.190	0.111	0.077	0.078	0.895	0.014

Notes: Superstars are the four largest firms of a four-digit NAICS industry-year with sales of at least 10% of the leader's, in industry-years with at least six Compustat firms. Regulatory costs (licenses, reporting duties, labeling, privacy): a firm lobbies in a quarter if any of its reports in the quarter lobbies on them, 2008–2022. Profit tax (all corporate income tax instruments) and procurement contracts: lobbying in the last quarter of the fiscal year, fiscal years 2008–2022. Monitoring counts. Stay: probability of lobbying on the instrument next quarter given lobbying this quarter; join: given not lobbying.

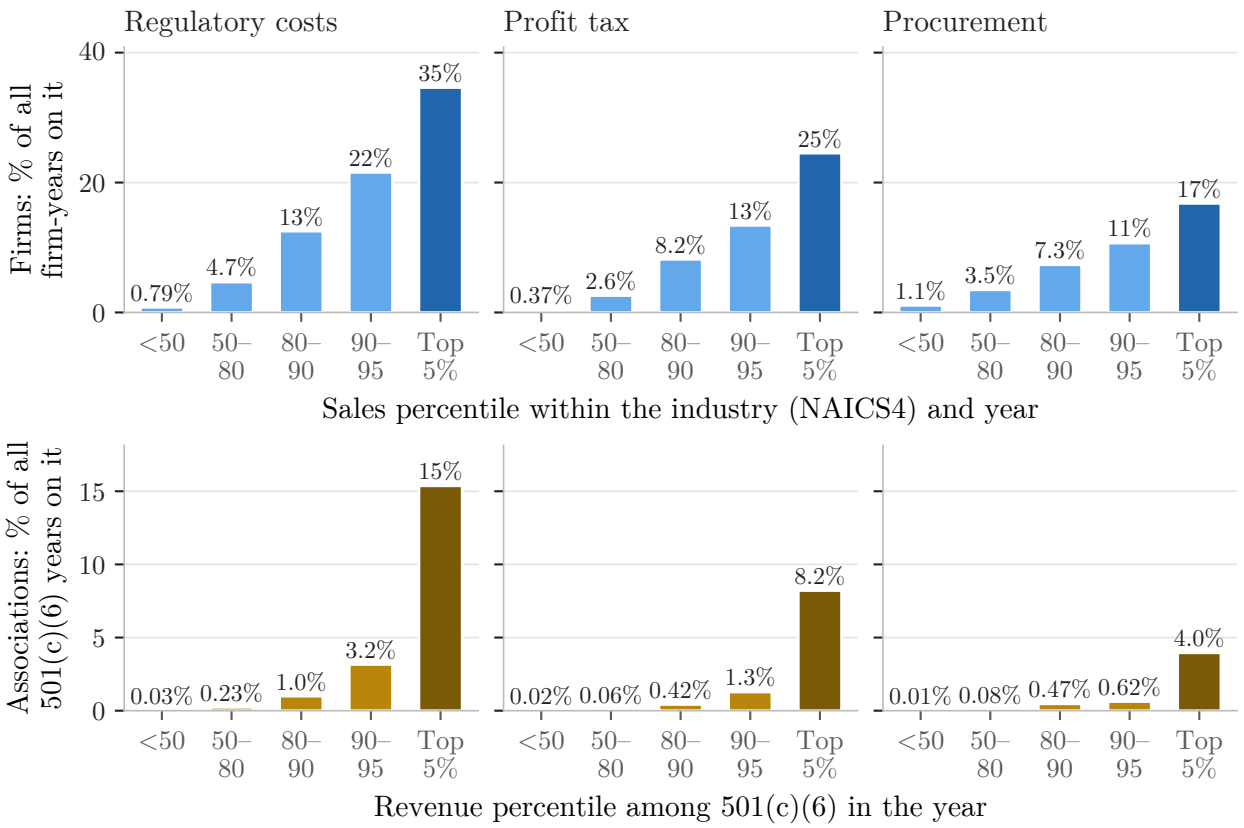


Figure 6. Participation in Lobbying by Firm Size

Notes: Share of all firm-years in which the firm lobbies on the instrument, by the firm's sales percentile within its industry (four-digit NAICS) and year, 75,957 firm-years of Compustat firms headquartered in the United States; for associations, the share of the 3,735 industry-years in which at least one industry association of the industry lobbies on it. 2008–2022. Appendix Figure 13 reports the shares among the years with a lobbying report.

3.4 Fact 4: Lobbying is persistent

Nine in ten firms that lobby on an instrument in one quarter lobby on it in the next: 91% on regulatory costs, 89% on the profit tax, and 89% on procurement among all Compustat firms, and 93, 91, and 91% among the largest 5%. Few firms join in a given quarter: 0.4 to 0.6% of all firms and 1.6 to 3.4% of the largest. Trade associations are as persistent (92% on regulatory costs). Lobbying thus looks like a costly organizational commitment, made at infrequent times and maintained, rather than a flow chosen anew each quarter.

3.5 Associations and superstars

Trade associations and superstars lobby on regulatory costs side by side. In a given industry-quarter of 2008–2022, the association lobbies without any superstar in 28.5% of cases, superstars lobby without the association in 11%, both lobby in 28%, and neither in

32%; the association is active in 57% of quarters, and in 84% of the quarters in which anyone in the industry lobbies on regulatory costs.

3.6 From facts to the model

The facts shape the model in four ways. Because tax and procurement lobbying is one-sided and firm-specific, the model treats it as each firm's own instrument, with an outcome observed in the data: the firm's tax rate and its contracts. Because lobbying over regulatory costs is two-sided and the rule applies to every firm in an industry, the model determines regulatory cost as a contest between firms with opposing stakes, and each firm's stake, and hence its side, is an equilibrium object. Because lobbying is concentrated among the largest firms while small firms lobby through associations, each industry has a few strategic superstars that lobby individually and a competitive fringe that lobbies only through its association. Because participation is persistent and rises with size, the decision to lobby carries a fixed cost that is revised only infrequently. The model computes the same moments as the data: participation by rank, persistence, side shares, association activity, and the outcomes of tax and procurement lobbying.

4 The model

We build a continuous-time economy with multiple industries and a representative household that expands on Cavenaile et al. (2026). Each industry consists of a few large, strategic firms (superstars) and a competitive fringe. Superstars can increase their productivity through incremental innovation, fringe firms can become superstars through breakthrough innovation, and entrepreneurs create fringe firms. Firms can lobby on three policy instruments: the firm-specific profit tax, firm-specific procurement contracts, and the industry-wide regulatory fixed cost. We introduce the key new margin of selection in the fringe: a higher regulatory cost forces the least efficient fringe firms to stop producing, which raises superstars' profits and lowers the rate at which fringe firms break into the superstar tier.

4.1 Environment

Household. The lifetime utility of the representative household is a function of consumption of the final good and leisure

$$U = \int_0^\infty e^{-\rho t} \left[\ln C_t \; - \; \xi \frac{L_t^{1+1/\epsilon_L}}{1+1/\epsilon_L} \right] dt, \qquad C_t \; = \; C_t^p + G_t, \tag{1}$$

C_t is per-period total consumption: private consumption C_t^p, chosen by the household, plus government purchases of the final good G_t, which the household takes as given. L_t is per period labor supply and ϵ_L the Frisch elasticity of labor supply.

Final good. The final good Y_t is produced competitively by aggregating over the continuum of industries $j \in [0,1]$:

$$\ln Y_t \;=\; \int_0^1 \ln(y_{jt})\, dj. \tag{2}$$

We normalize the price of the final good to one. With Cobb–Douglas aggregation, the expenditure share of the final good is constant across industries.

Industry structure. In each period t, each industry j is composed of $N_{jt} \leq \bar{N}$ *superstar* firms, who produce differentiated varieties and compete in quantities, and a continuum of competitive fringe firms with total mass m_t. The output of industry j in period t is a CES aggregate of superstar varieties and the fringe variety:

$$y_{jt} \;=\; \left(\sum_{i=1}^{N_{jt}} y_{ijt}^{\frac{\eta-1}{\eta}} + \tilde{y}_{fjt}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \qquad \eta > 1, \tag{3}$$

where $\tilde{y}_{fjt} = \int_0^{m_t} y_{fkjt}\, dk$ is aggregate fringe output.

Technology. In each period t, each firm, superstar or fringe, produces output with a decreasing returns to scale technology in labor:

$$y_{ijt} \;=\; q_{ijt}\, l_{ijt}^{\alpha}, \qquad \alpha \in (0,1). \tag{4}$$

q_{ijt} is the quality of firm i, α parametrizes the decreasing returns, and l_{ijt} is labor inputs.

Each superstar's quality follows a quality ladder. Denote by $n_{ijt} \in \{0, 1, \ldots, \bar{n}\}$ the quality gap of superstar i relative to the leader in industry j (the leader's gap is zero), with $\bar{n}$ the maximum quality gap to stay an oligopoly. The ladder depth $\bar{n}$ is not a free parameter: it is the largest gap at which the smallest superstar, in every configuration of its rivals' gaps, sells at least 10% of the leader's sales in the static equilibrium with $\bar{m}^{\mathrm{a}}$ active fringe firms, the data's definition of a superstar. Superstar i's quality is

$$q_{ijt} \;=\; q_{jt}^{\text{leader}} \cdot (1+\lambda)^{-n_{ijt}}, \qquad \lambda > 0, \tag{5}$$

and q_{jt}^{leader} is the quality of the leader in industry j in period t.

Quality is constant across firms in the competitive fringe of industry j. The quality level

in the fringe is inherited from j's superstars through technology diffusion given by

$$q_{fjt} \;=\; \zeta\Big(\sum_{i=1}^{N_{jt}} q_{ijt}^{\gamma}\Big)^{1/\gamma}, \qquad \gamma > 0, \tag{6}$$

The elasticity of fringe quality with respect to the quality of superstar i is $q_{ijt}^{\gamma}/\sum_{k=1}^{N_{jt}} q_{kjt}^{\gamma}$: the fringe learns the most from the most productive superstars. Each superstar i internalizes that its innovation spills over to the fringe in proportion to i's quality, eroding part of the innovator's advantage over its fringe competitors. The curvature γ governs how concentrated these spillovers are. As $\gamma \to \infty$, fringe productivity tracks the leader's productivity, as in Cavenaile et al. (2026); a lower γ spreads the spillovers more evenly across the superstars. Together with γ, ζ sets the level of fringe productivity and the fringe's share of industry sales.

Every active firm incurs a fixed cost of regulatory compliance, with R_{jt} set by lobbying: a fringe firm pays $R_{jt}Y_t$ and superstar i pays $\kappa_{ijt}R_{jt}Y_t$, where $\kappa_{ijt} = \bar{\kappa}\,(1+\lambda)^{-\gamma_R n_{ijt}}$, so that a superstar's compliance cost rises with its size. Both $\bar{\kappa}$ and γ_R follow from one parameter, the elasticity $e_R \in [0,1)$ of compliance costs to size: with $a + b\,n$ the least-squares line of $\ln(S_i/S_f)$ on the gap n_i over every superstar seat of every technology state, where S_i/S_f is the sales of superstar i relative to one fringe firm in the static equilibrium with $\bar{m}^{\mathrm{a}}$ active fringe firms, $\bar{\kappa} = e^{e_R a}$ and $\gamma_R = -e_R b/\ln(1+\lambda)$, so that $\kappa_{ijt} \approx (S_{ijt}/S_{fjt})^{e_R}$. In addition, each *active* fringe firm k incurs an idiosyncratic operating cost $\varepsilon_k Y_t$ with ε_k drawn at the firm's birth according to

$$\varepsilon_k \;\sim\; \mathrm{LogN}(\mu_\varepsilon, \sigma_\varepsilon^2), \qquad \mu_\varepsilon = \ln\bar{\varepsilon} - \sigma_\varepsilon^2/2, \tag{7}$$

so that $\mathbb{E}[\varepsilon_k] = \bar{\varepsilon}$. The cumulative distribution and density of ε are denoted $F(\cdot)$ and $f(\cdot)$. A fringe firm can stop and restart producing at no cost. Every instant it operates if and only if $\varepsilon_k \le \varepsilon_{jt}^*$. We call the operating firms *active*; their mass is $m_{jt}^{\mathrm{a}} = m_{jt}\,F(\varepsilon_{jt}^*)$. The draw ε_k is the only source of heterogeneity among fringe firms, and it generates the *selection margin* that we exploit throughout: a higher R_{jt} lowers ε_{jt}^*, and the marginal fringe firms stop producing. Heterogeneous fringe fixed costs and a positive R_{jt} are the central structural deviation from Cavenaile et al. (2026): they create the margin through which regulation, and lobbying over it, operates.

Entry and exit. Entry into the fringe is governed by a unit mass of entrepreneurs who pay $\psi e_t^2 Y_t$ to generate a Poisson rate e_t of new fringe firms. New fringe firms are randomly allocated across industries, implying $m_{jt} = m_t$. Entrepreneurs sell each new firm at its expected value before the draw of ε_k from (7) and remain in the entrepreneur population.

New firms enter only the fringe; a new superstar arises only from a successful breakthrough by a fringe firm. Fringe firms die exogenously at Poisson rate δ; firms that are not producing stay in the pool until they die, then resume production when the industry state makes it profitable. A superstar exits when a rival's innovation would put it more than $\bar{n}$ steps behind the leader, or when it is displaced by an entrant.

Same as in Cavenaile et al. (2026), modeling entry into and exit from the fringe explicitly lets us map entry in the model to new business creation in the data. This contrasts with models in which entrants immediately become industry leaders or are otherwise large enough to interact strategically with incumbents. Our model instead generates the following life cycle: each industry has a large pool of fringe firms, only some of which produce at any time, and a much smaller number of breakthroughs create new superstars.

4.2 Innovation

Innovation occurs through two distinct technologies: incremental innovation by incumbent superstars on the quality ladder, and breakthrough innovation, through which active fringe firms become superstars.

Superstar incremental innovation. Each superstar i in industry j chooses an R&D intensity $z_{ijt} \geq 0$ that generates a Poisson arrival rate z_{ijt} of a one-step quality improvement, at cost

$$\text{cost}_i^{\text{R\&D}} \;=\; \chi z_{ijt}^{\phi} Y_t, \qquad \phi > 1. \tag{8}$$

Upon success, the firm's productivity rises by a factor $(1+\lambda)$. If a follower innovates, its gap falls by one. If the leader innovates, the frontier moves up by one step, and every other superstar's gap rises by one; a superstar whose gap would exceed $\bar{n}$ exits.

Fringe breakthrough innovation. Each active fringe firm k chooses the arrival rate $x_{fkjt} \geq 0$ of a breakthrough that would create a new superstar, at cost

$$\text{cost}_k^{\text{breakthrough}} \;=\; \chi_f \, x_{fkjt}^{\phi_f} Y_t, \qquad \phi_f > 1. \tag{9}$$

A breakthrough creates a new superstar owned by the innovating firm, whose fringe operation continues. The entrant lands $k \geq 0$ steps above the *entry margin*, the lowest position a superstar can hold, with probability

$$\Pr(k) \;=\; (1-p)\, p^k. \tag{10}$$

In an industry with the maximum number $\bar{N}$ of superstars, the margin is the position of the least productive incumbent, whom the entrant displaces; otherwise it is $\bar{n}$ steps behind the leader. An entrant that climbs past the leader becomes the new leader; it lands at most $\bar{n}$ steps above the former leader (landings beyond are placed at $\bar{n}$ steps), and incumbents pushed more than $\bar{n}$ steps behind exit. Unlike in Cavenaile et al. (2026), where entrants join $\bar{n}$ steps behind the leader and only industries with fewer than $\bar{N}$ superstars admit entry, entrants in our model, as in the data, also join full industries and can land anywhere above the margin, including at the top.

4.3 Lobbying

Firms can lobby to affect three policy instruments $I \in \{T, P, R\}$: their profit tax (T), their procurement contracts (P), and the industry-wide regulatory fixed cost (R). A superstar lobbying for instrument I chooses an intensity $b_{Ii} \geq 0$ at cost $\kappa_I \, b_{Ii}^{\phi_I} \, Y_t$, with $\phi_I > 1$, and those costs are not tax deductible.[1] $a_{Iijt} \in \{0, 1\}$ indicates that the firm lobbies on the instrument I.

Profit tax (T). Lobbying on the tax instrument reduces the firm's own effective profit tax rate from the statutory τ_0:

$$\tau_{ijt} \;=\; \frac{\tau_0}{1 + a_{Tijt} \, b_{Tijt}}. \tag{11}$$

Procurement contracts (P). Superstar i holds government contracts worth $g_{ijt} \, Y_t$, with

$$g_{ijt} \;=\; \left(g_0 + a_{Pijt} \, b_{Pijt}\right) (1 + \lambda)^{-\kappa_g n_{ijt}}, \tag{12}$$

where g_0 is the baseline contract, common to all superstars, b_{Pijt} is the increment won by lobbying, and $\kappa_g \geq 0$ governs how contract amounts relate to the firm's distance to the leader, which is a proxy for the firm's sales. Contracts are cost-plus: of the payment $g_{ijt} Y_t$, the share $1 - \theta_g$ covers the cost of the goods the firm delivers to the government, and the share θ_g is the firm's fee, which is taxed as profit.

Regulatory fixed cost (R): the contest. The industry-wide regulatory cost is set by an influence contest among the firms that lobby on it. Each participant l lobbies with intensity b_{Rljt} either for a higher cost (the pro side) or against it (the anti side). Individual intensities aggregate to the total influence of side s according to $B_{jt}^{s} = \sum_{l \in s} b_{Rljt}^{\theta}$ with $\theta \in (0, 1)$. The

[1] Internal Revenue Code §162(e) disallows the deduction of lobbying expenditures for federal income tax purposes.

lobbying outcome, regulatory cost R_{jt}, is determined by the ratio

$$R_{jt} \;=\; R_0 \,\frac{1+\beta^p\, B_{jt}^{pro}}{1+\beta^a\, B_{jt}^{anti}} \tag{13}$$

which is common knowledge to all lobbying participants. With $\theta < 1$, each participant faces diminishing marginal returns to its own effort, and β^p and β^a measure how effective lobbying is for and against the cost. Without lobbying, $R_{jt} = R_0$. The baseline R_0 is set by the compliance level c: $R_0\,(\bar{K} + \bar{m}^{\mathrm{a}}) = c$, where $\bar{K}$ is the average of $\sum_i \kappa_i$ over technology states with $\bar{N}$ superstars, so that compliance costs are a share c of output in a full industry with the observed number of active fringe firms. Each participant joins the side given by the sign of its marginal value of R, derived in Section 4.5. This contest resembles the legislative process, where firms and associations lobby for or against the bills that increase or decrease regulatory fixed costs.

This lobbying contest represents the distinction between lobbying by agents who are relatively large versus those who are relatively small (Olson, 1982, 1965). Superstars have an incentive to lobby individually, as each can affect regulatory fixed costs by itself. For the small fringe firms, however, lobbying is a public good: no fringe firm has an incentive to lobby individually, and would rather free ride on the lobbying efforts of the other fringe firms who share its interests. The role of a trade association, as in Olson (1982), is to solve the collective action problem. Our model is a reduced-form representation of the idea in Olson (1982) that associations sell private benefits—such as trade publications and conventions—to individual firms and produce lobbying insofar as it is a complement to those sales.

Participation in lobbying. Lobbying on an instrument requires a commitment, such as retaining lobbyists and maintaining relationships with legislators. A superstar that lobbies on instrument I pays a flow participation cost $\varphi_{Ii}\, Y_t$, where

$$\ln \varphi_{Ii} \;\sim\; N(\mu_I, \sigma_I^2), \qquad \text{i.i.d. across firms and instruments.} \tag{14}$$

Participation is sticky: the firm draws a new cost at Poisson rate λ_I and can switch its status $a_{Ii} \in \{0, 1\}$ on instrument I only when a new draw arrives; in between, a participating firm keeps paying its current draw. This timing makes lobbying status persistent, as in the data (Section 3.4), and, because the arrivals are independent across firms and instruments, firms revise their participation one at a time. Participation costs are partly shared across instruments. A firm that lobbies on instrument I pays $\varphi_{Ii} \prod_{J \neq I} (1 - \iota_{IJ})^{a_{Ji}}$ on it: each other instrument J on which it also lobbies lowers the cost of I by the share $\iota_{IJ} = \iota_{JI}$, which measures the economies of scope between the two instruments, such as lobbyists,

relationships, and information that serve both. Each discount lies in $[0, 1/4]$, which ensures that, at equal costs, lobbying on a third instrument never lowers a firm's total participation cost.

Fringe trade association. The association organizes with the same timing as a superstar participates: at Poisson rate λ_A it draws a flow organizing cost $\varphi_A Y_t$, with $\ln \varphi_A \sim N(\mu_A, \sigma_A^2)$, and if it organizes, $a_{Ajt} = 1$, it pays this cost until the next draw. An organized association enters the contest (13) as a single participant and chooses the side that maximizes the value of the industry's fringe firms. The cost of the intensity b_{Ajt} for the association is

$$C_A\big(b_{Ajt}\big) \;=\; \kappa_A \, b_{Ajt}^{\phi_A} \, Y_t, \qquad \phi_A > 1. \tag{15}$$

Members of the association, the fringe firms, pay the organizing and intensity costs through dues proportional to their surplus from producing: a fringe firm with cost ε pays $d_{jt}(\varepsilon)\, Y_t$ if it produces, $\varepsilon \le \varepsilon_{jt}^*$, and nothing otherwise, with

$$d_{jt}(\varepsilon) \;=\; \omega_{jt}(\varepsilon)\left[\kappa_A \, b_{Ajt}^{\phi_A} + \varphi_A\right], \qquad \omega_{jt}(\varepsilon) \;=\; \frac{\varepsilon_{jt}^* - \varepsilon}{m_t \int_0^{\varepsilon_{jt}^*} \left(\varepsilon_{jt}^* - \tilde{\varepsilon}\right) dF(\tilde{\varepsilon})}, \qquad \varepsilon \le \varepsilon_{jt}^*, \tag{16}$$

where ε_{jt}^* is the production cutoff (Section 4.5). A producing firm's surplus is $(1-\tau_0)(\varepsilon_{jt}^* - \varepsilon)\, Y_t$, so $\omega_{jt}(\varepsilon)$ is its share of the members' total surplus; the shares integrate to one, and dues cover the association's spending exactly. Every member pays the same fraction of its surplus, and the marginal producer pays nothing, so dues do not affect which firms produce as long as they are feasible: the spending must not exceed the members' after-tax surplus, $(1-\tau_0)\, m_t \int_0^{\varepsilon_{jt}^*} (\varepsilon_{jt}^* - \varepsilon)\, dF(\varepsilon)$.

4.4 Government problem

The government buys goods through the superstars' procurement contracts in the amount of $G_t^S = Y_t \int_0^1 \sum_i g_{ijt}\, dj$, so a contract that a superstar wins by lobbying is an additional purchase. Contracts held by other firms, including fringe firms, are outside the model.[2] For a superstar i it costs $(1-\theta_g) g_{it} Y_t$ to produce the good, and the firm keeps $\theta_g g_{it} Y_t$ as its fee. Thus, the consumer's value of government purchases is its fair value: $G_t = (1-\theta_g)\, G_t^S$. Corporate taxes are collected from every firm and returned to households: the government rebates lump-sum the revenue that the statutory rate τ_0 raises on every firm's tax base. It also pays households baseline lump-sum transfers $\bar{T} Y_t$, a constant share of output, set in the economy without lobbying so that the labor income tax equals 24.4% there (the average

[2]Federal contracts are about one to two percent of the sales of listed fringe firms.

US labor income tax, OECD Taxing Wages) and held at that share in every economy we compare. The labor income tax τ_w adjusts to finance the transfers, the contracts, and the revenue lost to tax lobbying, so that the government budget balances in each period:

$$\tau_w \, w_t \, L_t \;=\; \bar{T} \, Y_t \;+\; G_t^S \;+\; \int_0^1 \sum_{i=1}^{N_{jt}} \left(\tau_0 - \tau_{ijt}\right) \Pi_{ijt} \, Y_t \, dj, \tag{17}$$

where Π_{ijt} is the tax base of superstar i in units of Y_t, defined below in (22). Fixed compliance costs are a resource cost, not government revenue.

4.5 Decisions in the decentralized economy

Households. The representative household owns all firms and chooses private consumption and labor supply, taking government purchases as given:

$$\max_{\{C_t^p,\, L_t\}} \int_0^\infty e^{-\rho t} \left[\ln\left(C_t^p + G_t\right) - \xi \, \frac{L_t^{1+1/\epsilon_L}}{1 + 1/\epsilon_L} \right] dt \quad \text{s.t.} \quad \dot{A}_t = r_t A_t + (1 - \tau_w) \, w_t L_t + T_t - C_t^p,$$

where A_t is the value of all firms, r_t the return on it, w_t the wage, and T_t the government's lump-sum payments: the rebate of corporate tax revenue at the statutory rate, τ_0 times every firm's tax base, and the transfers $\bar{T}Y_t$. Firms' profits, including the superstars' contract fees and net of their R&D, fixed and compliance costs and taxes, reach the household as the return on A_t. The Euler equation and the labor supply condition are:

$$\frac{\dot{C}_t}{C_t} \;=\; r_t - \rho, \qquad \xi \, L_t^{1/\epsilon_L} \;=\; \frac{(1 - \tau_w) \, w_t}{C_t}, \qquad C_t = C_t^p + G_t. \tag{18}$$

Private consumption and government purchases are both shares of output, so on the balanced growth path both grow at rate g and $r_t = \rho + g$. Combining the household's budget with the government's budget (17) and the firms' cash flows gives the resource constraint (45), so the goods and labor markets clear. Because all firm payoffs are proportional to Y_t, firm decisions do not depend on the wage, labor supply, or the labor tax; they affect only the level of output and welfare.

Final good producer. A representative final good producer chooses the quantity of each variety in each industry competitively, which determines the inverse demand functions for

superstars and the price of fringe output:

$$p_{ijt} = \frac{y_{ijt}^{-\frac{1}{\eta}} Y_t}{\sum_{k=1}^{N_{jt}} y_{kjt}^{\frac{\eta-1}{\eta}} + \tilde{y}_{fjt}^{\frac{\eta-1}{\eta}}}, \qquad p_{fjt} = \frac{\tilde{y}_{fjt}^{-\frac{1}{\eta}} Y_t}{\sum_{k=1}^{N_{jt}} y_{kjt}^{\frac{\eta-1}{\eta}} + \tilde{y}_{fjt}^{\frac{\eta-1}{\eta}}}. \tag{19}$$

Product market. Superstars compete à la Cournot. Each superstar chooses y_{ijt} to maximize operating profit $\pi_{ijt} = p_{ijt}\, y_{ijt} - w_t\, l_{ijt}$, taking rivals' quantities and the fringe aggregate $\tilde{y}_{fjt}$ as given, subject to (4), (5) and (19). Fringe firms take p_{fjt} as given. Appendix C derives the best responses and profits. A superstar's markup and operating profit are functions of its share of industry sales $\sigma_{ijt} = p_{ijt} y_{ijt} / Y_t$:

$$M_{ijt} \;=\; \frac{\eta}{(\eta-1)(1-\sigma_{ijt})}, \qquad \pi_{ijt} \;=\; \sigma_{ijt}\Big(1 - \frac{\alpha}{M_{ijt}}\Big)\, Y_t, \tag{20}$$

and each active fringe firm earns

$$\pi_{fjt} \;=\; \frac{(1-\alpha)\,\sigma_{fjt}\, Y_t}{m_{jt}^{\mathrm{a}}}, \tag{21}$$

where σ_{fjt} is the fringe's share of industry sales. Fringe firms price at marginal cost, but decreasing returns leave them with positive operating profits. Given productivities, all sales shares depend only on the industry's productivity gaps and the active fringe mass, and $\partial \pi_{ijt} / \partial m_{jt}^{\mathrm{a}} < 0$, $\partial \pi_{fjt} / \partial m_{jt}^{\mathrm{a}} < 0$.

Taxable income. A superstar's tax base, in units of Y_t, is its operating profit plus its contract fee, minus compliance and R&D costs:

$$\Pi_{ijt} \;=\; \frac{\pi_{ijt}}{Y_t} \;+\; \theta_g\, g_{ijt} \;-\; \kappa_{ijt} R_{jt} \;-\; \chi z_{ijt}^{\phi}. \tag{22}$$

The tax applies to the signed base: a firm whose base is negative receives $\tau_{ijt} |\Pi_{ijt}| Y_t$ (full loss offset), and the same holds for fringe firms at the rate τ_0. Fringe firms pay the statutory rate τ_0 on operating profit minus compliance, their fixed cost ε_k, and their breakthrough R&D. Lobbying intensities, participation costs, and association dues are not deductible.

States. The *technology state* of an industry is $\Theta = (N, \mathbf{n})$, where N is the number of superstars and $\mathbf{n} = (n_1, \ldots, n_{N-1})$, with $0 \le n_1 \le \cdots \le n_{N-1} \le \bar{n}$, is the vector of follower gaps behind the leader. The *industry state* adds every participation status: $S = (\Theta, \mathbf{a}_T, \mathbf{a}_P, \mathbf{a}_R, a_A)$, where $\mathbf{a}_I = (a_{I1}, \ldots, a_{IN})$ collects the superstars' statuses on instrument I and a_A is the association's status. A superstar's *own state* $s_i = (i, \boldsymbol{\varphi}_i)$ consists of its rank i in S, which gives its gap and its statuses, and the participation costs $\boldsymbol{\varphi}_i = (\varphi_{Ti}, \varphi_{Pi}, \varphi_{Ri})$

it has locked on the instruments it lobbies on. Locked costs are private, so a rival's policies enter a firm's problem through their expectation given S under the stationary distribution, which we mark with a bar. Values of the firms are normalized by output on the balanced growth path, $v \equiv V/Y_t$, and discounted at $r - g = \rho$; we drop time and industry subscripts where this creates no ambiguity.

Transitions. The industry state changes with the following events. After innovation by superstar i, the state is S^{+i}: a follower moves one step closer to the leader, while a superstar at the frontier advances it by one step, every other superstar falls one step further behind, and a superstar pushed more than $\bar{n}$ steps behind exits. After a breakthrough whose entrant lands k steps above the margin, the state is $S^{e,k}$. After a redraw of superstar i's cost on instrument I, the state is $S^{I,i,a}$, with its status on I set to $a \in \{0,1\}$; after a redraw of the association's organizing cost, it is $S^{A,a}$. Statuses and locked costs move with their firms when ranks change: tied superstars keep their order, an entrant ranks below the incumbents it ties and starts with every status off, and a superstar's own state after an event, such as s_i^{+i}, is its new rank with the same locked costs.

Superstar problem. A superstar chooses its R&D intensity and its lobbying intensities on the instruments it participates in at every instant, and its participation on instrument I whenever its cost on that instrument is redrawn. Its Hamilton–Jacobi–Bellman equation is

$$
\begin{aligned}
\rho\, v(S, s_i) \;=\; & \max_{z_i,\, b_{Ti},\, b_{Pi},\, b_{Ri}} \Big\{ (1-\tau_i)\,\Pi_i \;-\; \sum_I a_{Ii} \Big[\kappa_I\, b_{Ii}^{\phi_I} + \varphi_{Ii} \prod_{J \neq I} (1-\iota_{IJ})^{a_{Ji}} \Big] \\
& + z_i \left[v(S^{+i}, s_i^{+i}) - v(S, s_i) \right] \;+\; \sum_{i' \neq i} \bar{z}_{i'}(S) \left[v(S^{+i'}, s_i^{+i'}) - v(S, s_i) \right] \\
& + X(S) \sum_k P_k(S) \left[v(S^{e,k}, s_i^{e,k}) - v(S, s_i) \right] \\
& + \sum_{i' \neq i} \sum_I \lambda_I \Big[\bar{G}_{Ii'}(S)\, v(S^{I,i',1}, s_i) + \big(1 - \bar{G}_{Ii'}(S)\big)\, v(S^{I,i',0}, s_i) - v(S, s_i) \Big] \\
& + \lambda_A \Big[G_A(S)\, v(S^{A,1}, s_i) + \big(1 - G_A(S)\big)\, v(S^{A,0}, s_i) - v(S, s_i) \Big] \\
& + \sum_I \lambda_I \Big(\mathbb{E}_{\varphi'} \max \left\{ v(S^{I,i,1}, s_i^{I,\varphi'}),\; v(S^{I,i,0}, s_i^{I,0}) \right\} - v(S, s_i) \Big) \Big\},
\end{aligned}
\tag{23}
$$

subject to (13). The first line is the after-tax flow net of lobbying and locked participation costs. The second line contains innovation by the firm and by its rivals. The third contains fringe entry at rate $X(S)$, with landing probabilities $P_k(S)$ from (10), the probability beyond the cap placed at the cap, and the value averaged over the displaced incumbent when several are tied at the margin. The fourth and fifth lines contain the redraws of rivals, after which

rival i' is on instrument I with probability $\bar{G}_{Ii'}(S)$, and of the association, after which it is organized with probability $G_A(S)$ defined in (40). The last line contains the firm's own redraws: it either joins instrument I, locking the new draw φ', which gives the own state $s_i^{I,\varphi'}$, or stays out, which gives $s_i^{I,0}$. A firm that exits has value zero. The first-order conditions for innovation, procurement, and tax lobbying are

$$(1-\tau_i)\,\chi\phi\,z_i^{\phi-1} \;=\; v(S^{+i},s_i^{+i})-v(S,s_i), \tag{24}$$

$$(1-\tau_i)\,\theta_g\,(1+\lambda)^{-\kappa_g n_i} \;=\; \kappa_P\,\phi_P\,b_{Pi}^{\phi_P-1}, \tag{25}$$

$$\frac{\tau_0\,\Pi_i}{(1+b_{Ti})^2} \;=\; \kappa_T\,\phi_T\,b_{Ti}^{\phi_T-1}, \tag{26}$$

where the tax intensity is zero when the base Π_i is not positive. A higher locked cost lowers the firm's value at every date until its next redraw on the instrument, so the value after joining falls with the new draw. At a redraw, the firm therefore joins instrument I if and only if its draw is below a cutoff:

$$a'_{Ii}=1 \iff \varphi' \;\le\; \varphi_I^*(S,s_i), \qquad v\big(S^{I,i,1},s_i^{I,\varphi_I^*}\big) \;=\; v\big(S^{I,i,0},s_i^{I,0}\big), \tag{27}$$

and a rival's probability of being on the instrument after a redraw is $\bar{G}_{Ii'}(S)=\mathbb{E}\big[\Phi\big((\ln\varphi_I^*(S,s_{i'})-\mu_I)/\sigma_I\big)\mid S\big]$, where Φ is the standard normal distribution function.

The stake in regulation. A superstar participating in the R instrument chooses b_{Ri} to satisfy

$$\Delta_i(S,s_i)\,\frac{\partial R(S)}{\partial b_{Ri}} \;=\; \kappa_R\,\phi_R\,b_{Ri}^{\phi_R-1}, \tag{28}$$

where Δ_i is the marginal value of a higher R to the firm:

$$\Delta_i \;=\; \underbrace{-(1-\tau_i)\,\kappa_i}_{\text{(i) own compliance}} \;+\; \underbrace{(1-\tau_i)\,\frac{\partial(\pi_i/Y)}{\partial m^{\mathrm{a}}}\,\frac{dm^{\mathrm{a}}}{dR}}_{\text{(ii) fringe exit: Cournot gain}} \;+\; \underbrace{\frac{dX}{dR}\sum_k P_k\left[v(S^{e,k},s_i^{e,k})-v(S,s_i)\right]}_{\text{(iii) fewer entrants: displacement risk}}. \tag{29}$$

Equation (29) is the central object of the paper: it is the shadow value of regulation to an incumbent, and its sign determines which side of the contest the firm joins. Channel (i) is the firm's own compliance cost, κ_i per unit of R; it is negative and largest for the leader. Channel (ii) is the static Cournot gain as marginal fringe firms stop producing; it is positive, since $dm^{\mathrm{a}}/dR<0$ and $\partial\pi_i/\partial m^{\mathrm{a}}<0$. Channel (iii) is the lower rate of fringe breakthroughs, which would push the firm down in the rank or displace it; it is positive, since $dX/dR<0$ and an entry lowers the incumbent's value. In a Markov-perfect equilibrium, each agent

takes rivals' policies, the fringe's breakthrough intensity and continuation values as given, but internalizes the effect of its lobbying on R and, through R, on which fringe firms produce. The only fringe firm whose status responds to R is the marginal one at ε^*, so

$$\frac{dm^{\mathrm{a}}}{dR} \;=\; m\, f(\varepsilon^*)\,\frac{d\varepsilon^*}{dR}, \qquad \frac{d\varepsilon^*}{dR} \;=\; \frac{-1}{1 - m\, f(\varepsilon^*)\,\dfrac{\partial(\pi_f/Y)}{\partial m^{\mathrm{a}}}}, \qquad \frac{dX}{dR} \;=\; x_f\,\frac{dm^{\mathrm{a}}}{dR}. \tag{30}$$

The denominator exceeds one, since $\partial\pi_f/\partial m^{\mathrm{a}} < 0$: as R removes marginal fringe firms, the survivors earn higher profits, which partially offsets the direct compliance burden. The firm lobbies for a higher cost if $\Delta_i > 0$ and against it if $\Delta_i < 0$:

$$\mathcal{P}(S) = \{i : a_{Ri} = 1,\ \Delta_i > 0\}, \qquad \mathcal{A}(S) = \{i : a_{Ri} = 1,\ \Delta_i < 0\}. \tag{31}$$

Given the influence function (13), the marginal influence of b_{Ri} is

$$\frac{\partial R(S)}{\partial b_{Ri}} \;=\; \begin{cases} \dfrac{R_0\,\beta^p\,\theta\,b_{Ri}^{\theta-1}}{1+\beta^a B^{\mathrm{anti}}} & \text{if } i \in \mathcal{P}(S), \\[2ex] -\dfrac{R\,\beta^a\,\theta\,b_{Ri}^{\theta-1}}{1+\beta^a B^{\mathrm{anti}}} & \text{if } i \in \mathcal{A}(S), \end{cases} \tag{32}$$

so that (28) gives $b_{Ri} = \big[\Delta_i R_0 \beta^p \theta/(\kappa_R \phi_R (1+\beta^a B^{\mathrm{anti}}))\big]^{1/(\phi_R-\theta)}$ on the pro side and $b_{Ri} = \big[|\Delta_i| R \beta^a \theta/(\kappa_R \phi_R (1+\beta^a B^{\mathrm{anti}}))\big]^{1/(\phi_R-\theta)}$ on the anti side, solved as a fixed point.

Fringe problem. Within an industry, all fringe firms share the same productivity and earn the same operating profit $\pi_f(S)$; they differ only in ε. A firm's dues (16) also depend on the organizing cost φ_A that an organized association has locked. The HJB equation of a fringe firm with cost draw ε is

$$\begin{aligned}(\rho+\delta)\,v^e(S,\varphi_A,\varepsilon) = \max_{a,\,x}\; a\Big\{(1-\tau_0)\Big[\frac{\pi_f(S)}{Y} - R(S) - \varepsilon - \chi_f x^{\phi_f}\Big] - a_A\, d(S,\varphi_A,\varepsilon) + x\, v^{\mathrm{entry}}(S)\Big\} \\ +\; \sum_{S'\neq S} \Gamma(S,S')\,\big[\mathbb{E}\, v^e(S',\varphi_A',\varepsilon) - v^e(S,\varphi_A,\varepsilon)\big],\end{aligned} \tag{33}$$

where $a \in \{0,1\}$ indicates that the firm produces, $x \geq 0$ is its breakthrough intensity, and $\Gamma(S,S')$ are the industry's transition rates, induced by superstars' innovation, breakthroughs, and the redraws of the superstars and the association. The organizing cost changes only when the association reorganizes at a redraw, and the expectation is over the new cost φ_A' in that case. A breakthrough creates a new superstar with all lobbying statuses off, owned by the

innovating firm, whose expected value is

$$v^{\text{entry}}(S) \;=\; \sum_{k} P_k(S)\; v\big(S^{e,k}, \text{entrant's rank}\big). \tag{34}$$

The first-order condition gives the same breakthrough intensity for every active firm in the industry,

$$(1-\tau_0)\,\chi_f\,\phi_f\,x_f^{\phi_f-1} \;=\; v^{\text{entry}}(S), \tag{35}$$

so breakthroughs arrive in the industry at rate $X(S) = m^{\text{a}}(S)\,x_f(S)$. Because switching is costless and ε is permanent, the decision to produce affects only the current flow, and because the marginal producer pays no dues, a fringe firm produces if and only if $\varepsilon \le \varepsilon^*(S)$, where

$$\varepsilon^*(S) \;=\; \frac{\pi_f(S)}{Y} - R(S) + \frac{\Omega(S)}{1-\tau_0}, \qquad \Omega(S) \;=\; \max_{x\ge 0}\big\{x\,v^{\text{entry}}(S) - (1-\tau_0)\,\chi_f x^{\phi_f}\big\}. \tag{36}$$

Because π_f depends on $m^{\text{a}} = mF(\varepsilon^*)$, (36) is a fixed point at each state; it has a unique solution, since π_f falls with m^{a}.

Let $\bar{v}^e(S,\varphi_A) \equiv \mathbb{E}_\varepsilon[v^e(S,\varphi_A,\varepsilon)]$ be the value of a firm in the industry's fringe pool. No decision depends on the organizing cost the association has locked, and each pool firm pays on average a share $1/m$ of it until the association's next redraw or the firm's death, so

$$\bar{v}^e(S,\varphi_A) \;=\; \bar{v}_0^e(S) \;-\; \frac{a_A\,\varphi_A}{m\,(\rho+\delta+\lambda_A)}, \tag{37}$$

where the pool value gross of the current organizing cost solves

$$\begin{aligned}(\rho+\delta)\,\bar{v}_0^e(S) \;=\; &(1-\tau_0)\,\mathbb{E}\big[(\varepsilon^*(S)-\varepsilon)^+\big] \;-\; \frac{a_A\,\kappa_A\,b_A(S)^{\phi_A}}{m} \;-\; \frac{\lambda_A\,\mathbb{E}\big[\varphi_A\,\mathbf{1}\{\varphi_A \le \varphi_A^*(S)\}\big]}{m\,(\rho+\delta+\lambda_A)} \\ &+ \sum_{S'\neq S}\Gamma(S,S')\,\big[\bar{v}_0^e(S') - \bar{v}_0^e(S)\big].\end{aligned} \tag{38}$$

Its flow terms are the producers' after-tax surplus, the dues that pay for lobbying, and the members' share of any organizing cost committed at the next redraw. Entrepreneurs sell new firms at the pool value averaged over the stationary distribution μ; with $e = \delta m$ on the balanced growth path, free entry requires

$$2\psi\delta\,m \;=\; \sum_{S}\mu(S)\,\Big[\bar{v}_0^e(S) - \frac{a_A\,\bar{\varphi}_A(S)}{m\,(\rho+\delta+\lambda_A)}\Big], \tag{39}$$

where $\bar{\varphi}_A(S)$ is the expected organizing cost locked by an association organized in state S.

Association problem. At each redraw, the industry's current fringe firms decide whether the association organizes. They organize if their gain covers their share of the new cost:

$$\varphi_A \;\le\; \varphi_A^*(S) \;=\; (\rho+\delta+\lambda_A)\, m\,\big[\bar v_0^e(S^{A,1}) - \bar v_0^e(S^{A,0})\big], \qquad G_A(S) \;=\; \Phi\Big(\frac{\ln\varphi_A^*(S)-\mu_A}{\sigma_A}\Big), \tag{40}$$

where $S^{A,1}$ and $S^{A,0}$ denote S with the association organized and not; firms that join the pool later pay the rest, which free entry prices in. While organized, the association chooses b_A at every instant to maximize the total value of the industry's fringe firms net of its lobbying cost, taking rivals' intensities as given:

$$m^{\mathrm{a}}\,\Delta_A\,\frac{\partial R}{\partial b_A} \;=\; \kappa_A\,\phi_A\,b_A^{\phi_A-1}, \tag{41}$$

where $\partial R/\partial b_A$ takes the form (32) and Δ_A is the marginal value of a higher R per producing member:

$$\Delta_A \;=\; \underbrace{-(1-\tau_0)}_{\text{own compliance}} \;+\; \underbrace{(1-\tau_0)\,\frac{\partial(\pi_f/Y)}{\partial m^{\mathrm{a}}}\,\frac{dm^{\mathrm{a}}}{dR}}_{\text{fringe exit: survivors' profit}} \;+\; \underbrace{\frac{m}{m^{\mathrm{a}}}\,\frac{dX}{dR}\sum_k P_k\,\big[\bar v_0^e(S^{e,k}) - \bar v_0^e(S)\big]}_{\text{breakthroughs: the pool's value}}. \tag{42}$$

By (36), the first two terms sum to $(1-\tau_0)\,d\varepsilon^*/dR < 0$: the exit of marginal rivals returns only part of a member's higher cost. Only the last term, which captures how rarer breakthroughs change the value of every firm in the pool, can make the association favor a higher cost; it lobbies on the side given by the sign of Δ_A.

Within-state political conflict. The contest at state S splits the industry whenever the participants' stakes differ in sign, for example when some superstars gain from a higher cost while the association loses. The two sides push R in opposite directions through (13), and $R(S)$ solves the joint fixed point of the superstar conditions (28) and the association condition (41).

4.6 Equilibrium

Definition 1 (*Balanced-Growth-Path Equilibrium*)**.** *A balanced-growth-path (BGP) equilibrium consists of*

- superstar values $v(S,s_i)$ and policies $\{z_i, b_{Ri}, b_{Ti}, b_{Pi}\}$ with participation cutoffs $\{\varphi_I^*\}$; fringe values $v^e(S,\varphi_A,\varepsilon)$, cutoffs $\varepsilon^*(S)$ and breakthrough intensities $x_f(S)$; the association's cutoff $\varphi_A^*(S)$ and intensity $b_A(S)$;
- the regulatory cost $R(S)$ at every industry state;

- a fringe pool m and active masses $m^{\mathrm{a}}(S) = mF(\varepsilon^*(S))$;
- a stationary distribution μ over industry states, superstars' locked costs, and the association's locked cost;
- a wage ratio $\omega = w/Y$, labor supply L, a labor tax τ_w, and a growth rate g;

such that:

1. **Product market.** Given $m^{\mathrm{a}}(S)$, quantities and prices satisfy the superstars' Cournot best responses and the fringe's price-taking conditions.
2. **Superstars.** Values and policies solve (23) given $R(\cdot)$ and the rivals' strategies; participation follows (27).
3. **Fringe.** $\varepsilon^*(S)$ satisfies (36), $x_f(S)$ satisfies (35), and the association's decisions satisfy (40) and (41).
4. **Contest.** At every state, intensities satisfy (28) for every superstar on the R instrument, with sides (31), and (41) for an organized association, and $R(S)$ is given by (13).
5. **Free entry.** m satisfies (39).
6. **Consistency.** μ is the invariant distribution of the process induced by the policies, and the rivals' expected policies $\bar{z}_{i'}$ and $\bar{G}_{Ii'}$ are computed under μ.
7. **Labor market and government.** Labor demand equals labor supply (18), and the government budget (17) balances.

Growth. The frontier of an industry advances when its leader, or a firm tied with it, innovates, and when an entrant passes the leader. The aggregate growth rate is

$$g \;=\; \ln(1+\lambda) \sum_{S} \mu(S) \Big[\sum_{i:\, n_i = 0} z_i(S) \;+\; X(S) \sum_{k} P_k(S)\, \ell_k(S) \Big], \tag{43}$$

where $\ell_k(S) \geq 0$ is the number of steps by which an entrant that lands k steps above the margin passes the former leader.

Labor market and output. Within each industry, producers' labor demands follow from the static stage; each is proportional to Y/w. Labor market clearing requires

$$\int_0^1 \left(\sum_{i=1}^{N_{jt}} l_{ijt} + \int_0^{m_t} l_{fkjt}\, dk \right) dj \;=\; L_t, \tag{44}$$

with L_t given by the household's labor supply (18). Output then follows from (2).

Resource constraint. Final-good output is allocated between private consumption, government purchases, R&D, entry, fixed costs, compliance and lobbying. Dividing by Y_t and aggregating across industries with μ, the BGP resource constraint is

$$1 \;=\; \frac{C_t^p}{Y_t} \;+\; (1-\theta_g)\,\mathbb{E}_\mu\Big[\sum_i g_i\Big] \;+\; \mathcal{R}_{\text{R\&D}} \;+\; \psi\,(\delta m)^2 \;+\; \mathcal{R}_{\text{fixed}} \;+\; \mathcal{R}_{\text{comp}} \;+\; \mathcal{L}, \qquad (45)$$

where

$$\begin{aligned}
\mathcal{R}_{\text{R\&D}} &= \mathbb{E}_\mu\Big[\sum_i \chi\, z_i^{\phi}\Big] \;+\; \sum_S \mu(S)\, m^{\text{a}}(S)\, \chi_f\, x_f(S)^{\phi_f},\\
\mathcal{R}_{\text{fixed}} &= m \sum_S \mu(S)\, \mathbb{E}\big[\varepsilon\, \mathbf{1}\{\varepsilon \le \varepsilon^*(S)\}\big],\\
\mathcal{R}_{\text{comp}} &= \sum_S \mu(S)\, \Big[\sum_{i=1}^{N(S)} \kappa_i(S) + m^{\text{a}}(S)\Big]\, R(S),\\
\mathcal{L} &= \mathbb{E}_\mu\Big[\sum_i \sum_I a_{Ii}\big(\kappa_I\, b_{Ii}^{\phi_I} + \varphi_{Ii} \prod_{J\neq I} (1-\iota_{IJ})^{a_{Ji}}\big) \;+\; a_A\Big(\frac{C_A}{Y} + \varphi_A\Big)\Big].
\end{aligned}$$

The expectations $\mathbb{E}_\mu$ are over industry states and over the locked costs. Compliance is a pure resource cost: paying it generates no output, consistent with the interpretation of regulation as administrative friction. Tax revenue, contract fees, transfers, and association dues are transfers and do not appear in (45).

Computing the equilibrium. With $\bar{N} = 4$ and $\bar{n} = 5$, the full state has 481,040 industry states and 1.9 million superstar states. We compute the equilibrium with two simplifications. First, a firm forms beliefs about its rivals' tax and procurement statuses, equal to their stationary frequencies among superstars of the same rank in the same technology state; the state keeps the regulation statuses, the association's status, and each firm's own statuses, which leaves 2,180 industry states and 33,104 superstar states. Decisions use the beliefs, while the stationary distribution, growth, and every moment use the actual statuses. Second, a superstar's policies are evaluated at the expected cost of its current spell, $\hat{\varphi}_I = \mathbb{E}[\varphi \mid \varphi \le \varphi_I^*]$, so that values are affine in the locked costs and the cutoff (27) has a closed form; spending uses the exact distribution of locked costs. Appendices D–F give the details.

4.7 Welfare and Counterfactual Analysis

We compare each equilibrium with a reference economy without lobbying. The reference economy is identical except that every participation cost is infinite, so $R = R_0$, every superstar pays τ_0, and contracts equal their baseline $g_0(1+\lambda)^{-\kappa_g n_i}$. It has the same statutory tax

rate τ_0, baseline contracts, transfers $\bar{T}$, and entry cost ψ, and its fringe pool m is determined by free entry. A counterfactual ban on an instrument sets that instrument's participation costs to infinity. We compare balanced growth paths and report two kinds of results: the changes in output, resources, growth and hours, which do not depend on preferences, and a consumption-equivalent measure that aggregates them.

Changes without preferences. Consumption at date zero, $c_0 = C_0^p + G_0$, equals output times the share of output that remains after compliance, fixed costs, entry, R&D, and lobbying, and it grows at rate g. Every producer's technology is homogeneous of degree α in labor, so output splits exactly as

$$\ln Y_0 \;=\; \alpha \ln L + \ln A^* + \ln E, \quad \ln A^* \;=\; \sum_S \mu(S)\,\frac{1}{e}\ln \sum_k A_k(S)^e, \quad e = \frac{s}{1-\alpha s}, \quad s = \frac{\eta-1}{\eta}. \tag{46}$$

Here A^* is the productivity that an efficient allocation of the same labor would achieve given each industry's state and its operating fringe, with $A_k = q_k$ for a superstar and $A_f = (m^{\mathrm{a}})^{1-\alpha} q_f$ for the fringe taken as one producer, and $\ln E \leq 0$ is the loss from the allocation of labor across producers that markups distort. The share of output consumed splits by item of the resource constraint (45), and growth by source (43). None of these changes requires an assumption on preferences, and together they bound the welfare cost from below: when lobbying lowers both the level and the growth of consumption and raises hours, any preferences that value consumption and leisure put the cost at no less than the fall in consumption at date zero.

Consumption-equivalent welfare. On the balanced growth path, welfare is

$$W \;=\; \frac{\ln c_0}{\rho} \;+\; \frac{g}{\rho^2} \;-\; \frac{D}{\rho}, \qquad D \;=\; \xi\,\frac{L^{1+1/\epsilon_L}}{1+1/\epsilon_L}. \tag{47}$$

We summarize welfare by the consumption-equivalent variation of Lucas (1987), as in Jones and Klenow (2016): the permanent percentage increase in consumption, at every date of the equilibrium with lobbying, that would make the household as well off as in the reference economy. It solves

$$\ln(1+\mathrm{CEV}) \;=\; \rho\left(W^{\mathrm{ref}} - W\right) \;=\; \underbrace{\ln c_0^{\mathrm{ref}} - \ln c_0}_{\text{level}} \;+\; \underbrace{\frac{g^{\mathrm{ref}} - g}{\rho}}_{\text{growth}} \;+\; \underbrace{D - D^{\mathrm{ref}}}_{\text{hours}}, \tag{48}$$

so that $\mathrm{CEV} > 0$ means that lobbying lowers welfare. As in Jones and Tonetti (2020), the measure splits additively into the level of consumption (which (46) and the consumption

share decompose further), the growth rate, and hours. Preferences enter only through the weights on growth and on hours, and both weights are prices. For any balanced-growth preferences $u = [c\,e^{-D}]^{1-\sigma}/(1-\sigma)$, which reduce to ours at $\sigma = 1$, the growth term is $\frac{1}{1-\sigma}\ln\frac{r-g}{r^{\text{ref}}-g^{\text{ref}}}$, where $r - g = \rho - (1-\sigma)\,g$ is the gap between the interest rate and the growth rate. To first order, it is $(g^{\text{ref}} - g)/(r^{\text{ref}} - g^{\text{ref}})$, the growth loss capitalized at the interest rate, so the curvature and the discount rate matter only through $r - g$, which equals ρ with log utility. By the labor supply condition (18), the hours term is, to first order, $[(1-\tau_w)\,wL/C]\,(\ln L - \ln L^{\text{ref}})$: hours valued at the after-tax wage. Following Perla et al. (2021), who write welfare gains from trade as a weighted sum of changes in the growth rate and the initial level of consumption, we group the terms by the channel through which lobbying acts. Appendix H gives the details of the decomposition.

5 Estimation

We estimate the model by the simulated method of moments. The parameters fall into three blocks, each disciplined by a different kind of data: the technology block by firm dynamics and industry structure; the tax and procurement block by the outcomes that lobbying buys, which the data reveal directly; and the regulation block, whose outcome is not observed, by participation, sides, the trade associations, and the enactment of bills. All blocks are estimated jointly, because lobbying changes the industry structure that the technology moments describe and the industry structure determines the stakes that drive lobbying.

5.1 Strategy

Objective. Let $\hat{m}_k$ be the data value of moment k and $m_k(\vartheta)$ its model counterpart at parameters ϑ. The estimator minimizes the sum of squared deviations $\sum_k d_k(\vartheta)^2$ with equal weights. For the technology moments $d_k = m_k/\hat{m}_k - 1$. For the lobbying moments, $d_k = \ln(m_k/\hat{m}_k)$ when the moment is positive by construction (shares, rates, ratios, and levels) and $d_k = (m_k - \hat{m}_k)/\max(|\hat{m}_k|, \text{se}_k)$ when it can take either sign (regression coefficients, elasticities, and differences); these deviations are capped at a hundredfold miss in either direction, so that no single moment dominates. There are 107 targeted moments: 35 in the technology block and 72 lobbying moments, eight of which come from the simulation of bills; two further moments enter as upper bounds, which count only when exceeded.

Model moments. Every model moment is computed with the definition of its data counterpart. Moments of the industry structure are computed from the stationary distribution

over industry states. Moments that the data measure on a panel of firms, such as participation by rank, the persistence of lobbying, the participation of firms by decile of their stake, and the regressions of tax rates and contracts on past lobbying, are computed on a simulated panel of superstars of the same length, with the same regressions and fixed effects. Bill moments are computed by simulating every real proposal (Section 5.5).

Search. An equilibrium takes about two and a half hours to compute, and the objective is not smooth in all parameters, so we estimate the model by a global search. We draw 8,192 points from a Sobol sequence over the 43 parameters: the technology parameters within eight standard errors of an estimate of the technology block alone, and the lobbying parameters over wide ranges, with the tax and procurement parameters drawn in units of the outcomes they produce. The estimate is the point with the lowest objective among the 336 points whose equilibrium is certified and whose moments are all computed (Appendix F).

5.2 Parameters set outside the model

Table 5 lists the parameters set outside the model. The discount rate is $\rho = 0.04$, as in Cavenaile et al. (2026), the Frisch elasticity is $\epsilon_L = 0.5$, and ξ normalizes hours to one in the economy without lobbying, taxes, and contracts. An industry has at most $\bar{N} = 4$ superstars, each selling at least 10% of the leader's sales; the ladder depth $\bar{n}$ is the largest gap at which the smallest superstar meets this threshold in every configuration of its rivals' gaps, recomputed at every parameter point ($\bar{n} = 5$ at the estimate). The number of active fringe firms per industry without lobbying or contracts, $\bar{m}^{\mathrm{a}} = 4{,}665$, is the number of employer firms in the median industry: the Business Dynamics Statistics count averaged over 2008–2022, with the median taken across the 132 industries of our panel in the Economic Census, weighted by industry-years. Free entry sets ψ to deliver it. As in Cavenaile et al. (2026), we compare the model's labor share with the data after scaling by one minus a capital share of 0.25. Of the lobbying parameters, the contractor's margin $\theta_g = 0.101$ is the sales-weighted operating margin of firms whose federal contracts are at least 70% of their sales, and $\beta^p = 1$ is a normalization that lets κ_R carry the scale of influence.

5.3 Technology

We estimate the thirteen technology parameters jointly with the lobbying parameters (Section 5.1) on 35 moments of industry structure, firm dynamics, and aggregates. Each model moment is computed in the equilibrium with lobbying, with the definition of its data counterpart (Appendix I). Table 6 reports the estimates and Table 7 the moments. The fringe's technology is expressed in aggregate units, $\zeta = \tilde{\zeta}\,(\bar{m}^{\mathrm{a}})^{\alpha-1}$ and $\chi_f = \tilde{\chi}_f\,(\bar{m}^{\mathrm{a}})^{\phi_f-1}$: $\tilde{\zeta}$

Table 5. Parameters Set Outside the Model

Parameter		Value	Source or target
Discount rate	ρ	0.04	Cavenaile et al. (2026)
Frisch elasticity of labor supply	ϵ_L	0.5	standard
Weight on the disutility of labor	ξ	0.519	hours equal one without lobbying, taxes, and contracts
Maximum number of superstars	$\bar{N}$	4	superstar definition
Smallest superstar, share of the leader's sales		0.10	superstar definition
Maximum gap behind the leader	$\bar{n}$	5	smallest superstar sells 10% of the leader's sales
Active fringe firms per industry, no lobbying or contracts	$\bar{m}^{\mathrm{a}}$	4,665	employer firms in the median industry (BDS)
Capital share (labor-share mapping)		0.25	Barkai (2020)
Contractor's margin	θ_g	0.101	Compustat, federal contractors
Effectiveness of pro-side lobbying	β^p	1	normalization

Notes: The maximum gap is recomputed at every parameter point. The entry cost ψ is set by free entry so that $\bar{m}^{\mathrm{a}}$ fringe firms produce in the economy without lobbying and without contracts. BDS: Business Dynamics Statistics.

is the productivity of the fringe as a single producer with $\bar{m}^{\mathrm{a}}$ active firms, and $\tilde{\chi}_f X^{\phi_f}$ is the cost of an industry-wide breakthrough rate X shared equally among them.

Unlike Cavenaile et al. (2026), who discipline the quality ladder with an innovation–size regression on patents and the leader's share of citations, we use firm dynamics measured in sales and the cost structure of the largest firms. Most firms in the patent regression have no patents, so it is driven by the extensive margin of patenting; the citation share implies more dominant leaders than the sales data show; and a fringe that responds to regulation must be disciplined by its entry, exit, and losses, which patents do not measure. Patents serve as an untargeted check.

Identification. Every moment depends on every parameter, but some moments are particularly informative about each parameter (last column of Table 6).

Quality step, demand, and returns to scale. For given quality gaps, a larger step λ and a higher elasticity η both widen the sales gaps between superstars, so the distribution of sales gaps pins down a combination of the two. Growth separates them: it equals $\ln(1+\lambda)$ times the rate at which the frontier advances (43). So does the payroll ratio, which measures the markup: a superstar pays a share α/M_i of its sales in wages and a fringe firm a share α, so the ratio of the two is the inverse of the superstar markup, $M_i = \eta/((\eta-1)(1-\sigma_i))$, which falls with η. A superstar's sales-to-cost ratio equals M_i/α, so the two cost ratios together pin down α; in the data they imply $\alpha \approx 1/(0.764 \times 1.497) = 0.87$.

Superstars' innovation. Growth and the gap dynamics measure how often superstars innovate. By (24), a superstar's innovation rate rises with the value of an innovation with elasticity $1/(\phi-1)$; because this value differs among leaders, tied firms, and followers, ϕ

Table 6. Technology Parameters

Parameter		Estimate	Most informative moments
Quality ladder, demand, and production			
Quality step	λ	0.140	sales gaps among superstars; growth
Elasticity of substitution	η	9.31	sales gaps among superstars; payroll ratio
Returns to scale in labor	α	0.862	sales-to-cost ratio, with the payroll ratio
Superstars' innovation			
R&D cost: level	χ	2.30×10^3	growth; gap dynamics
R&D cost: convexity	ϕ	6.09	R&D/GDP; gap dynamics; industries with fewer than four superstars
Breakthroughs			
Breakthrough cost: level, aggregate units	$\tilde{\chi}_f$	2.00	new superstars per industry-year
Breakthrough cost: convexity	ϕ_f	2.97	new superstars by #4/#1; industries with fewer than four superstars
Climb of new superstars	p	0.300	landing steps; new superstars' sales relative to the leader
Fringe productivity			
Fringe productivity, aggregate units	$\tilde{\zeta}$	0.729	CR4
Curvature of spillovers to the fringe	γ	2.18	CR4; R&D/GDP
Fringe fixed costs and turnover			
Fixed cost: mean, per firm-year	$\bar{\varepsilon}$	8.24×10^{-5}	applications that become employers; losses
Fixed cost: dispersion of logs	σ	1.43	dispersion of fixed costs; applications that become employers
Death rate	δ	0.075	firm exit rate; applications per employer firm

Notes: Best point of the joint global search (Section 5.1). Costs are in units of aggregate output Y_t. Aggregate units: $\zeta = \tilde{\zeta}\,(\bar{m}^{\mathrm{a}})^{\alpha-1}$ and $\chi_f = \tilde{\chi}_f\,(\bar{m}^{\mathrm{a}})^{\phi_f-1}$, so that per firm $\zeta = 0.228$ and $\chi_f = 3.26 \times 10^7$. All parameters are estimated jointly; the last column lists the most informative moments.

shapes how the gap dynamics vary with the initial gap and how often followers fall out of the superstar set. R&D intensity disciplines the cost of innovation: at the optimum, after-tax R&D spending is a share $1/\phi$ of the expected gain from innovation, and breakthrough spending a share $1/\phi_f$ of the expected value of entry.

Breakthroughs. The rate at which new superstars appear pins down a combination of the level $\tilde{\chi}_f$ and the convexity ϕ_f of the breakthrough cost. The convexity also sets the elasticity $1/(\phi_f - 1)$ of breakthroughs to the value of entry (35), which varies across industry states; entry by the previous year's #4/#1 ratio and the share of industries with fewer than four superstars carry this information. By (10), a share $(1-p)p^k$ of entrants lands k steps above the entry margin, so the landing shares and the entrants' sales relative to the leader identify p.

Fringe productivity. CR4 pins down $\tilde{\zeta}$: a more productive fringe takes a larger share of industry sales. The curvature γ determines whether the fringe learns mostly from the leader (high γ) or from all superstars (low γ) (6). It moves CR4 together with $\tilde{\zeta}$ and changes how much of an innovation's value superstars keep, which R&D intensity reflects; the two

moments pin down a combination of $\tilde{\zeta}$ and γ more tightly than either parameter.

Fringe fixed costs and turnover. The dispersion of fringe fixed costs, measured as selling, general, and administrative expenses net of R&D and of firm size, pins down σ. Given σ, the mean $\bar{\varepsilon}$ sets the share of new firms that produce, measured by the share of business applications that become employer firms. In the steady state, applications per employer firm equal δ divided by the producing share of the pool, and the exit rate adds the firms that stop producing; together they pin down δ. The loss share and the exit premium of loss-making firms measure how close producers are to the cutoff: by (36), a fringe firm produces at a loss as long as the option value of a breakthrough covers it.

Estimates. An innovation raises a superstar's quality by 14%. With $\eta = 9.3$, a superstar with a negligible share charges a markup of 1.12, and the cost-weighted superstar markup is 1.32. With $\alpha = 0.86$, fringe firms keep 14% of their revenue as operating profit (21). The elasticity of a superstar's innovation to its value is 0.20 ($\phi = 6.1$), and that of breakthroughs to the value of entry 0.51 ($\phi_f = 2.97$). A new superstar lands at the entry margin with probability 0.70 and one step above it with probability 0.21. Forty percent of the fringe pool produces at any time, and fringe firms die at a rate of 0.075 per year.

Fit. The median moment is within 15% of its data value, and 22 of the 35 are within 25%. The model matches the aggregates (growth of 1.85% against 1.66, R&D of 2.2% of GDP against 2.0, and a labor share of 0.59 against 0.60), the cost structure of the largest firms, the number of superstars per industry (3.35 against 3.64), the rate at which new superstars appear (0.124 against 0.132), the firm exit rate, and the dispersion of fixed costs. It misses in four respects. Industries are more concentrated (CR4 of 0.37 against 0.22), and more have fewer than four superstars (0.46 against 0.23). New superstars land too high: 35% land at the bar, against 64%. The gap dynamics are too slow, with four of the six rates 40 to 52% below the data, because sales ratios in the model change in discrete steps of the ladder, while in the data they also drift. Fewer fringe firms have losses (0.14 against 0.39), because the model's fringe profit is operating profit before depreciation and interest, while the tax data count negative net income; in Compustat, 26% of fringe firms have negative operating income before depreciation. The share of applications that become employers (0.40 against 0.27) and the exit premium of loss-making firms (0.050 against 0.035) are also too high.

Untargeted moments. Table 8 reports moments the estimation does not target. The cost-weighted superstar markup is 1.32, against 1.27 for production-function markups of the same firms computed as in De Loecker et al. (2020). Industries with at most two superstars are somewhat more common than in the data (0.16 against 0.11). Because patents are proportional to innovation and R&D spending is χz^{ϕ}, the model's within-firm elasticity of patents to R&D is $1/\phi = 0.16$, against 0.26 (standard error 0.055) in the data, which points

to less convex R&D costs.

5.4 Tax and procurement

On the tax and procurement instruments the outcome of lobbying is in the data: we observe each firm's tax rate and its federal contracts before and after it lobbies. The block has sixteen parameters: the statutory tax rate τ_0; the baseline contract g_0 and its decline with the firm's gap κ_g; for each instrument, the lobbying cost κ_I and its convexity ϕ_I, and the mean, dispersion, and redraw rate of the participation cost $(\mu_I, \sigma_I, \lambda_I)$; and the three economies of scope ι_{IJ} between the instruments.

Identification. Table 9 summarizes the moments. What lobbying buys identifies the lobbying cost: within a firm, a year of tax lobbying lowers the next three years' book tax rate by 1.1 percentage points, and a year of procurement lobbying raises the firm's prime contracts by 0.27 log points. The model computes the same regressions, with firm and year fixed effects, on its simulated panel. The level of the outcomes without lobbying identifies τ_0 (superstars' effective rate, 0.263), g_0 (contracts over sales, 0.019), and κ_g (the elasticity of contracts to sales across firms, 0.75). Who lobbies identifies the participation costs: participation by rank, and across deciles of the firm's stake (pretax income for the tax instrument, past contracts for procurement), rises from 1 to 69% on the tax instrument and from 3 to 59% on procurement. How long firms lobby identifies the redraw rates: 80% of tax lobbyers and 78% of procurement lobbyers still lobby a year later. How much they spend identifies the convexity of the lobbying cost: at an interior optimum, spending on an instrument equals the gain from lobbying divided by ϕ_I, and spending is 0.014% of the stake on the tax instrument and 0.078% on procurement. Which instruments firms combine identifies the economies of scope: among superstars that lobby on regulatory costs, 50% also lobby on taxes and 28% on procurement, and among those that lobby on procurement, 43% also lobby on taxes.

Estimates and fit. The statutory rate is $\tau_0 = 0.283$. A lobbying leader lowers its tax rate by 0.71% of τ_0, and 1.3% of a lobbying leader's contracts are due to lobbying; both lobbying costs are convex ($\phi_T = 40$, $\phi_P = 4.9$). Participation costs are redrawn on average every four years on the tax instrument and every year on procurement, and economies of scope are largest between regulation and taxes ($\iota_{RT} = 0.18$). The model matches participation on the tax instrument across deciles of pretax income in the lower half of the distribution, the share of tax lobbyers still lobbying a year later (0.85 against 0.80), the stay rate of procurement lobbyers from one quarter to the next (0.87 against 0.90), and participation in procurement across the middle deciles of contracts. Within a firm, the model's tax rate falls by 0.13 percentage points after a year of lobbying, against 1.12 in the data, and contracts rise by

0.003 log points against 0.27. Participation is more concentrated at the top than in the data: the model's leader lobbies more often on taxes (0.35 against 0.27) and on procurement (0.39 against 0.19), and its fourth firm less often (Figure 7). Section 6.4 shows why transfers of this kind cost little in the model.

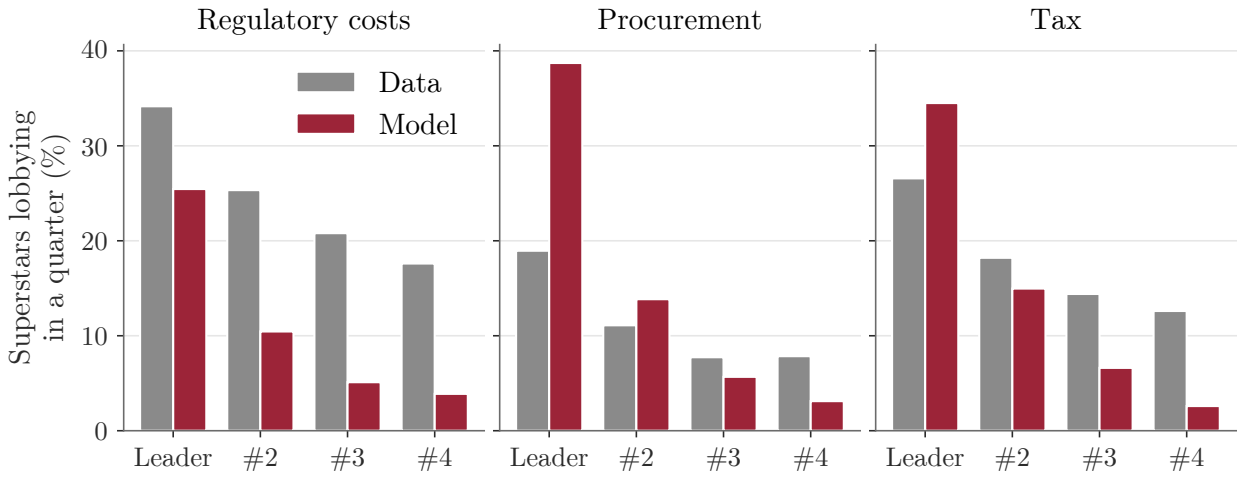


Figure 7. Who Lobbies: Superstars' Participation by Rank, Model and Data

Notes: Share of superstars lobbying on each instrument in a quarter, by rank within the industry. Data: Compustat superstars headquartered in the United States; regulatory costs 2008–2022; tax and procurement, last quarter of fiscal years 2008–2022. Model: simulated panel at the estimate.

5.5 Regulation

The regulation block raises four difficulties, and we address each with its own source of data. First, the regulatory fixed cost R_j is not observed by industry and year; we identify the contest from the bills that set it. Second, firms take both sides, and the model determines each firm's side; we use the sides firms state in their reports. Third, many industries lobby each bill; we simulate each real proposal with every industry that lobbied it. Fourth, fringe firms lobby through associations; we use the census of industry associations: their activity, their spending, and their sides. The block has fourteen parameters: the compliance level c, which sets R_0; the elasticity e_R of superstars' compliance costs to their size; the contest curvature θ and the effectiveness of the anti side β^a; the price of influence κ_R and the convexities ϕ_R and ϕ_A of the superstars' and the association's lobbying costs; the participation cost of superstars $(\mu_R, \sigma_R, \lambda_R)$; and the association's organizing cost $(\mu_A, \sigma_A, \lambda_A)$ and lobbying cost κ_A.

Participation, sides, and the association. Participation of superstars by rank (34, 25, 21, and 18% of the first to fourth firms lobby on regulatory costs in a quarter) and its gradient in the firm's share of industry sales identify the mean and dispersion of their

participation cost, and its persistence (93% continue the next quarter) the redraw rate. The share of stated positions that ask for higher compliance costs, 53% among superstars and 37% among associations, identifies the elasticity e_R: the faster a superstar's compliance bill rises with its size, the more its own cost weighs against the Cournot and deterrence gains, and the more the larger superstars ask for lower costs. The elasticity of lobbying spending to sales (0.81) and the leader's share of superstars' spending (0.48) discipline how fast influence rises with the stake, and hence the curvature θ and the convexity ϕ_R, together with the bills; total lobbying relative to industry sales disciplines the price κ_R. The level of compliance costs, 0.58% of output in the measure of Trebbi et al. (2023), identifies c; their broader measure, 1.64%, enters as an upper bound. The association's activity (active in 57% of industry-quarters, 95% of active associations still active the next quarter) identifies its organizing cost; its spending relative to industry sales (4.5×10^{-5}) and the difference in superstars' participation between quarters with and without an active association (1.6 percentage points) identify its strength.

Bills. The enactment of bills identifies how lobbying force translates into regulation. A *proposal* is a set of bills of one Congress that are the same legislation; in 2008–2022, 2,244 proposals raise regulatory fixed costs and 698 lower them. Following Kang (2016), proposal k passes with probability

$$p_k \;=\; \frac{\pi_k + x_k}{1 + x_k + y_k}, \qquad x_k = \sum_{i \in \text{pushers}} \beta^{s_i}\, b_{Ri}^{\theta}\, \omega_{ik}, \qquad y_k = \sum_{i \in \text{blockers}} \beta^{s_i}\, b_{Ri}^{\theta}\, \omega_{ik}, \qquad (49)$$

where π_k is its probability of passing without lobbying, $\beta^{s_i} b_{Ri}^{\theta}$ is firm i's force (β^p on the pro side, β^a on the anti side), and ω_{ik} is the share of its force the firm puts on proposal k, chosen optimally and larger where more is at stake. On a raising proposal, pro firms push and anti firms block; on a lowering proposal, the roles flip. Enacted proposals add to or subtract from the stock of rules, which lapse at a constant rate, and Appendix G shows that in the steady state the stock reproduces the contest (13) to first order. We simulate each of the real proposals: every industry that lobbied it in the data is drawn from the model's stationary distribution conditional on one of its firms lobbying, forces add up across industries, and the baseline probability π_k depends on the proposal's observable features (sponsor, cosponsors, whether it is must-pass legislation, and Congress) and an unobserved component, fitted at each parameter point. We then run the data's enactment regressions on the simulated proposals.

Table 10 summarizes what the bill moments identify. Each log point of lobbying superstars raises the enactment of raising proposals by 3.7 percentage points, which identifies the price of influence κ_R: cheaper influence means more force per firm. How this effect grows

with the size of the lobbying firms disciplines the curvature θ: force grows with size as size$^{\varrho}$, with $\varrho = e\theta/(1-\theta)$ where e is the elasticity of lobbying to size, and the data give $\varrho = 0.10$ (standard error 0.19). On lowering proposals the roles flip, which identifies the anti side's effectiveness β^a: because unlobbied proposals rarely pass, blocking matters only when β^a is large, and a single firm alone on a raising proposal changes its enactment by $f(1-\pi)/(1+f)$ by pushing but only by $\pi\beta^a f/(1+\beta^a f)$ by blocking.

Estimates. Table 11 reports the estimates. Compliance without lobbying is 0.33% of output. A superstar's compliance bill rises with its size with elasticity $e_R = 0.47$, so compliance weighs far more on a fringe firm relative to its size. The contest curvature is $\theta = 0.89$, and lobbying against the cost is more effective per unit of force than lobbying for it ($\beta^a = 23$). Superstars' lobbying cost is more convex ($\phi_R = 3.8$) than the association's ($\phi_A = 2.9$). Participation costs on regulatory costs are redrawn on average every 2.7 years, and the association's organizing cost every 2.0 years.

Fit. Table 12 compares the main lobbying moments with the data. The model reproduces the persistence of superstars' participation in regulation (0.93 against 0.93), the gradient of participation in the firm's share of industry sales (0.17 against 0.15), the level of compliance costs (0.53 against 0.58% of output), the share of stated positions of associations that ask for higher costs (0.25 against 0.37), the effect of lobbying superstars on the enactment of raising proposals (4.1 against 3.7 percentage points per log point), and the association's effect on raising proposals (-2.2 against -0.8). In the model, every superstar has a positive stake in higher compliance costs, so 96% of superstars' positions ask for higher costs, against 53% in the data; the association is active in fewer industry-quarters than in the data (0.14 against 0.57); and participation in regulation falls faster with rank (0.25 for the leader and 0.04 for the fourth firm, against 0.34 and 0.18).

Untargeted moments. Table 13 compares moments that the estimation does not target. The model reproduces the share of superstars that lobby on no instrument (0.69 against 0.66), the share of industries with at most two superstars (0.16 against 0.11), and the sales of new superstars relative to the leader (0.22 against 0.25 in the middle tercile of the fourth firm's relative size). Superstars in the model lobby on several instruments at once less often than in the data (0.020 against 0.042 lobby on all three), and in more industry-quarters superstars lobby without the association (0.31 against 0.11).

6 The Consequences of Lobbying

This section reports the political equilibrium at the estimate, the effect of lobbying on regulation, industry structure, and growth, its welfare cost, and which kind of lobbying

causes it. Every comparison is between the estimated economy and the same economy without lobbying: every participation cost set to infinity, so that the regulatory fixed cost is R_0 in every industry and contracts and tax rates are at their baseline, with the same statutory tax rate, baseline contracts, and entry cost, and with the fringe pool given by free entry.

6.1 Who lobbies, on which side, and why

Superstars lobby for higher regulatory costs. At the estimate, the stake in regulation (29) is positive for every superstar in every industry state, so superstars lobby on the side of a higher regulatory fixed cost; the association mostly lobbies against it. Figure 8 decomposes the stake by rank, in units of the firm's own after-tax compliance cost. Every superstar gains many times its own compliance cost from higher regulatory costs, but the source of the gain changes down the ranks. The leader gains 17 times its compliance cost from the profits it takes over when marginal fringe firms stop producing, and 10 times from the lower rate of fringe breakthroughs: 63% of its gross gain is static. For the fourth superstar, the shares reverse: it gains 5 times its compliance cost from fringe exit and 11 times from deterrence, so 68% of its gain is dynamic. The net stake falls from 26 times the compliance cost for the leader to 17 for the second superstar and 14 and 15 for the third and fourth.

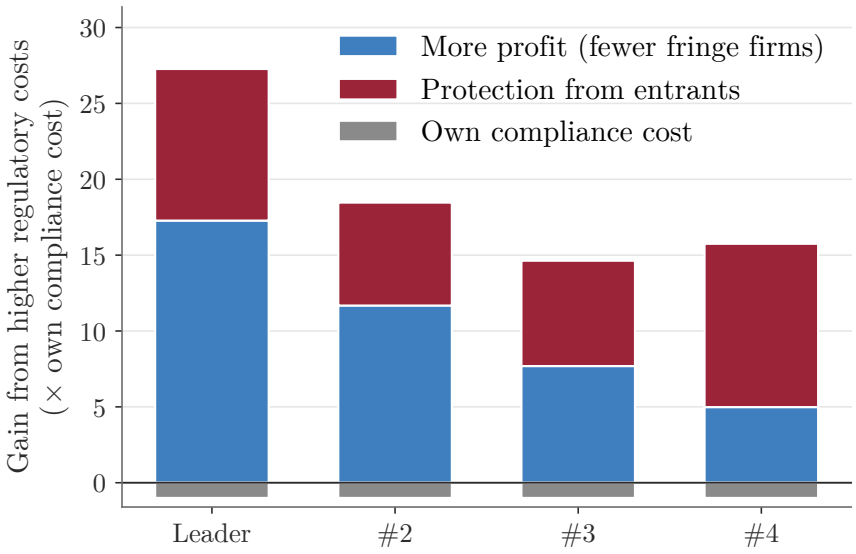


Figure 8. Why Superstars Lobby for Higher Regulatory Costs

Notes: The three components of a superstar's marginal value of a higher regulatory fixed cost (29), by rank, in units of the firm's own after-tax compliance cost, averaged over the stationary distribution of industry states at the estimate. Blue: Cournot gain as marginal fringe firms stop producing. Red: value of fewer fringe breakthroughs. Grey: own compliance cost.

The fringe opposes the regulatory costs. A fringe firm bears the compliance cost without the protection. By (36), the first two terms of the association's stake (42) sum to the after-tax change in the operating cutoff, which is negative: the survivors' higher profits only partly offset the compliance cost. The third term, the value of a breakthrough to the industry's fringe pool, changes the sign only in some industry states, and 75% of the association's positions ask for lower costs. The association's stake is spread over about 4,600 producing members, and it is organized in 14% of industry-quarters: the superstars win the contest.

The regulatory fixed cost. Averaged over industries, lobbying raises the regulatory fixed cost by 65% relative to its level without lobbying. The effect is concentrated where superstars lobby: in 55% of industry-quarters no one lobbies on regulation and the cost stays at R_0, while at the 90th percentile across industries it is 3.4 times R_0. It is largest where few superstars compete: the regulatory cost is 2.11 times R_0 on average in industries with one superstar, 1.97 with two, 1.71 with three, and 1.51 with four.

6.2 Lobbying, industry structure, and growth

Table 14 compares the economy with and without lobbying. The higher regulatory costs lower the operating cutoff of the fringe and the expected value of a firm in the fringe pool, so that the pool shrinks under free entry (by 1.5%) and the mass of operating fringe firms falls by 1.8%. Fewer operating fringe firms mean fewer breakthroughs into the superstar set, and the stationary distribution of industries shifts toward states with fewer superstars: the share of industries with four superstars falls from 54.9 to 54.3%, and the share of the four largest firms in industry sales rises from 37.1 to 37.3%. Compliance costs rise from 0.33 to 0.53% of output, and lobbying spending is 0.036% of output.

Composition, not effort. Growth falls by 0.17 basis points, from 1.8472 to 1.8456%. The decline does not come from lower innovation effort by a given firm. Superstars innovate *more* with lobbying: at every rank and for every number of superstars, a superstar's innovation rate is higher than in the same industry states without lobbying, by 0.10 to 0.21% (Figure 9), consistent with the regulatory costs protecting the rents that innovation secures. Growth falls because the composition of industries changes. Industries with four superstars grow fastest, 2.06% a year, against 1.77% with three superstars and 1.54% with two; a full top four requires a steady supply of challengers, and the regulatory costs reduce it. The shift away from industries with four superstars lowers growth by 0.0038 percentage points, and the superstars' higher innovation within each state raises it by 0.0022 (Figure 10).

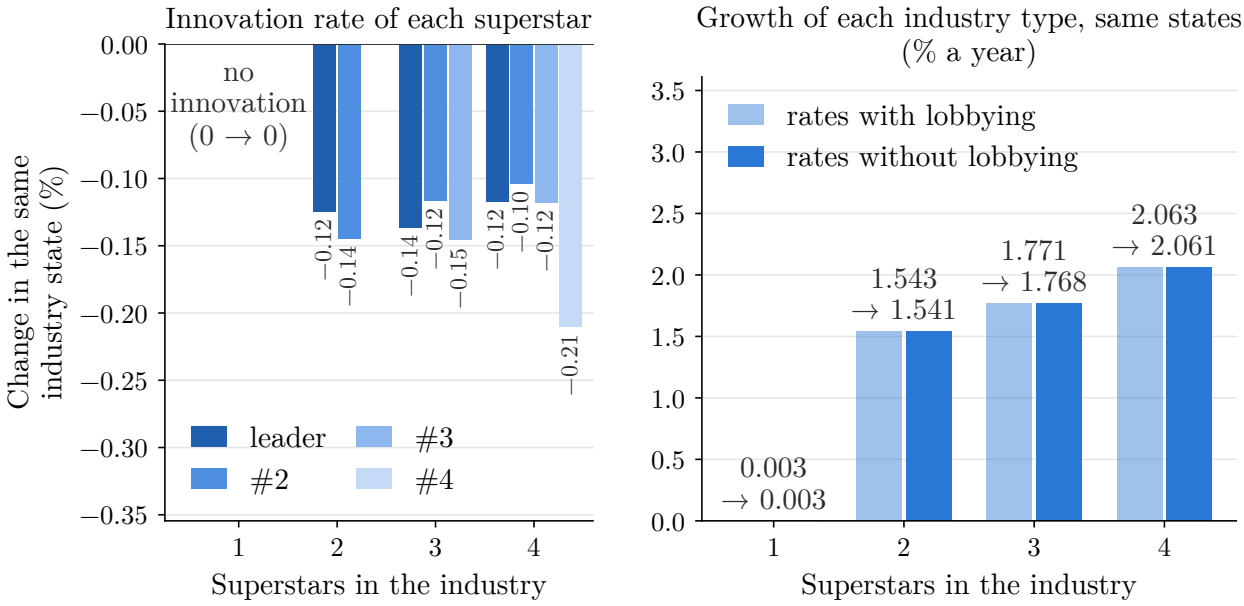


Figure 9. Innovation and Growth Within Industry States

Notes: Left: change in each superstar's innovation rate without lobbying, in the same industry state, by rank and number of superstars, percent. Right: growth rate of each industry type, with lobbying and without lobbying in the same industry states, percent per year. At the estimate.

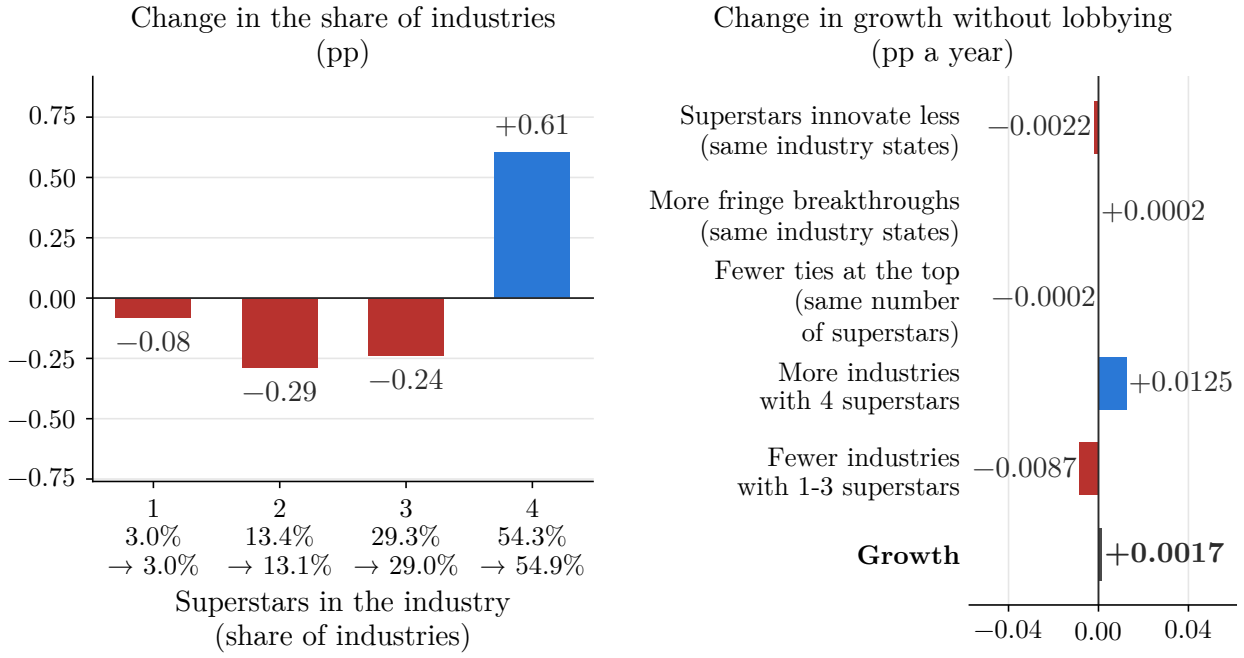


Figure 10. Industry Composition and Growth Without Lobbying

Notes: Left: change in the share of industries by number of superstars without lobbying, percentage points. Right: change in growth without lobbying by source, percentage points a year: superstars' innovation and fringe breakthroughs in the same industry states, ties at the top, and the shift across numbers of superstars. At the estimate.

6.3 The welfare cost of lobbying

Lobbying lowers consumption by 0.35% at every date, lowers its growth rate by 0.0017 percentage points, from 1.8472 to 1.8456% a year, and lowers hours by 0.006%. Table 15 decomposes these changes by the channel through which lobbying acts. Its first column measures each change without reference to preferences; its second converts the changes into the consumption-equivalent cost (48), which with log utility and $\rho = 0.04$ values a permanent change in growth at $1/(r - g) = 25$ times the same change in the level of consumption. The cost is 0.39% of lifetime consumption.

Three findings stand out. First, two thirds of the cost is direct waste, and most of it is compliance: higher compliance costs absorb 0.22% of consumption, while lobbying spending, including participation and organizing costs, absorbs 0.04. A rent-seeking calculation that counts the dollars spent on lobbying would miss nine tenths of the cost.

Second, pushing marginal fringe firms out of production costs nothing in itself. The smaller operating fringe produces 0.10% less, and markups distort the allocation of labor by a further 0.03, but the firms that stop producing are those with the highest fixed costs, and the fixed and entry costs that are no longer paid return 0.15%. These firms were close to indifferent between producing and not, so their exit is a wash.

Third, two fifths of the cost comes from fewer breakthroughs. With fewer entrants, industries have fewer superstars and the fringe learns from a weaker set of leaders, which lowers the output that an efficient allocation of labor could produce by 0.08%; markups across industry states cost a further 0.02; and growth slows, because fewer industries have four superstars (0.10), although each superstar innovates more (-0.06). Figure 11 shows the resource side: compliance rises by 0.20% of output while the fixed and entry costs of the fringe fall by 0.13.

Preferences change only the value of the growth loss. With $r - g$ of 2, 3, 4, and 6%, the cost is 0.43, 0.41, 0.39, and 0.38% of consumption, whatever the curvature and discount rate that produce each value.[3]

These calculations treat compliance as pure waste: a rule that raises the regulatory fixed cost absorbs resources and yields nothing in return. The rules that make up R are those whose compliance costs do not vary with output, such as licenses, product certification, reporting and disclosure, data privacy, and surety bonds; rules whose costs scale with output, including technology and performance standards and most environmental regulation, belong to other wedges (Appendix K). Some of these rules may nevertheless protect consumers, so we ask how valuable they would have to be for lobbying to be harmless. Suppose that

[3]Exactly with log utility and to first order otherwise. Valuing government purchases at 0 or 0.5 instead of 1 leaves the cost at 0.393%.

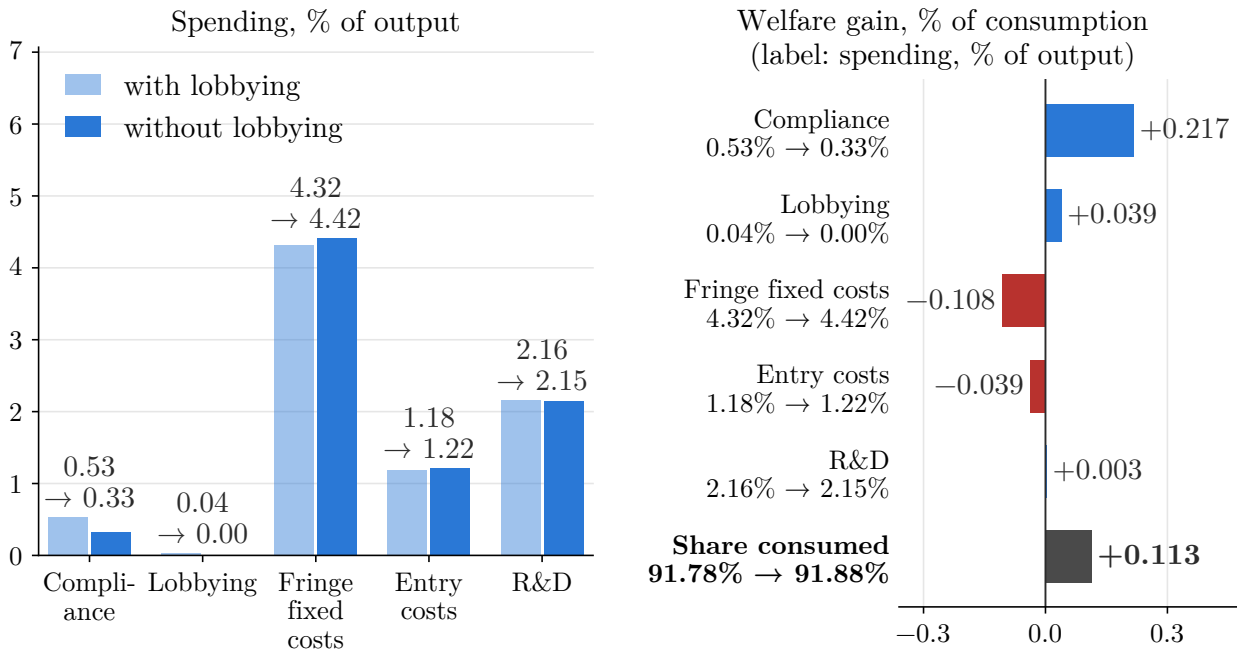


Figure 11. Resources With and Without Lobbying

Notes: Left: spending on compliance, lobbying, fringe fixed costs, entry, and R&D, percent of output, with and without lobbying. Right: contribution of each item to the welfare cost of lobbying through the share of output consumed, percent of consumption. At the estimate.

each dollar of compliance spending yields benefits worth v dollars of consumption to households, which leaves every decision of firms unchanged. Lobbying lowers welfare unless v exceeds 1.81 (Table 16). The threshold exceeds one because a higher regulatory cost also pushes marginal fringe firms out of production, which lowers productivity and the rate of breakthroughs: even if every dollar of compliance returned a dollar of benefits, lobbying would cost 0.18% of lifetime consumption. The evidence on the rules in R falls short of this threshold. Occupational licensing, the clearest case, raises prices and wages, and its welfare costs exceed its benefits (Kleiner and Soltas, 2023); the large benefit-cost ratios reported for federal regulation come mainly from air-quality rules (Office of Management and Budget, 2013), which lie outside R.

6.4 Which lobbying causes the cost?

Table 17 shuts down lobbying over regulation by setting its participation cost to infinity and solving for the new equilibrium. Regulatory costs lobbying accounts for almost the entire welfare cost. With lobbying allowed only for taxes and contracts, the regulatory costs are at their level without lobbying, growth is unchanged, and the welfare cost is 0.0025% of consumption, against 0.39% when all three kinds of lobbying are allowed.

Why tax and procurement lobbying cost little. In the model, a tax cut won by lobbying is a transfer from the household as taxpayer to the household as owner of the firm, and the government finances it with a slightly higher labor tax. A contract won by lobbying is an additional government purchase: the household consumes the goods it delivers, which the government pays at their cost, and the contractor's fee is a transfer financed by the same labor tax. The only efficiency costs are the lobbying resources, 0.003% of output, and the distortion of labor supply by the labor tax, which rises by 0.008 percentage points. In basis points of consumption, lower output costs 0.13 and the resources spent on lobbying 0.34, while slightly faster growth gains 0.07 and more leisure 0.15, for a total of 0.25: a tax cut on own profits and a contract that grows as the firm closes the gap to the leader make climbing the ladder slightly more valuable. Even if the goods bought with lobbied contracts were worth nothing to the household, the cost of lobbying would rise by only 0.1 basis points.

This result inverts the emphasis of the rent-seeking view and of public debate, which focus on firm-specific tax breaks and contracts. These are the most visible products of lobbying, and in the model they are close to pure transfers. The costly product is the least visible one: the contest over the requirements that set the cost of operating in an industry, which changes how many firms compete and how often the leaders are challenged.

6.5 Policy implications

Three implications follow. First, lobbying reforms should target the contest over regulation rather than lobbying in general: banning regulatory-cost lobbying removes more than 99% of the welfare cost, while lobbying for tax cuts and contracts on its own costs almost nothing. Second, the cost of capture runs through entry and the number of firms that compete, so procedural reforms of industry-wide rule-making, such as requirements to assess the burden of a rule on small firms and on entrants, address the margin on which the cost arises. Third, the outcome of the contest depends on the organization of the side that opposes the regulatory costs. The fringe's association is organized in a minority of industry-quarters, and the contest is decided by the few firms whose stakes are concentrated; measures that lower the cost of organizing small firms would shift the contest without restricting political activity.

7 Conclusion

We measured what firms lobby for, instrument by instrument, and quantified the consequences in a dynamic model of competition and innovation. Firms agree on transfers and

disagree on the regulatory fixed costs of their industries, and the largest firms ask most often for higher ones. In the estimated model, superstars lobby for higher regulatory costs, the leaders to take over the business of the marginal small firms, and the smaller superstars to keep entrants out; the association of small firms opposes them but is organized too rarely to prevail. Lobbying raises the regulatory fixed cost by 65%, makes industries more concentrated, shrinks the competitive fringe by 1.8%, slows growth, and costs 0.39% of lifetime consumption. Regulatory costs lobbying accounts for almost the entire cost, while lobbying for tax cuts and contracts, the most visible and most criticized form of corporate political activity, is close to a pure transfer.

The instrument-level classification opens two directions. It makes it possible to evaluate reforms of individual instruments, for example licensing requirements separately from reporting duties, and to extend the model to the other wedges firms lobby on, such as government demand and administered prices. And because it records how compliance costs scale with firm size instrument by instrument, it can discipline heterogeneity in the stakes of large firms, which would let firms of similar size take opposite sides on the same rule.

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Table 7. Technology Moments: Data and Model

Moment	Data	Model	Source
Sales among superstars			
Industries with fewer than four superstars	0.229	0.457	Compustat
Superstars per industry	3.64	3.35	Compustat
Leader's share of superstars' sales, median	0.458	0.520	Compustat
Leader's share of superstars' sales, 90th percentile	0.735	0.731	Compustat
#2 / #1 sales, 10th percentile	0.235	0.265	Compustat
#2 / #1 sales, median	0.613	0.529	Compustat
Industries with #2 / #1 below 0.3	0.162	0.162	Compustat
Industries with #2 / #1 above 0.8	0.252	0.156	Compustat
#4 / #1 sales, median	0.288	0.205	Compustat
Cost structure of the largest firms			
Sales / cost of goods sold, superstars	1.497	1.536	Compustat
Payroll per dollar of sales, four largest firms relative to the rest	0.764	0.745	Economic Census
New superstars			
New superstars per industry-year	0.132	0.124	Compustat
lowest tercile of the previous year's #4 / #1	0.105	0.117	Compustat
highest tercile of the previous year's #4 / #1	0.171	0.199	Compustat
Landing at the entry bar	0.636	0.346	Compustat
Landing one step above the bar	0.247	0.210	Compustat
Landing two steps above the bar	0.069	0.208	Compustat
Log sales relative to the leader, mean	−1.255	−1.424	Compustat
Log sales relative to the leader, standard deviation	0.639	0.490	Compustat
Gap dynamics of the same two superstars over two years			
Tied (sales ratio at least 0.9): apart	0.303	0.183	Compustat
Close (0.7 to 0.9): apart	0.146	0.127	Compustat
Close: closer	0.128	0.076	Compustat
Separated (0.4 to 0.7): apart	0.045	0.055	Compustat
Separated: closer	0.098	0.054	Compustat
Far (below 0.4): closer	0.079	0.038	Compustat
Fringe			
CR4	0.224	0.373	Economic Census
Firms with losses	0.387	0.144	IRS Statistics of Income
Exit premium of loss-making fringe firms	0.035	0.050	Compustat
Dispersion of log fixed costs, active fringe firms	0.670	0.628	Compustat
Firm exit rate	0.079	0.079	BDS
Business applications per employer firm	0.242	0.189	BFS, BDS
Applications that become employers within two years	0.273	0.404	BFS
Aggregates			
Growth of real GDP per capita	0.0166	0.0185	BEA
Labor share	0.596	0.587	Karabarbounis and Neiman (2014), OECD
R&D funded by business / GDP	0.0197	0.0216	NSF

Notes: All 35 moments are targeted with equal weights; model values in the equilibrium with lobbying at the estimate. Superstars are the four largest Compustat firms of a four-digit NAICS industry-year with at least 10% of the leader's sales, in industry-years with at least six Compustat firms, 2008–2022. In the data, a step of the ladder is a sales ratio of 1.27. Gap dynamics: share of pairs whose sales ratio is below (apart) or above (closer) its initial value both one and two years later. BDS: Business Dynamics Statistics; BFS: Business Formation Statistics. Appendix I defines each moment.

Table 8. Technology: Untargeted Moments

Moment	Data	Model	Source
Markup of superstars, cost-weighted	1.268	1.324	Compustat, as in De Loecker et al. (2020)
Industries with at most two superstars	0.111	0.164	Compustat
Industries with one superstar	0.023	0.030	Compustat
Fringe firms with negative operating income before depreciation	0.262	0.144	Compustat
Within-firm elasticity of patents to R&D intensity	0.262	0.164	Compustat, patents

Notes: Model values at the estimate. The model's loss share is the targeted moment of Table 7, here compared with the Compustat measure. Patent elasticity: Poisson regression of patents, by filing year, on the log of R&D over industry sales, with firm and year fixed effects, on superstar firm-years with positive R&D, 2008–2020 (2,356 firm-years of 380 firms); the standard error, 0.055, is clustered by firm. In the model, the elasticity equals $1/\phi$. Appendix I gives the definitions.

Table 9. Tax and Procurement: Moments and Identification

	Tax	Procurement	Identifies
What lobbying buys	book tax rate −1.12 pp (0.74) after a year of lobbying	contracts +0.273 log points (0.101)	κ_T, κ_P
No lobbying	superstars' effective rate 0.263	contracts / sales 0.019; elasticity to sales 0.75 (0.23)	τ_0; g_0, κ_g
Who lobbies	by rank 0.27 / 0.18 / 0.14 / 0.13; 1% to 69% across profit deciles	by rank 0.19 / 0.11 / 0.08 / 0.08; 3% to 59% across contract deciles	μ_I, σ_I
How long	80% still lobbying a year later	78%	λ_I
How much	spending 0.014% of the stake	0.078%	ϕ_I
Together	given regulation: 50%; given procurement: 43%	given regulation: 28%	ι_{IJ}

Notes: Payoffs: regressions of the three-year book tax rate and of log prime contracts on last year's share of quarters lobbied on the instrument, with firm and year fixed effects; standard errors in parentheses. Stake: pretax income (tax) and prime contracts (procurement). All ten deciles and quarterly and annual stay and join rates are targeted. Together: share of superstar firm-quarters lobbying on the instrument among those lobbying on the instrument named.

Table 10. Bills: Moments and Identification

In the bill data	Identifies	Data	Model
Lobbying superstars (log), raising proposals	price of influence κ_R	+3.7 pp	+4.1 pp
Any lobbying superstar, raising proposals		−1.3 pp	+1.3 pp
Association lobbying, raising proposals	association's strength	−0.8 pp	−2.2 pp
Lobbying superstars (log), lowering proposals	effectiveness β^a	−0.3 pp	−1.6 pp
Association lobbying, lowering proposals	effectiveness β^a	−1.3 pp	+2.4 pp
Lobbying superstars (log), balanced proposals		+2.7 pp	+1.9 pp
change per unit of the direction index		+0.1 pp	+2.1 pp
Size exponent ϱ	curvature θ	0.10	0.29
Enacted, lobbied, no stated side: raising / lowering	baseline π	6.4 / 9.4%	5.7 / 4.3%
Lobbying firms stating a side		1.0%	0.7%

Notes: Enactment effects per log point of lobbying firms; linear probability regressions with bill features and Congress fixed effects, on the model's industries; the same code on real and simulated proposals. Direction index: (provisions raising − provisions lowering compliance costs) / all provisions. The first eight rows are targeted; the last two are reported.

Table 11. Lobbying Parameters

Parameter		Estimate
Regulation		
Compliance without lobbying, share of output	c	0.0033
Size elasticity of superstars' compliance costs	e_R	0.47
Contest curvature	θ	0.89
Effectiveness of anti-side lobbying	β^a	22.8
Price of influence	κ_R	6.89×10^{-6}
Convexity of lobbying cost: superstars, association	ϕ_R, ϕ_A	3.78, 2.94
Participation cost of superstars: log mean, s.d.	μ_R, σ_R	−6.53, 1.53
Participation cost redraw rate (per year)	λ_R	0.37
Association organizing cost: log mean relative to superstars', s.d.	$\mu_A - \mu_R$, σ_A	1.59, 1.31
Association cost redraw rate (per year)	λ_A	0.50
Association lobbying cost: spending / industry sales	κ_A	1.60×10^{-4}
Profit tax		
Statutory tax rate	τ_0	0.283
Lobbying cost: leader's tax cut, share of τ_0	κ_T	0.71%
Lobbying cost convexity	ϕ_T	40.4
Participation cost: log mean, s.d.; redraw rate (per year)	μ_T, σ_T; λ_T	−8.16, 1.55; 0.23
Procurement		
Baseline contracts: contracts / sales	g_0	0.010
Decline of contracts with the gap	κ_g	2.86
Lobbying cost: leader's contracts due to lobbying	κ_P	1.3%
Lobbying cost convexity	ϕ_P	4.94
Participation cost: log mean, s.d.; redraw rate (per year)	μ_P, σ_P; λ_P	−12.24, 1.30; 1.00
Economies of scope in participation		
Regulation and tax; regulation and procurement; tax and procurement	ι_{RT}; ι_{RP}; ι_{TP}	0.18; 0.03; 0.001

Notes: Best point of the joint global search. The lobbying costs of the association and of the tax and procurement instruments are searched over, and reported, in the units of the outcomes they produce (association spending relative to industry sales; a lobbying leader's tax cut; the share of a lobbying leader's contracts due to lobbying), which map one-to-one into the structural cost parameters at each point; the baseline contract is reported as contracts over sales.

Table 12. Lobbying Moments: Model and Data

	Data	Model
Regulation: superstars		
Lobbying in a quarter, leader / #4	.342 / .176	.254 / .039
Still lobbying next quarter	0.928	0.933
Starting next quarter	0.026	0.010
Participation on log share of industry sales	0.149	0.167
Ask for higher compliance costs (share of positions)	0.533	0.957
Spending–sales elasticity	0.81	1.24
Leader's share of superstars' spending	0.48	0.67
Superstars' lobbying / industry sales ($\times 10^{-5}$)	4.2	18.5
Compliance costs / output	0.58%	0.53%
Regulation: association		
Active	0.569	0.144
Still active next quarter	0.955	0.905
Ask for higher compliance costs (share of positions)	0.374	0.253
Association lobbying / industry sales ($\times 10^{-5}$)	4.5	2.7
Superstars lobbying: association active − inactive	0.016	0.104
Tax		
Effective tax rate	0.263	0.282
Book tax rate change after a year of lobbying	−0.0112	−0.0013
Spending / stake	0.014%	0.101%
Still lobbying a year later	0.796	0.855
Procurement		
Contracts / sales	0.019	0.010
Elasticity of contracts to sales	0.75	1.04
Log contracts change after a year of lobbying	0.273	0.003
Spending / stake	0.078%	0.121%
Still lobbying a year later	0.781	0.607
Across instruments		
Lobbying on taxes, given lobbying on regulatory costs	0.496	0.306
Lobbying on procurement, given lobbying on regulatory costs	0.281	0.471
Lobbying on taxes, given lobbying on procurement	0.429	0.303

Notes: Selected targeted moments; Appendix I lists all of them. Model: simulated panel and stationary distribution at the estimate.

Table 13. Untargeted Moments

	Data	Model
Who lobbies on regulatory costs in an industry-quarter		
Association only	0.285	0.077
Superstars only	0.109	0.311
Both	0.284	0.067
Neither	0.322	0.545
Instruments a superstar lobbies on (share of quarters)		
None	0.661	0.689
Regulatory costs only	0.096	0.046
Tax only	0.040	0.099
Procurement only	0.038	0.070
All three	0.042	0.020
Industry structure		
Industries with at most two superstars	0.111	0.164
New superstar / leader sales, median (middle #4/#1 tercile)	0.254	0.223

Table 14. The Economy With and Without Lobbying

	No lobbying	Lobbying	Change
Regulatory fixed cost, $\mathbb{E}[R]/R_0$	1	1.650	+65.0%
Operating fringe firms per industry	4,667	4,581	−1.8%
Fringe pool per industry	11,665	11,495	−1.5%
Superstars per industry	3.359	3.349	−0.3%
Industries with four superstars	0.549	0.543	−0.6 pp
CR4	0.371	0.373	+0.2 pp
Markup of superstars (cost-weighted)	1.322	1.324	+0.1%
Compliance costs / output	0.33%	0.53%	+0.20 pp
Fringe fixed costs / output	4.42%	4.32%	−0.10 pp
Entry costs / output	1.22%	1.18%	−0.04 pp
Lobbying spending / output	0	0.036%	
R&D / output	2.153%	2.156%	+0.003 pp
Labor tax	24.40%	24.43%	+0.03 pp
Growth rate	1.8472%	1.8456%	−0.17 bp
Welfare cost (% of lifetime consumption)		**0.39**	

Notes: No lobbying: every participation cost set to infinity, at the same statutory tax rate, baseline contracts, and entry cost; the fringe pool solves free entry. Lobbying spending includes participation and organizing costs. Welfare cost: consumption-equivalent variation (48); positive values are losses.

Table 15. Where the Welfare Cost Comes From

Channel	Change (no preferences)	Welfare cost (% of consumption)	Share
Direct waste		*0.257*	*65%*
Compliance spending	share consumed −0.217%	0.217	
Lobbying, participation, and organizing	share consumed −0.039%	0.039	
Exit of marginal fringe firms		*−0.018*	*−5%*
Output of the smaller fringe	output −0.098%	0.098	
Markups within industry states	output −0.031%	0.031	
Fixed costs saved	share consumed +0.108%	−0.108	
Entry costs saved	share consumed +0.039%	−0.039	
Fewer breakthroughs		*0.148*	*38%*
Fewer, more unequal superstars	output −0.034%	0.034	
Fringe technology	output −0.049%	0.049	
Markups across industry states	output −0.023%	0.023	
Growth: fewer industries with four superstars	growth −0.0038 pp	0.096	
Growth: superstars' innovation	growth +0.0022 pp	−0.055	
Growth: fringe breakthroughs in the same states	growth −0.0002 pp	0.005	
Growth: ties at the top	growth +0.0002 pp	−0.004	
Other: R&D and hours	hours −0.006%	*0.005*	*1%*
Total	consumption −0.355% growth −0.0017 pp	**0.392**	100%

Notes: Balanced growth paths of the economies with and without lobbying at the estimate. Change: in percent for output and for the share of output consumed, whose changes add up to the change in consumption at every date, and in percentage points for the growth rate, by source (43); none requires an assumption on preferences. Welfare cost: contribution to $100\ln(1+\text{CEV})$ (48) with log utility and $\rho = 0.04$; level terms equal the change in consumption they cause, growth terms are the change in growth divided by $r - g = 0.04$, and hours are valued at the after-tax wage. The total equals $100\ln(1.00392)$. The output terms split the change in efficient productivity A^* and in allocative efficiency E (46) (Appendix H).

Table 16. The Welfare Cost of Lobbying When Regulation Has Benefits

Benefits per dollar of compliance spending	CEV, %
$v = 0$ (baseline)	0.392
$v = 0.5$	0.283
$v = 1$	0.175
$v = 1.5$	0.068
$v = 1.81$ (break-even)	0.000

Notes: Consumption-equivalent welfare cost of lobbying (48), in% of lifetime consumption, when regulation yields benefits that households value as consumption, equal to v times compliance spending in both economies. Allocations, growth, and hours are held at the estimated equilibria.

Table 17. Which Lobbying Causes the Cost?

Lobbying allowed on	Regulatory costs, $\mathbb{E}[R]/R_0$	Growth	Welfare cost (%)
None	1.000	1.8472%	0
Tax and procurement only	1.000	1.8472%	0.0025
All three	**1.650**	**1.8456%**	**0.392**

Notes: Each row solves the equilibrium with the participation cost of the excluded instruments set to infinity, at the estimated parameters. Welfare cost: consumption-equivalent variation relative to the economy without lobbying; positive values are losses.

A Example of the lobbying report

Clerk of the House of Representatives Legislative Resource Center 135 Cannon Building Washington, DC 20515 http://lobbyingdisclosure.house.gov	Secretary of the Senate Office of Public Records 232 Hart Building Washington, DC 20510 http://www.senate.gov/lobby

LOBBYING REPORT

Lobbying Disclosure Act of 1995 (Section 5) - **All Filers Are Required to Complete This Page**

1. Registrant Name ☑ Organization/Lobbying Firm ☐ Self Employed Individual
APPLE INC.

2. Address
Address1 700 K Street NW Address2
City WASHINGTON State DC Zip Code 20001 Country USA

3. Principal place of business (if different than line 2)
City Cupertino State CA Zip Code 95014 Country USA

4a. Contact Name: Mr. Timothy Powderly
b. Telephone Number: 2027729500
c. E-mail: gholger@apple.com
5. Senate ID# 4152-12

7. Client Name ☑ *Self* ☐ *Check if client is a state or local government or instrumentality*
APPLE INC.
6. House ID# 314500000

TYPE OF REPORT 8. Year 2026 Q1 (1/1 - 3/31) ☑ Q2 (4/1 - 6/30) ☐ Q3 (7/1 - 9/30) ☐ Q4 (10/1 - 12/31) ☐
9. Check if this filing amends a previously filed version of this report ☐
10. Check if this is a Termination Report ☐ Termination Date _____ 11. No Lobbying Issue Activity ☐

INCOME OR EXPENSES - YOU MUST complete either Line 12 or Line 13

12. Lobbying
INCOME relating to lobbying activities for this reporting period was:
Less than $5,000 ☐
$5,000 or more ☐ $ _____
Provide a good faith estimate, rounded to the nearest $10,000, of all lobbying related income for the client (including all payments to the registrant by any other entity for lobbying activities on behalf of the client).

13. Organizations
EXPENSE relating to lobbying activities for this reporting period were:
Less than $5,000 ☐
$5,000 or more ☑ $ 2,480,000.00

14. REPORTING Check box to indicate expense accounting method. See instructions for description of options.
☑ **Method A.** Reporting amounts using LDA definitions only
☐ **Method B.** Reporting amounts under section 6033(b)(8) of the Internal Revenue Code
☐ **Method C.** Reporting amounts under section 162(e) of the Internal Revenue Code

Signature Digitally Signed By: Timothy Powderly, Senior Director, Government Affairs **Date** 4/20/2026 10:26:49 PM

LOBBYING ACTIVITY. Select as many codes as necessary to reflect the general issue areas in which the registrant engaged in lobbying on behalf of the client during the reporting period. Using a separate page for each code, provide information as requested. Add additional page(s) as needed.

15. General issue area code CPT

16. Specific lobbying issues

General patent policy, including issues related to the Patent Trial and Appeal Board, S. 1553/H.R. 3160 the PREVAIL Act, and S. 708/H.R. 1574 the RESTORE Patent Rights Act
Issues related to standard essential patents
Issues related to anti-counterfeit policy
General trademark and copyright policy
Issues related to patent proceedings and the International Trade Commission, including the Advancing Americas Interests Act
General overview of music licensing legal framework

Figure 12. Sample lobbying report

B Participation: All Observations and Lobbying Periods

The participation figures of Section 3 use two denominators. Figures 2 and 6 divide by all observations: all firm-years and, for associations, all industry-years; they measure how often a client lobbies, the object of the model's participation targets. Figure 3 divides by the quarters with a lobbying report; it measures what lobbying clients lobby on. Figure 13 reports the shares from Figure 6 for years with a lobbying report.

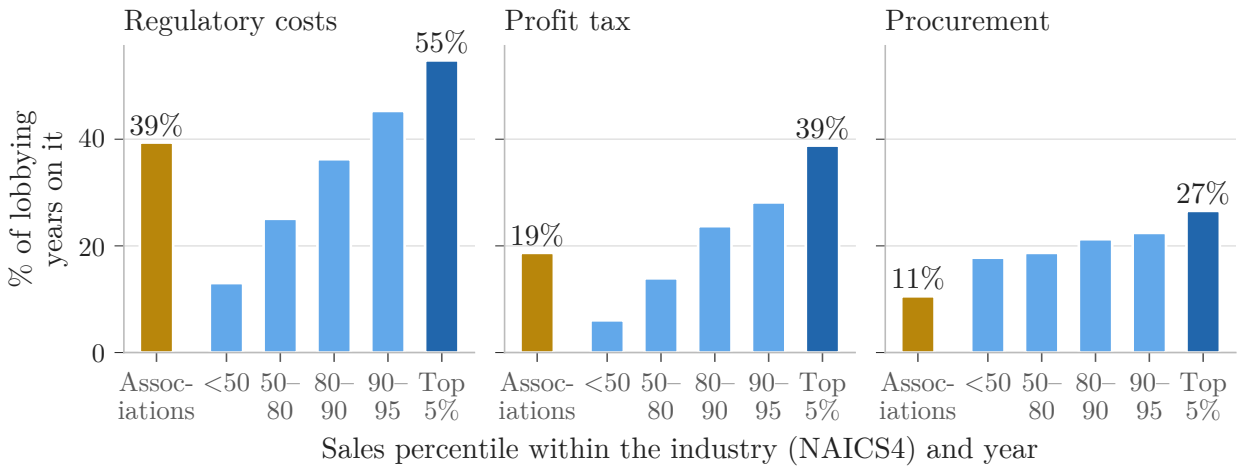


Figure 13. Participation in Lobbying by Firm Size, Among Lobbying Years

Notes: Share of the years with a lobbying report in which the client lobbies on the instrument: the 14,439 firm-years in which the firm lobbies and the 13,070 association-years with a report, by the firm's sales percentile within its industry (four-digit NAICS) and year. A year with a lobbying report has a report with a positive amount that names a classified instrument. 2008–2022.

C Static Cournot Equilibrium under Decreasing Returns to Scale

This appendix derives the static product-market objects used in Section 4: the superstar best responses, markups, and profits; the fringe output and profits; and the relative-output system solved at each industry state. The derivation follows Cavenaile et al. (2026), Appendix C.1, adapted to our notation. We fix an industry and a point in time and drop the j and t subscripts throughout. The industry contains N superstars with productivities $\{q_i\}_{i=1}^{N}$, an active fringe of mass m^{a} with common productivity q_f, the wage is w, and aggregate output is Y. Government contracts do not enter: government purchases are purchases of the final good, and a firm's contracts depend on its gap, not on its output choice.

Demand and notation. The final-good producer's problem yields the inverse demands

$$p_i \;=\; \frac{y_i^{-1/\eta}\,Y}{D}, \qquad p_f \;=\; \frac{\tilde{y}_f^{-1/\eta}\,Y}{D}, \qquad D \;\equiv\; \sum_{k=1}^{N} y_k^{\frac{\eta-1}{\eta}} + \tilde{y}_f^{\frac{\eta-1}{\eta}}. \tag{50}$$

It is convenient to define, for each superstar i and for the fringe aggregate,

$$\Sigma_i \;\equiv\; \sum_{k=1}^{N}\left(\frac{y_k}{y_i}\right)^{\frac{\eta-1}{\eta}} + \left(\frac{\tilde{y}_f}{y_i}\right)^{\frac{\eta-1}{\eta}} \;=\; \frac{D}{y_i^{\frac{\eta-1}{\eta}}}, \qquad \Sigma_{-i} \;\equiv\; \Sigma_i - 1, \qquad \Sigma_f \;\equiv\; \frac{D}{\tilde{y}_f^{\frac{\eta-1}{\eta}}}, \tag{51}$$

where Σ_i includes firm i's own term (equal to one). Revenues then satisfy $p_i y_i = Y/\Sigma_i$ and $p_f \tilde{y}_f = Y/\Sigma_f$: each producer's share of industry expenditure is the reciprocal of its own-normalized CES aggregator, $\sigma_i = 1/\Sigma_i$ and $\sigma_f = 1/\Sigma_f$, with $\sum_i \sigma_i + \sigma_f = 1$.

Superstar best response. Superstar i chooses y_i taking rivals' quantities $\{y_k\}_{k\neq i}$ and the fringe aggregate $\tilde{y}_f$ as given, internalizing the effect of y_i on D. Inverting the production function (4), $l_i = (y_i/q_i)^{1/\alpha}$, so operating profit is

$$\pi_i \;=\; \frac{y_i^{\frac{\eta-1}{\eta}}\,Y}{D} \;-\; w\left(\frac{y_i}{q_i}\right)^{1/\alpha}. \tag{52}$$

The first-order condition equates marginal revenue with marginal cost:

$$\frac{\eta-1}{\eta}\; y_i^{-1/\eta}\, Y\; \frac{D - y_i^{\frac{\eta-1}{\eta}}}{D^2} \;=\; \frac{w}{\alpha}\; y_i^{\frac{1-\alpha}{\alpha}}\; q_i^{-1/\alpha}. \tag{53}$$

The factor $(D - y_i^{(\eta-1)/\eta})/D^2$ on the left is the Cournot internalization: expanding output depresses the industry price index in proportion to the firm's own weight in D. Multiplying both sides of (53) by $(\alpha/w)\, q_i^{1/\alpha}\, y_i^{(\eta-1)/\eta}$ collects all powers of y_i into $y_i^{1/\alpha}$; raising to the power α and substituting (51) gives the best response

$$y_i \;=\; \alpha^{\alpha}\left(\frac{Y}{w}\right)^{\alpha}\left(\frac{\eta-1}{\eta}\right)^{\alpha} q_i \left[\frac{\Sigma_{-i}}{\Sigma_i^2}\right]^{\alpha}. \tag{54}$$

Markups. From (50) and (51), $p_i = Y/(y_i\Sigma_i)$. Substituting this into the right side of (53) shows that marginal cost equals $\frac{\eta-1}{\eta}\, p_i\, \Sigma_{-i}/\Sigma_i$, so the markup is

$$M_i \;=\; \frac{p_i}{\mathrm{MC}_i} \;=\; \frac{\eta}{\eta-1}\cdot\frac{\Sigma_i}{\Sigma_{-i}} \;=\; \frac{\eta}{(\eta-1)(1-\sigma_i)}. \tag{55}$$

The markup formula as a function of relative outputs is identical to the constant-returns baseline of Cavenaile et al. (2026); decreasing returns alter the level of quantities but not the mapping from market shares to markups.

Superstar profits. The wage bill satisfies $wl_i = w\, y_i^{1/\alpha} q_i^{-1/\alpha} = \alpha\, \mathrm{MC}_i\, y_i$, so using (55),

$$\pi_i \;=\; p_i y_i - wl_i \;=\; \frac{Y}{\Sigma_i}\left[1 - \alpha\,\frac{\eta-1}{\eta}\,\frac{\Sigma_{-i}}{\Sigma_i}\right] \;=\; \frac{\big(\eta + [(1-\alpha)\eta + \alpha]\,\Sigma_{-i}\big)\,Y}{\eta\,\Sigma_i^2} \;=\; \sigma_i\Big(1 - \frac{\alpha}{M_i}\Big)Y,$$

and the wage bill is $wl_i = \alpha\,\sigma_i Y/M_i$. The term $(1-\alpha)\eta + \alpha$ multiplying Σ_{-i} captures the additional rent that decreasing returns generate on inframarginal units.

Relative outputs. Taking the ratio of the pre-exponentiation form of (53) for two superstars i and k and raising to the power α yields

$$\left(\frac{y_i}{y_k}\right)^{\frac{\eta(1-\alpha)+\alpha}{\eta}} \;=\; \frac{q_i}{q_k}\left[\frac{\Sigma_i - 1}{\Sigma_i - \left(\frac{y_k}{y_i}\right)^{\frac{\eta-1}{\eta}}}\right]^{\alpha}, \tag{56}$$

where both sums are normalized by y_i. Combining the superstar best response (54) with the fringe aggregate output (58) below gives the superstar-to-fringe condition

$$\left(\frac{y_i}{\tilde{y}_f}\right)^{\frac{\eta(1-\alpha)+\alpha}{\eta}} \;=\; \left(\frac{\eta-1}{\eta}\right)^{\alpha}\frac{q_i}{q_f}\,(m^{\mathrm{a}})^{\alpha-1}\left[\frac{\Sigma_{-i}}{\Sigma_i}\right]^{\alpha}. \tag{57}$$

Fringe output and profits. Each fringe firm is a price taker and equates price to marginal cost, $p_f = \frac{w}{\alpha}\, y_{fk}^{\frac{1-\alpha}{\alpha}}\, q_f^{-1/\alpha}$. Using $p_f = Y/(\tilde{y}_f \Sigma_f)$ and $\tilde{y}_f = m^{\mathrm{a}}\, y_{fk}$, all powers of y_{fk} again collect into $y_{fk}^{1/\alpha}$, giving per-firm and aggregate fringe output

$$y_{fk} \;=\; \frac{\alpha^{\alpha}\, Y^{\alpha}\, q_f}{w^{\alpha}\,(m^{\mathrm{a}})^{\alpha}\,\Sigma_f^{\alpha}}, \qquad \tilde{y}_f \;=\; m^{\mathrm{a}}\, y_{fk} \;=\; \frac{(m^{\mathrm{a}})^{1-\alpha}\,\alpha^{\alpha}\, Y^{\alpha}\, q_f}{w^{\alpha}\,\Sigma_f^{\alpha}}. \tag{58}$$

Since $wl_{fk} = \alpha\, p_f\, y_{fk}$, per-firm fringe profit is

$$\pi_{fk} \;=\; (1-\alpha)\, p_f\, y_{fk} \;=\; \frac{(1-\alpha)\, Y}{m^{\mathrm{a}}\,\Sigma_f}. \tag{59}$$

Fringe firms charge no markup, yet decreasing returns ($\alpha < 1$) leave them positive operating profits. This is essential for the lobbying extension: it is these profits, net of the regulatory cost $R(S)$ and the idiosyncratic fixed cost ε, that give the cutoff $\varepsilon^*(S)$ its content.

Solving the static block. Productivities enter only through ratios. By (5) and (6),

$$\frac{q_i}{q_f} \;=\; \frac{(1+\lambda)^{-n_i}}{\zeta\left(\sum_{k=1}^{N}(1+\lambda)^{-\gamma n_k}\right)^{1/\gamma}},$$

where n_k is the gap of superstar k behind the leader. Given the technology state Θ and the active mass m^{a}, equations (56)–(57) form a system in the N relative outputs $\{y_i/\tilde{y}_f\}_{i=1}^{N}$, which we solve by Newton's method with the analytic Jacobian. Its solution determines the aggregators $\{\Sigma_i, \Sigma_f\}$, hence sales shares, normalized profits π_i/Y and π_f/Y, and markups (55); these depend only on (Θ, m^{a}), not on the wage, which the dynamic solver exploits. The derivatives $\partial(\pi_i/Y)/\partial m^{\mathrm{a}}$ and $\partial(\pi_f/Y)/\partial m^{\mathrm{a}}$ in the stakes (29) and (42) follow from the implicit function theorem on the same system. Labor demands $l_i = (y_i/q_i)^{1/\alpha}$ and $l_{fk} = (y_{fk}/q_f)^{1/\alpha}$ are each proportional to Y/w, and enter only labor-market clearing (44).

D Participation in Lobbying

This appendix gives the closed form of the participation cutoff (27) in the computed model of Section 4.6, in which a superstar's policies are evaluated at the expected costs of its current spells. Fix a superstar and write $e = (S, s_i)$ for its state without its locked costs; $e^{I,\mathrm{on}}$ and $e^{I,\mathrm{off}}$ denote its state after joining and after staying out of instrument I at a redraw.

Durations. For each instrument I on which the firm is active, $D_I(e)$ is the expected discounted time over which its current draw is paid, weighted by the scope discount. It solves, on states where $a_{Ii} = 1$,

$$\begin{aligned}\Big(\rho + \sum_{I'} \lambda_{I'} + q(e)\Big)\, D_I(e) \;=\; &\prod_{J \neq I} \big(1 - \iota_{IJ}\big)^{a_{Ji}(e)} \;+\; \sum_{e'} q(e,e')\, D_I(e') \\ &+ \sum_{I' \neq I} \lambda_{I'} \Big[\big(1 - G_{I'}(e)\big)\, D_I\big(e^{I',\mathrm{off}}\big) + G_{I'}(e)\, D_I\big(e^{I',\mathrm{on}}\big)\Big],\end{aligned} \tag{60}$$

where $q(e,e')$ are the rates of the events in (23) other than the firm's own redraws (innovation, entry, and the redraws of rivals and of the association), $q(e) = \sum_{e'} q(e,e')$, $G_{I'}(e) = \Phi\big((\ln \varphi^*_{I'}(e) - \mu_{I'})/\sigma_{I'}\big)$ is the probability of being on instrument I' after its next redraw, and $D_I = 0$ after the firm exits. The firm's own redraw on I ends the spell, so it appears in the discount rate with no continuation term.

Values. When the policies do not depend on the locked costs, the value is affine in them, $v(e, \boldsymbol{\varphi}_i) = v^0(e) - \sum_I a_{Ii}\, \varphi_{Ii}\, D_I(e)$, where v^0 solves (23) without the locked costs in the flow

and with the own-redraw terms

$$\lambda_I\Big[\big(1-G_I(e)\big)\,v^0\big(e^{I,\text{off}}\big)+G_I(e)\,v^0\big(e^{I,\text{on}}\big)-D_I\big(e^{I,\text{on}}\big)\,\mathbb{E}\big[\varphi\,\mathbf{1}\{\varphi\le\varphi_I^*(e)\}\big]-v^0(e)\Big]. \quad (61)$$

The expectation is the partial mean of the lognormal distribution (14),

$$\mathbb{E}\big[\varphi\,\mathbf{1}\{\varphi\le\varphi^*\}\big] \;=\; e^{\mu_I+\sigma_I^2/2}\,\Phi\Big(\frac{\ln\varphi^*-\mu_I-\sigma_I^2}{\sigma_I}\Big).$$

Cutoffs and policies. Comparing the values after joining and after staying out, with the new draw subtracted, gives the cutoff

$$\varphi_I^*(e) \;=\; \frac{v^0(e^{I,\text{on}})-v^0(e^{I,\text{off}})-\sum_{I'\neq I}a_{I'i}\,\varphi_{I'i}\,\big[D_{I'}(e^{I,\text{on}})-D_{I'}(e^{I,\text{off}})\big]}{D_I(e^{I,\text{on}})}. \quad (62)$$

The policies (z_i, b_{Ii}), the cutoffs, and the stake (29) are evaluated at the firm's expected locked costs, $\hat{\varphi}_I(e)=\mathbb{E}[\varphi \mid \varphi\le\varphi_I^*(e)]$, so that the locked costs need not be tracked as a continuous state. Spending and resource costs use the exact distribution of locked costs: on each instrument, the stationary mass of locked costs at every state solves a linear balance in which a spell starts with a draw accepted at the cutoff of the state where it starts and moves with the firm until its next redraw or its exit.

E Beliefs about Rivals' Tax and Procurement Statuses

This appendix describes the beliefs that replace rivals' tax and procurement statuses in the computed model of Section 4.6. The reduced industry state $\tilde{S}=(\Theta,\mathbf{a}_R,a_A)$ keeps the technology state, the superstars' statuses on regulation, and the association's status, and each superstar's own state keeps its own tax and procurement statuses $b=(a_T,a_P)$. Let $\nu(\tilde{S},j,b)$ be the stationary measure of superstars at rank j of reduced state $\tilde{S}$ with own statuses b. The belief about the statuses of the superstar at rank j in technology state Θ is

$$\pi(b\mid\Theta,j) \;=\; \frac{\sum_{\tilde{S}\in\Theta}\nu(\tilde{S},j,b)}{\sum_{\tilde{S}\in\Theta}\sum_{b'}\nu(\tilde{S},j,b')}, \quad (63)$$

where $\tilde{S}\in\Theta$ runs over every configuration of the regulation statuses and the association's status in technology state Θ. Decision objects use the beliefs: rivals' innovation rates and redraw probabilities in (23) are averages of rivals' policies at the actual reduced state over (63), and so are the fringe's pool value, the association's cutoff, and the contest stakes. A superstar's own stake is averaged over its own tax and procurement statuses with the same weights,

so that its intensity on regulation, and the regulatory cost, depend on $\tilde{S}$ alone. The process uses the actual conditional mix, $w(\tilde{S}, j, b) = \nu(\tilde{S}, j, b)/\sum_{b'} \nu(\tilde{S}, j, b')$: the stationary distribution, growth, the locked-cost balances, and every moment. Because no participation status enters the conditioning, a firm's participation choice cannot change what it predicts about its rivals.

F Computing the Equilibrium

At the estimated ladder ($\bar{n} = 5$), the computed model has 84 technology states, 2,180 reduced industry states once the regulation statuses and the association's status are included, and 33,104 superstar states once each firm's rank and own tax and procurement statuses are included. The equilibrium is computed by three nested fixed points.

1. **Inner**, given the pool m and the contest outcome at every state: superstars' values, durations, policies and participation cutoffs; the fringe cutoffs, active masses and breakthrough intensities; the association's cutoffs; the beliefs; and the stationary distribution.

2. **Middle**, given m: the stakes from the inner solution, the contest at every state (the regulatory cost, dues and lobbying costs), iterated to a fixed point with Anderson acceleration (Walker and Ni, 2011).

3. **Outer**: the pool m that satisfies free entry (39), by a secant method.

An equilibrium is accepted only if one further evaluation at full precision certifies it: the contest residual is below 10^{-6}, the free-entry residual below 10^{-7} (relative), and the inner fixed point is solved to 10^{-9}. At the estimate, the lobbying equilibrium takes about two and a half hours on one core, and the certificate gives a contest residual of 8.5×10^{-7} and a free-entry residual of 1.1×10^{-9}. The economy without lobbying is computed with the same code, with every participation cost set to infinity. The global search of Section 5.1 solves each point in a fresh process with the same certificate; points whose fixed points do not certify within the time limit are discarded.

G From Bills to the Contest

This appendix describes how the enactment of bills maps into the regulatory fixed cost of the contest (13), and how the bill moments are simulated.

Rules add up. The regulatory fixed cost of an industry is the stock of rules in force. Proposals that raise it arrive at rate Λ_U; if enacted, proposal k adds r_k to the stock. Proposals that lower it arrive at rate Λ_L; if enacted, proposal k removes a share s_k of the stock. Rules lapse at rate δ_R. Proposal k is enacted with probability p_k, given by (49), and with probability π_k without lobbying. In the steady state, $\Lambda_U \mathbb{E}_U[rp] = \big(\delta_R + \Lambda_L \mathbb{E}_L[sp]\big) R$, so that

$$\frac{R}{R_0} \;=\; \frac{\mathbb{E}_U[r\,p]}{\mathbb{E}_U[r\,\pi]} \cdot \frac{1+\ell}{1+\ell\,\mathbb{E}_L[s\,p]/\mathbb{E}_L[s\,\pi]}, \qquad \ell \;=\; \frac{\Lambda_L\,\mathbb{E}_L[s\,\pi]}{\delta_R}, \tag{64}$$

where ℓ measures how fast lowering proposals erode the stock relative to lapsing; we fit $\ell = 0.20$. Without lobbying, $p = \pi$ and $R = R_0$. The result is exact when proposals arrive independently of the rules in force and the industry's lobbying is constant while a rule lasts.

How a firm splits its force. Firm i puts a share h_{ik} of its lobbying dollars on proposal k, with mean one across proposals. With the lobbying cost of convexity ϕ_R, its force on proposal k grows as h_{ik}^{θ/ϕ_R}. To first order in the forces, the value of force on proposal k for R/R_0 is

$$v_k^U \;=\; \frac{r_k}{\lambda_U\,\mathbb{E}_U[r\pi]} \times \begin{cases} 1-\pi_k & \text{push} \\ \pi_k & \text{block} \end{cases}, \qquad v_k^L \;=\; \frac{\ell}{1+\ell}\,\frac{s_k}{\lambda_L\,\mathbb{E}_L[s\pi]} \times \begin{cases} 1-\pi_k & \text{push} \\ \pi_k & \text{block} \end{cases},$$

where λ_U and λ_L are the shares of each type of proposal. Let e_k be the firm's efficiency on the proposal, one for pushing and β^a for blocking. The optimal split is $h_{ik} \propto (v_k e_k)^{1/(1-\theta/\phi_R)}$, and we scale the force so that the firm's force on proposal k is $f_{ik} = F_i\,e_k h_{ik}^{\theta/\phi_R} / \mathbb{E}[v\,e\,h^{\theta/\phi_R}]$, with $F_i = \beta^p b_{Ri}^{\theta}$ on the pro side and $\beta^a b_{Ri}^{\theta}$ on the anti side; the association splits its force in the same way with its own convexity ϕ_A. Then $\sum_k v_k f_{ik} = F_i$, and to first order the bills give $R/R_0 - 1 = \beta^p B^{\text{pro}} - \beta^a B^{\text{anti}}$, which is the first-order expansion of the contest (13).

Simulating each proposal. We simulate the 2,244 raising and 698 lowering proposals of 2008–2022 as follows. (1) Each proposal is lobbied by the industries that lobbied it in the data. (2) For each such industry, we draw a model industry whose state comes from the stationary distribution, conditional on one of its firms lobbying on regulatory costs. (3) Forces add up across industries; a firm is listed on the proposal if its spending on it passes a threshold, and a listed firm may state its side. (4) The baseline probability is $\pi_k = \Phi(a_{U/L} + c\,z_k + \sigma_\xi\,\xi_k)$, where z_k is the proposal's index in the data's enactment regression (sponsor, cosponsors, must-pass legislation, Congress) and ξ_k an unobserved component; a_U, a_L, c, σ_ξ, and ℓ are fitted at each parameter point to twelve facts of the bill data, including the enactment of proposals that no modeled firm lobbied and of lobbied proposals on which no firm states a side. The sizes r_k and s_k are not observed and are drawn from a lognormal

distribution with standard deviation 0.8. (5) The data's own regression code is run on the simulated proposals; industries outside the model's set enter as a control, the number of other lobbyists, as in the data. The persistence of enactment chances across re-filed versions of the same proposal, a correlation of 0.82 between the latent chances of consecutive versions, measures the scale of the unobserved component directly.

Pushing and blocking. A firm with force f alone on a raising proposal changes its enactment probability by $f(1-\pi)/(1+f)$ by pushing and by $-\pi\beta^a f/(1+\beta^a f)$ by blocking. Since π is small, blocking matters only if β^a is large: to first order, lobbying superstars raise enactment on average only if $\beta^a < s(1-\pi)/((1-s)\pi)$, where s is the pro share. The association's dues fix its blocking force rather than β^a: a larger β^a with less effort gives the same dues and the same blocking. Information on β^a comes from lowering proposals, on which the roles flip, from the direction index, and from the association's effect on lowering proposals.

H Welfare Decomposition

Exact decomposition. Superstar k of an industry produces $y_k = q_k l_k^\alpha$, and the fringe of m^{a} identical firms, efficient internally, produces as one producer $Y_f = (m^{\mathrm{a}})^{1-\alpha} q_f L_f^\alpha$. Industry output is $y_j = (\sum_k y_k^s + Y_f^s)^{1/s}$ with $s = (\eta-1)/\eta$, and $\ln Y_0 = \sum_S \mu(S) \ln y(S)$. Since every producer is of degree α in labor, $\ln Y_0 = \alpha \ln L + \ln A^* + \ln E$ exactly, with

$$\ln A^* \;=\; \sum_S \mu(S)\, \frac{1}{e} \ln \sum_k A_k(S)^e, \qquad e = \frac{s}{1-\alpha s},$$

the productivity of a planner who allocates the same labor equally across industries (all have the same expenditure share and the same α) and within each industry in proportion to A_k^e, given the industry states and the operating fringe. The term $\ln E = \ln Y_0 - \alpha \ln L - \ln A^* \le 0$ splits into a within-industry and an across-industry part. Welfare (47) is $W = \ln c_0/\rho + g/\rho^2 - D/\rho$ with $\ln c_0 = \ln Y_0 + \ln(c_0/Y_0)$, so that

$$100 \ln(1 + \mathrm{CEV}) \;=\; -100 \left[\alpha\, \Delta \ln L + \Delta \ln A^* + \Delta \ln E + \Delta \ln(c_0/Y_0) + \frac{\Delta g}{\rho} - \Delta D\right],$$

where Δ is the lobbying economy minus the economy without lobbying. The change in $\ln A^*$ is split into the change within industry states, at the stationary distribution of the economy with lobbying, and the change due to the shift in the industry states, which itself splits into the superstars' configuration and the fringe's technology. The change in c_0/Y_0 is split by item of the resource constraint (45), and the change in growth by source (43).

Components at the estimate. In percent of lifetime consumption, the components are: labor input +0.005; efficient productivity within industry states +0.098 and from the change in industry states +0.084, of which superstars +0.034 and the fringe's technology +0.049; allocative efficiency within industry states +0.031, from the change in industry states +0.023, and across industries +0.000; resources +0.113; growth +0.041; and the disutility of labor −0.004; their sum, 0.392, equals $100\ln(1.00392)$. The resource component splits into compliance (+0.217), lobbying, participation, and organizing costs (+0.039), fringe fixed costs (−0.108), entry costs (−0.039), and R&D (+0.003). Growth falls from 1.8472 to 1.8456%: the shift away from industries with four superstars lowers it by 0.0038 percentage points, ties at the top by 0.0002, and fewer fringe breakthroughs in the same states by 0.0002, while the superstars' higher innovation raises it by 0.0022.

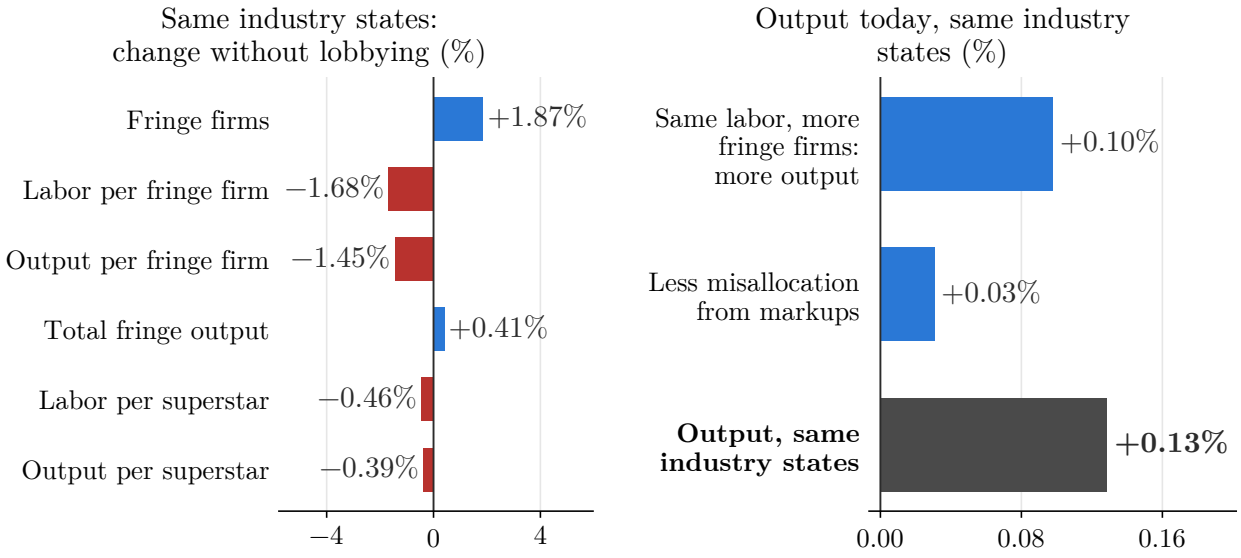


Figure 14. Producers Within Industry States, Without Lobbying

Notes: Left: change without lobbying, in the same industry states, in the number of operating fringe firms, labor and output per fringe firm, total fringe output, and labor and output per superstar, percent. Right: change in output within industry states from spreading the same labor over more fringe firms and from markups, percent. At the estimate.

I Target Moments

This appendix defines each targeted moment and its model counterpart. Table 7 reports the technology moments and Table 19 the lobbying moments. All moments receive equal weights (Section 5.1).

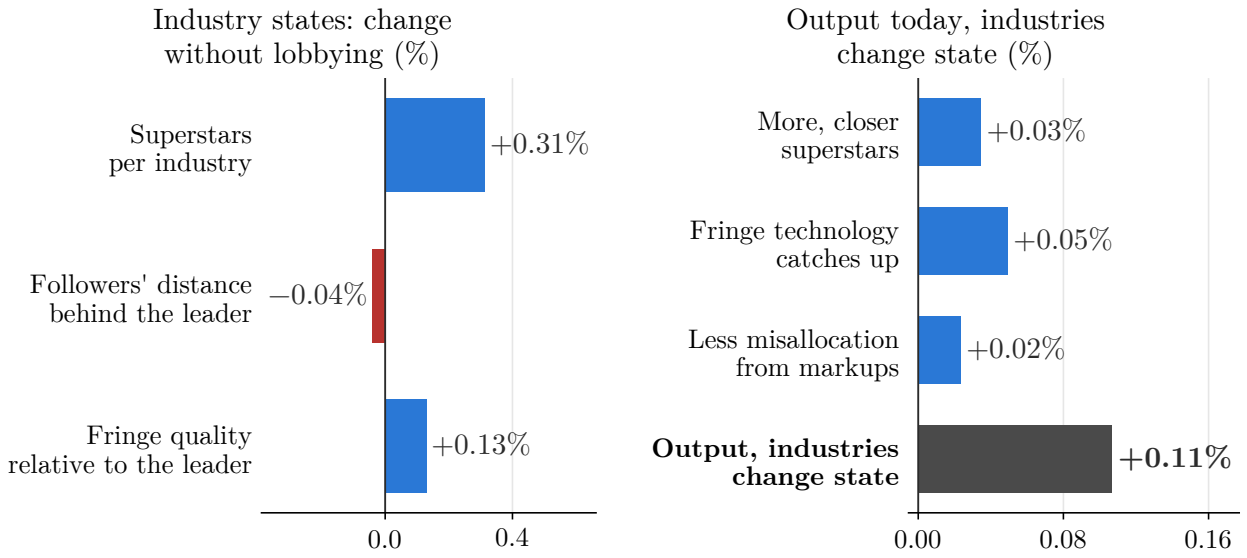


Figure 15. Industry States and Output, Without Lobbying

Notes: Left: change without lobbying in the number of superstars per industry, the gap of followers behind the leader, and the fringe's quality relative to the leader, percent. Right: change in output from the shift in industry states, by component, percent. At the estimate.

I.1 Technology moments

Sample. The firm sample is the Compustat panel of firms headquartered in the United States, 2008–2022. An industry is a four-digit NAICS code, and an industry-year enters if it has at least six Compustat firms: 2,131 industry-years in 170 industries. The superstars of an industry-year are the firms among its four largest by sales with at least 10% of the leader's sales (7,752 firm-years of 1,082 firms); all other firms form the fringe. For moments that follow firms across years, a change in a firm's reported industry code is treated as a re-coding, not a move, when its segment data show the same business in both years. A firm-year is affected by a material acquisition if cash acquisitions or the sales of acquired firms exceed 10% of the firm's sales, or goodwill rises by more than 5% of sales. In the model, statistics of industries are computed over the stationary distribution of industry states, and statistics of firms over time from the equilibrium transition rates or a simulated panel. Table 18 lists the sources.

Sales among superstars. The share of industries with fewer than four superstars is the share of industry-years in which the fourth-largest firm sells less than 10% of the leader's sales. The leader's share is its sales over the superstars' total sales (median and 90th percentile). The ratio #2/#1 is measured in industry-years with at least two superstars (10th percentile, median, and shares below 0.3 and above 0.8), and #4/#1 in industry-years with four superstars (median). The model computes each statistic with the same conditioning

Table 18. Technology Moments: Sources and Samples

Moments	Source	Years	Observations
Sales among superstars	Compustat	2008–2022	2,131 industry-years
Sales / cost of goods sold	Compustat	2008–2022	superstar firm-years
Payroll ratio	Economic Census	2017, 2022	132 industries
New superstars	Compustat	2008–2022	2,089 industry-years; 275 new superstars
Gap dynamics	Compustat	base years 2008–2020	3,825 pair-years
CR4	Economic Census	2012, 2017, 2022	132 industries
Firms with losses	IRS Statistics of Income, Corporation Source Book	2008–2013	936 industry-years
Exit premium of loss-making firms	Compustat	2008–2022	61,266 fringe firm-years
Dispersion of fixed costs	Compustat	2008–2022	5,801 fringe firms
Firm exit rate	Business Dynamics Statistics	2008–2022	all four-digit industries
Business applications	Business Formation Statistics; Business Dynamics Statistics	2008–2021	all sectors
Growth	BEA, real GDP per capita	2010–2019, 2022	annual
Labor share	Karabarbounis and Neiman (2014); OECD sector accounts	2008–2022	annual
R&D intensity	National Science Foundation	2008–2022	annual

over the stationary distribution.

Cost structure of the largest firms. The sales-to-cost ratio is the superstars' total sales over their total cost of goods sold, with each firm's dollars divided by the Compustat sales of its industry-year so that industries have equal size, as in the model; in the model, it is superstars' sales over their variable cost, which equals their cost-weighted markup divided by α. The payroll ratio is the payroll per dollar of receipts of the four largest firms of an industry over that of all its other firms in the Economic Census, averaged over 2017 and 2022 and across the 132 industries of our panel in the Census, weighted by industry-years; in the model, it is the superstars' labor cost per dollar of sales over the fringe's, which equals the inverse of the superstars' markup.

New superstars. A new superstar in year t is a superstar that was in the same industry in $t-1$ without being a superstar, has no material acquisition in t, and would remain a superstar if the firms that left the industry's four largest between $t-1$ and t were still present, with their sales grown at the industry's median rate. The entry rate is the number of new superstars per industry-year, overall and in the lowest and highest terciles of the previous year's #4/#1 (2,089 industry-years, 275 new superstars). A new superstar's landing is its real sales in t over the real entry bar of $t-1$, the larger of the fourth firm's sales and 10% of the leader's; its

number of steps above the bar is the log landing divided by $\ln 1.27$, rounded, with landings below the bar set to zero. The step of 1.27 is the median sales ratio of the third to the fourth superstar in industry-years with four superstars. We target the shares landing zero, one, and two steps above the bar and the mean and standard deviation of log sales relative to the leader in t, excluding the two new superstars that become leaders. The model computes the same statistics exactly over one year of the continuous-time process, with sales grown by one year of aggregate growth so that landings are in real terms.

Gap dynamics. For each pair of superstars in year t, the sales ratio is the smaller firm's sales over the larger firm's. Pairs are tied (ratio at least 0.9), close (0.7 to 0.9), separated (0.4 to 0.7), or far (below 0.4). A pair moves *apart* if its ratio is below its value in t in both $t+1$ and $t+2$, and *closer* if it is above in both years. The sample keeps pairs whose firms remain superstars of the industry in $t+1$ and $t+2$ without material acquisitions: 3,825 of the 9,521 pairs with base years 2008–2020. The model computes these probabilities exactly from the transition rates, for every pair of superstars in every state, weighted by the stationary distribution and conditional on both firms remaining superstars.

Concentration. CR4 is the receipts share of the four largest firms in the Economic Census, averaged over 2012, 2017, and 2022; we take its median across the 132 industries of our panel present in all three Censuses, weighted by industry-years. In the model, it is the superstars' share of industry sales, averaged over the stationary distribution.

Fringe firms. The loss share is the share of corporate returns without net income among corporations with assets of \$1 to \$500 million, by minor industry of the IRS Corporation Source Book, 2008–2013, each industry-year weighted equally; in the model, it is the share of producing fringe firms whose operating profit net of compliance, breakthrough spending, and fixed cost is negative. The exit premium is the coefficient from a regression of an indicator that a fringe firm's last Compustat year is t and it was deleted for bankruptcy, liquidation, or other reasons (deletion codes 2, 3, and 10) on an indicator of negative operating income before depreciation in t, with industry-year fixed effects and industry-years weighted equally; the model runs the same regression on a simulated panel in which a firm exits when it dies or stops producing. The dispersion of fixed costs is the standard deviation across 5,801 fringe firms of the firm average, over at least three years, of the residual from regressing log selling, general, and administrative expenses net of R&D on log sales with industry-year fixed effects (slope 0.667); in the model, it is the same statistic for the log of the fixed cost plus compliance of producing fringe firms, net of its industry-year mean. The exit rate is firm deaths over firms in the Business Dynamics Statistics, summed over industries and averaged over 2008–2022; in the model, it is the share of producing fringe firms that die or stop producing within a year. Applications per employer firm are high-propensity applications in

the Business Formation Statistics over employer firms in the Business Dynamics Statistics, averaged over 2008–2021, and the share that become employers is business formations within eight quarters over high-propensity applications. In the model, applications are entries into the pool, so the first moment is the entry flow per producing fringe firm and the second the share of new firms that produce within two years before dying.

Aggregates. Growth is the mean annual log growth of real GDP per capita (BEA) over 2010–2019 and 2022, excluding the recession years; in the model, it is the growth rate (43). The labor share is the corporate labor share of Karabarbounis and Neiman (2014), extended with the OECD accounts of nonfinancial and financial corporations, averaged over 2008–2022; in the model, it is the wage bill over output times one minus the capital share. R&D intensity is business-funded R&D over GDP from the National Science Foundation, averaged over 2008–2022; in the model, it is superstars' R&D plus fringe breakthrough spending, over output.

Untargeted moments. The superstar markup is the production-function markup computed as in De Loecker et al. (2020), weighted by cost of goods sold relative to industry-year sales; in the model, superstars' markups are weighted by their variable cost. The patent elasticity is the coefficient from a Poisson regression of a superstar's granted patents, by filing year, on the log of its R&D over industry-year sales, with firm and year fixed effects, on superstar firm-years with positive R&D, 2008–2020 (2,356 firm-years of 380 firms). Patents are matched to firms with the match of Dyèvre and Seager, extended to recent filing years with that of Kogan et al. (2017). In the model, innovation is $z = (\text{R\&D}/\chi)^{1/\phi}$, so the elasticity is $1/\phi$.

I.2 Lobbying moments

Table 19. Lobbying and Bill Moments: Data and Model

Moment	Data	Model
Regulation: superstars		
Regulatory costs: superstars lobbying in a quarter, rank 1	0.342	0.254
rank 2	0.253	0.105
rank 3	0.208	0.051
rank 4	0.176	0.039
Regulatory costs: still lobbying next quarter	0.928	0.933
Regulatory costs: starting next quarter	0.0262	0.0101

Moment	Data	Model
Participation on log share of industry sales	0.149	0.167
Ask for higher compliance costs (share of positions)	0.533	0.957
Regulation: association		
Association active	0.569	0.144
Association still active next quarter	0.955	0.905
Association starting next quarter	0.0669	0.0159
Association asks for higher compliance costs (share of positions)	0.374	0.253
Superstars lobbying: association active − inactive	0.0161	0.1037
Association lobbying / industry sales	4.46×10^{-5}	2.67×10^{-5}
Association lobbying / industry sales (upper bound)	1.83×10^{-4}	1.75×10^{-4}
Regulation: levels		
Regulatory costs spending–sales elasticity	0.811	1.242
Leader's share of superstars' regulatory costs spending	0.483	0.666
Superstars' regulatory costs lobbying / industry sales	4.18×10^{-5}	1.85×10^{-4}
Compliance costs / output	0.0058	0.0053
Compliance costs / output (upper bound)	0.0164	0.0053
Profit tax		
Effective tax rate of superstars	0.263	0.282
Book tax rate change after a year of lobbying	−0.0112	−0.0013
Tax lobbying spending / stake	1.40×10^{-4}	1.01×10^{-3}
Tax: superstars lobbying in a quarter, rank 1	0.266	0.345
rank 2	0.182	0.150
rank 3	0.144	0.066
rank 4	0.126	0.026
Tax: lobbying, pretax-income decile 1	0.012	0.022
decile 2	0.019	0.033
decile 3	0.034	0.055
decile 4	0.039	0.074
decile 5	0.087	0.112
decile 6	0.123	0.154
decile 7	0.162	0.201
decile 8	0.320	0.260
decile 9	0.480	0.339
decile 10	0.692	0.454

Moment	Data	Model
Tax: still lobbying a year later	0.796	0.855
Tax: starting within a year	0.0569	0.0317
Tax: still lobbying next quarter	0.910	0.961
Tax: starting next quarter	0.0219	0.0083
Procurement		
Contracts / sales	0.0195	0.0100
Elasticity of contracts to sales (PPML)	0.749	1.036
Log contracts change after a year of lobbying	0.273	0.0025
Procurement lobbying spending / stake	7.79×10^{-4}	1.21×10^{-3}
Procurement: superstars lobbying in a quarter, rank 1	0.190	0.387
rank 2	0.111	0.138
rank 3	0.077	0.057
rank 4	0.078	0.031
Procurement: lobbying, contracts decile 1	0.028	0.013
decile 2	0.031	0.020
decile 3	0.045	0.044
decile 4	0.062	0.073
decile 5	0.073	0.107
decile 6	0.111	0.151
decile 7	0.183	0.280
decile 8	0.204	0.288
decile 9	0.311	0.375
decile 10	0.588	0.513
Procurement: still lobbying a year later	0.781	0.607
Procurement: starting within a year	0.0310	0.0881
Procurement: still lobbying next quarter	0.895	0.872
Procurement: starting next quarter	0.0140	0.0279
Across instruments		
Lobbying on taxes, given lobbying on regulatory costs	0.496	0.306
Lobbying on procurement, given lobbying on regulatory costs	0.281	0.471
Lobbying on taxes, given lobbying on procurement	0.429	0.303
Bills		
Bills: lobbying superstars (log), raising	0.0370	0.0408
Bills: any lobbying superstar, raising	−0.0129	0.0125

Moment	Data	Model
Bills: association lobbying, raising	−0.0078	−0.0224
Bills: lobbying superstars (log), lowering	−0.0034	−0.0161
Bills: association lobbying, lowering	−0.0133	0.0235
Bills: lobbying superstars, balanced proposals	0.0266	0.0191
Bills: change per unit of the direction index	0.0011	0.0212
Bills: size exponent ϱ	0.10	0.29

Notes to Table 19: Compustat superstars headquartered in the United States; regulation and association moments 2008–2022 (industry associations: every report 2008–2022); tax and procurement participation at the last quarter of the fiscal year, fiscal years 2008–2022 (participation by decile: fiscal years 2008–2017). Model: point 3285, estimated on the previous targets; the data column shows the targets on 2008–2022. A firm lobbies on an instrument in a quarter if any of its reports lobbies on it, monitoring included, at high or medium confidence. Positions: share of a filer-year's stated positions on regulatory costs that ask for higher costs. Payoff regressions: last year's share of quarters lobbied on the instrument, firm and year fixed effects. Deciles: of pretax income (tax) and of last year's prime contracts (procurement). Bill coefficients: percentage points per log point of lobbying firms (or per indicator), enactment regressions with bill features and Congress fixed effects on the proposals of the model's industries. Upper bounds enter the objective only when exceeded.

J Issue-Text Splitting: Examples

Each example shows a raw LD-2 issue-text block and the subissues produced by the splitting stage. Deterministic pattern rules handle unambiguous structure (bill-led segments, enumerated lists, blank-line groups); blocks whose boundaries require semantic judgment are routed to the language-model stage. Original filers' typos are preserved.

Example 1: bill-led block, one subissue per bill.

> "S. 1816 and H.R. 3852: Chesapeake Reauthorization, H.R. 5088: America's Commitment to Clean Water Act, H.R. 5509: Chesapeake Bay Program Reauthorization and Improvement Act"

Split into three subissues: (1) S. 1816 / H.R. 3852, Chesapeake Reauthorization (a cross-chamber companion pair kept as one subissue); (2) H.R. 5088, America's Commitment to Clean Water Act; (3) H.R. 5509, Chesapeake Bay Program Reauthorization and Improvement Act.

Example 2: semicolon list mixing instruments.

> "Issues and incentives related to biodiesel, and the Renewable Fuel Standard; S. 1277/H.R. 2238, Biodiesel Tax Incentive Reform and Extension Act; S. 1294, Oil Independence for a Stronger America Act, Title V – Alternative Transportation Fuels; S. 1220, Fulfilling U.S. Energy Leadership Act, Title IIII – Alternative Fuel Deployment. Implementation of Budget Control Act. Extension of biodiesel tax credit. S. 3240, Agriculture Reform, Food and Jobs Act."

Split into seven subissues. The block mixes a regulatory quantity mandate (the Renewable Fuel Standard), a tax-credit companion pair (S. 1277 / H.R. 2238), title-restricted positions on two energy bills, statute implementation, a free-standing tax-extender request, and a farm bill. Keeping it whole would force a single label onto instruments from different wedges.

Example 3: blank-line group with a header.

> "Monitor the following:
>
> Public Law 116-136, Coronavirus Aid, Relief, and Economic Security (CARES) Act, Economic development grant issues; Assistance to Firefighter grant issues; and Municipality related issues
>
> Public Law 116-260, Consolidated Appropriations Act, 2021, [. . .]
>
> Public Law 116-127, Families First Coronavirus Response Act, Municipality related issues
>
> H.R.6800, Health and Economic Recovery Omnibus Emergency Solutions (HEROES) Act, Municipality related issues"

The header ("Monitor the following:") is distributed to each blank-line segment, and each public law or bill becomes its own subissue with its attached topic list.

Example 4: enumerated list of funding requests.

> "FY 2012 National Defense Authorization and Defense Appropriations Bills and competitive grants seeking funding for the following project initiatives: 1) High Power Laser Technology Initiative.; 2) advanced materials design processing development project; 3) photonics based healthcare technologies for battlefield medicine with the Army; and 4) Comprehensive Solider Fitness. 5) Cancer biomarker detection for early cancer screening."

Split into five subissues, one per numbered item, each inheriting the stem ("FY 2012 NDAA and Defense Appropriations . . . seeking funding for"). All five are labeled appropriations requests, an instrument of the demand wedge.

Example 5: language-model split required (no reliable delimiters).

> "Issues pertaining to intellectual property rights including, but not limited to, patent protection and enforcement, copyright protection and enforcement in international markets, and antitrust, S. 3197 – Platform Competition and Opportunity Act of 2021, S. 2992 – American Innovation and Choice Online Act"

Commas here separate both topics and bills, so pattern rules cannot place boundaries. The language model produces four subissues: (1) patent protection and enforcement; (2) copyright protection and enforcement in international markets; (3) antitrust, S. 3197; (4) antitrust, S. 2992. The intellectual-property topics and the antitrust bills belong to different wedges.

Example 6: title enumeration creates distinct subissues.

> "S 3240, Agriculture Reform, Food, and Jobs Act, and HR 6083, Federal Agriculture Reform and Risk Management Act of 2012 with focus on Titles II, VIII, and IX. HR 5873, Forest Products Fairness Act."

The title enumeration is a real structure: Titles II (conservation), VIII (forestry), and IX (energy) of the farm bills are distinct instruments, so the block yields the per-title subissues plus H.R. 5873 separately.

Example 7: block correctly kept whole.

> "H.R. 1427, Federal Housing Finance Reform Act of 2007. All Provisions. GSE Regulation Reform."

One bill, one position ("All Provisions"), one topic restatement: a single subissue. Internal punctuation alone does not trigger a split; the guards require boundary evidence, and uncertain cases route to the language-model stage rather than being split by rule.

K Policy-Instrument Taxonomy and Model Crosswalk

The policy objects of the model, the industry-wide regulatory fixed cost $R(S)$, the firm's own profit tax τ_i, and its own procurement contracts g_i, must be measured from text that refers to none of them. A lobbying report names a statute, a rule, a bill, or a program; it does not say whether the policy at issue raises a fixed cost of operating or a cost that scales with production. Asking a language model to supply that judgment directly would place the central economic distinction of the paper inside a single step that can be neither inspected nor revised. We therefore separate the reading of the text from its economic interpretation. A

closed codebook of policy instruments, each defined by the statutory or regulatory language that identifies it rather than by what it does to costs, is written in advance, and a crosswalk from instruments to the terms of the firm's problem is fixed before any text is classified. The language model assigns text to instruments; the assignment of instruments to economic channels is ours, and takes the form of a fixed table.

K.1 Design

Classification proceeds in three stages. In the first, a language model reads a single text segment and returns, for each distinct request the segment contains, the identifier of a codebook instrument, the conditions defining whom the policy covers, the direction and, where the text states one, the magnitude of the change requested relative to current law, and the position the filer takes. Each of these is a statement about what the text says. In the second stage the identifier is matched to the codebook, which supplies the economics: the wedge the instrument moves, the government account on which it falls, the payoff it typically delivers to a firm that qualifies for it, the base with which that payoff scales, and, for instruments delivered through the tax code, the Treasury provision numbers. The match is mechanical. The third stage derives the quantities the analysis requires: the breadth of the request, whether it favors the firms it covers, the side of the contest on which the filer stands, and the firm's portfolio across wedges.

Two consequences of this division matter. Because the economic content of the classification is held in a table written in advance, it can be examined, disputed, and replaced without reclassifying a single filing, and every result can be recomputed under an alternative crosswalk. Because the classifier's errors and the crosswalk's errors are of distinct kinds, they can also be measured separately. The unit of classification is the provision rather than the bill or the filing: an omnibus statute moves many wedges at once, so a bill enters the panel as a vector of classified provisions, and a subissue containing several distinct requests yields one observation for each. A segment consisting only of a bill reference is recorded as such together with the resolved bill number, and its instruments arrive through the bill's provisions; a segment that names no policy object, or names a policy area too vaguely to place, is recorded as uninformative and excluded. Neither is forced onto an instrument.

K.2 The codebook and its sources

The codebook is intended to be exhaustive over the policy space on which firms lobby, and its completeness is borrowed rather than asserted: every instrument traces to an official or established inventory of its domain. We draw on the Treasury tax-expenditure inventory,

whose provision numbers the codebook retains; the regulatory-forms taxonomy of Xie and Perez (2018); the UNCTAD International Classification of Non-Tariff Measures; the Harmonized Tariff Schedule of the United States and Title 19 of the U.S. Code; the federal Assistance Listings; the Federal Acquisition Regulation; the patent statute and the exclusivity provisions administered by the Food and Drug Administration; the Immigration and Nationality Act; and, as a check on the inventory as a whole, the tools-of-government catalog of Salamon (2002). Coverage failures are not random: they concentrate in the domains a researcher knows least well and bias the measured portfolio toward whatever the codebook covers. Deriving the list from external inventories makes omissions visible to any reader who consults the same sources.

Instruments are organized by what they do to the firm's problem, not by whom they reach. A research credit written for a single firm and one available to the whole economy are the same instrument, distinguished only by the eligibility conditions the classifier reads from the text. Breadth is thus a property of the observation rather than of the codebook, which keeps the observed breadth of lobbying falsifiable.

K.3 Wedges and the crosswalk to the model

A wedge names the term of the firm's problem an instrument moves. Table 20 lists the ten wedges, the margin each moves, and the object of the model to which each corresponds. Three objects of the model are measured from the classification: regulatory fixed costs are the industry-wide burden $R(S)$ of the contest (13); the profit-tax wedge is the object of tax lobbying (11); and procurement contracts, an instrument of the demand wedge, are the object of procurement lobbying (12). The other wedges are recorded rather than discarded, which allows the classification to be reused for extensions and for counterfactuals defined on individual instruments.

Instruments whose incidence is conditional on concentration, among them merger control, conduct rules for dominant platforms, and regulation triggered by size thresholds, are kept separate from R_j, since their burden depends on a firm's position in the industry rather than falling on every active firm alike; they are the natural object of an extension in which lobbying acts on the transitions of the industry state directly. A requirement enters R_j only if its compliance cost is quasi-fixed; any component of the same statute whose cost scales with output is assigned to the output wedge.

K.4 What the classifier records

Table 21 lists what the classifier records, one observation for each request it identifies.

Table 20. Wedges and the Crosswalk to the Model

Wedge	Margin moved	Model object
R_j	Industry-wide quasi-fixed cost of operating, entering, or introducing a product	$R(S)$: **regulatory costs lobbying**
b_T	Profit-tax level; scales after-tax profit with no first-order effect on marginal prices	τ_i: **tax lobbying**
s^d	Demand facing the firm: government purchases, buyer subsidies, trade protection	g_i (procurement contracts): **procurement lobbying**; other instruments recorded
s^y	Per-unit wedge on marginal cost or marginal revenue	recorded; extension
s^r	Price of R&D and the private return to innovation	R&D cost χz^{ϕ}; extension
s^k	User cost of capital, including financing	recorded; no capital in the model
s^w	Effective wage bill	labor cost; extension
s^p	Price the firm may charge	markup; extension
$A(\Theta)$	Concentration-conditional rules: act on the industry state or switch on above a size threshold	transitions of Θ; extension
s^i	Lump-sum transfer to, or fee from, the firm	profit level; extension

Table 21. What the Classifier Records for Each Request

Field	Content
Instrument	The codebook instrument the text concerns, or a label marking the segment as a bare bill reference or as uninformative
Legislation	Bill and public-law numbers named in the text, resolved to the Congress in which they were introduced
Granularity	Whether the text is finer than the nearest instrument and, if so, a verbatim statement of the finer target
Eligibility	Whom the policy covers, over and above what the instrument itself implies: industry, geography, product, occupation, legal form, and certified status, recorded in standard classification systems
Conditions	Thresholds the policy places on quantities the firm chooses, expressed as comparisons against a closed list of model variables
Effect	The direction of the change requested relative to current law and, when the text states them, its magnitude and starting level
Position	The position the filer takes: support, opposition, either with an amendment, the pursuit of a specific change, monitoring, or none stated
Confidence	The classifier's confidence in the instrument assignment; low-confidence observations are dropped

Eligibility records only the conditions the text adds beyond the instrument's own definition, and because it is recorded in standard classification systems it also serves as the key that joins an observation to the firm panel. Conditions that bind on quantities the firm chooses are recorded against a closed list of variables that map into the model: output, revenue, price, employment, wages, capital, investment, research spending, profit, market share and concentration, merger events, and firm age. The direction of the requested change is one of a small set of operations, distinguishing the creation of an instrument from its expansion, reduction, extension, or repeal; magnitudes are recorded when the text states them and never imputed. Positions are recorded as the filer states them, and a filing that reports monitoring an issue is kept distinct from one that states no position.

K.5 From classified text to model objects

Besides the wedge, each instrument carries attributes that attach to every observation assigned to it: the government account (expenditure, receipt, or a mandate whose cost is borne privately), the sign of the payoff to a qualifying firm, and its shape, which records whether the payoff scales with a flow or is a lump independent of scale. The shape enforces the rule that only quasi-fixed burdens are regulatory costs.

The scope of a request is determined by the width of its eligibility conditions: a named entity yields a firm-specific request, an industry code with qualifiers a narrow one, an industry branch an industry-wide one, and unrestricted eligibility an economy-wide one. Whether a request favors the firms it covers follows from its direction and the instrument's sign. Let $\xi \in \{-1, 0, +1\}$ denote the requested change in the level of instrument k, with creation, increase, extension, reauthorization, and funding taking the value $+1$, reduction and repeal -1, and restatements and studies 0, and let $\text{sgn}_k \in \{-1, +1\}$ be the instrument's sign. Favorability is

$$\text{fav} \;=\; \xi \cdot \text{sgn}_k. \tag{65}$$

The same quantities place an observation on one side of the contest. For an observation on a regulatory costs instrument, let $\text{st} = +1$ when the filer supports the request and $\text{st} = -1$ when it opposes it. The filer would move the regulatory fixed cost in the direction $\text{st} \cdot \xi$: supporting a bill that raises a compliance requirement and opposing the repeal of one are the same political act, and the classification records them on the same side without a further judgment. The share of a firm-year's regulatory costs requests with $\text{st} \cdot \xi = +1$ determines its side, and the share of positions that ask for higher regulatory costs is the moment that disciplines the elasticity of superstars' compliance costs to their size in Section 5.5.

For a few instruments the sign is not a property of the instrument: a tariff protects the

firms that produce the good and raises the costs of the firms that use it. Observations on such instruments are resolved filer by filer, first from the stated position, because supporting a duty reveals the protected side, and then from a match between the affected good and the firm's own products or inputs. Classified observations collapse to a panel of firm, quarter, instrument, wedge, and side, from which every moment of Sections 3 and 5 is computed.

K.6 The instruments

The list below groups the 87 instruments of the codebook by wedge.

Regulatory fixed costs *(wedge R_j, 18 instruments)*

- Licenses and charters required to operate; product certification and inspection; pre-market product approval; facility permitting and certificates of need.
- Recurring reporting, recordkeeping, and registration duties; labeling and warning mandates; securities disclosure and shareholder-rights rules; data privacy and cybersecurity rules; one-time outlays required by a standard; financial assurance (surety) bonds.
- Health, safety, and technical requirements on imports; services trade and data-flow restrictions; foreign investment screening; issuance of limited rights to public resources.
- Liability immunities and shields, limits on damages, expansion of liability exposure; conduct rules on how a transaction is made.

Profit tax *(wedge b_T, 7 instruments)*

- The general corporate income tax; deductions from taxable profit; reduced rates on classes of income; reduced tax on foreign and export income; exemption by legal form; deferral and basis rules; deductibility of losses and forgiven debt.

Demand *(wedge s^d, 14 instruments)*

- Government demand: program funding, appropriations, and authorizations; procurement contracts; public infrastructure and services used by firms; tax exemption for state and local government bonds.
- Buyer purchase subsidies and purchase mandates; domestic-content requirements for public purchases; markets reserved by law to designated sellers; advertising and marketing restrictions; government provision of competing services.
- Trade: import tariffs and tariff-rate quotas; limits on import quantities; antidumping and countervailing duties; duty-free trade under preference programs; export controls and sanctions.

Output *(wedge s^y, 17 instruments)*

- Per-unit production credits and support payments; percentage depletion; excise and sales taxes; taxes on emissions and discharges; regulatory levies on revenue; public-resource royalties.

– Technology and performance standards with per-unit costs; bank capital and liquidity requirements; limits on domestic production; tradable permits; portfolio and blending standards; prohibitions of products; obligations to serve; insurance coverage mandates.
– Duty suspensions and drawback; customs procedures and rules of origin; federal insurance backstops.

Innovation cost *(wedge s^r, 5 instruments)*

– R&D tax credits and expensing; R&D grants and technology transfer; patent, copyright, trademark, and trade-secret law; market exclusivity for approved products; enforcement of intellectual property against foreign infringement.

Capital cost *(wedge s^k, 6 instruments)*

– Investment tax credits and expensing; accelerated and bonus depreciation; tax-exempt private-activity bonds; federal lending and loan guarantees; creditor rights in bankruptcy; deductibility of business interest.

Labor cost *(wedge s^w, 8 instruments)*

– Hiring and wage credits; untaxed employee benefits; tax deferral on retirement contributions; payroll taxes; wage floors, hour rules, and benefit mandates; workplace safety and worker protections; employee status and collective bargaining; restrictions on hiring foreign workers.

Set prices *(wedge s^p, 5 instruments)*

– Government-set payment rates; mandatory discounts, rebates, and negotiated prices; price ceilings; price floors; allowed rates of return and revenue caps.

Competition *(wedge $A(\Theta)$, 4 instruments)*

– Merger control and conduct rules; platform gatekeeper regulation; antitrust exemptions; mandated access and interoperability.

Lump-sum transfers and fees *(wedge s^i, 2 instruments)*

– Grants and direct payments to firms; regulatory user fees.

Place-based bundles *(wedge 1 instrument)*

– Zone regimes that combine an employment credit, a financing subsidy, and a capital-gains preference; decomposed into their components, with the geographic restriction recorded as a condition of eligibility.

K.7 Validation protocol

Since the crosswalk is deterministic, validation concerns the classifier's output alone. Section 2.2 reports the completed checks: agreement with an independent classification by a second language model, agreement with the positions recorded by MapLight, and the stability of repeated classification. The protocol in progress adds a hand-coded sample. Both authors code a stratified sample of segments independently, with sampling weighted toward the segments carrying the most lobbying dollars, and adjudicate disagreements in writing; agreement between independent coders bounds what any automated procedure can reach, and separates the fields that are intrinsically difficult, chiefly the filer's position and the direction of the change, from those that are close to mechanical, such as the identification of a named statute. Against this sample we will report precision, recall, and F_1 for the assignment of instruments, by instrument and by wedge, per-field accuracy for eligibility, conditions, direction, and position, and the confusion matrix across wedges, and use it to debias the estimates following Carlson and Dell (2025). The final checks operate on the moments rather than the labels: re-computing the target moments without low-confidence observations, with ambiguous observations reassigned to their runner-up class, and at the level of wedges rather than instruments.

K.8 Examples

Consider first a statutory provision: "Producers of electricity from wind receive a credit of 1.5 cents per kilowatt-hour of electricity produced." The classifier assigns it to per-unit production credits, records wind generation as the finer target, sets eligibility to the electric-power industry nationally, and reads the effect as the creation of a credit of \$0.015 per kilowatt-hour from a base of zero. The codebook supplies the output wedge and a favorable sign; aggregation supplies a narrow breadth, a change of $+1$, and, through (65), a favorable request.

Consider next a provision granting manufacturers with annual revenue below one million dollars a one-time grant of five thousand dollars. The classifier assigns it to grants and direct payments, sets eligibility to manufacturing, records the revenue threshold as a condition on a variable the firm chooses, and reads the effect as the creation of a one-time payment. Because the threshold is recorded as a condition, the observation implies a notch in the firm's problem at one million dollars rather than a transfer to small firms in general.

Consider finally a filing that reads "Support extension of the renewable electricity production tax credit; seek inclusion of energy storage." This is two requests and therefore two observations: an extension of the existing credit with a supporting position, and an expan-

sion of its qualifying technologies to energy storage, with the pursuit of a specific change as the position. Both carry $\xi = +1$ on a favorably signed instrument, so both are favorable requests: an extension and a widening of scope point in the same direction, and the classification recovers this without a further judgment.